

Multi-scale attention fusion for enhanced transformer models in intrusion detection systems

Samia Saidane^{a,*}, Francesco Telch^b, Kussai Shahin^b, Fabrizio Granelli^a

^a DISI, University of Trento, Trento, Italy

^b Trentino Digitale Spa, Trento, Italy

ARTICLE INFO

Keywords:

Intrusion detection systems (IDS)
Transformer models
MEGA
FNet
Multi-scale attention fusion (MAF)
Cybersecurity
Class imbalance
Attention mechanisms
Experience replay

ABSTRACT

Modern network environments generate vast streams of complex and evolving traffic data, presenting significant challenges for accurate and real-time intrusion detection. Conventional deep learning approaches often fail to effectively capture the multi-scale temporal patterns and are frequently hampered by severe class imbalance and catastrophic forgetting when faced with non-stationary data streams. To overcome these limitations, this paper introduces a novel Multi-scale Attention Fusion (MAF) module, a general-purpose architectural enhancement for transformer-based models designed to achieve synergistic optimization across three critical dimensions: computational efficiency, continual learning adaptability, and multi-scale temporal perception. We present two instantiations of this approach: **MEGA + MAF** and **FNet + MAF**. These models synergistically combine short-term local context, long-range dependencies, and global sequence information through an adaptive, learnable gating mechanism. A comprehensive evaluation across four diverse benchmark datasets—*ToN-IoT*, *X-IIoTID*, *CIC-CEVS2024*, and *CSE-CIC-IDS2018*—demonstrates state-of-the-art performance with balanced optimization across all three dimensions: (1) **FNet + MAF** achieved superior computational efficiency with up to 8.5× **lower memory** and 5.3× **faster inference** while maintaining high accuracy; (2) **MEGA + MAF** demonstrated exceptional continual learning capability, achieving **99.10%** accuracy in dynamic streaming environments while effectively eliminating backward forgetting (**0.00%**) and minimizing forward forgetting (**0.06%**); and (3) Both models exhibited robust multi-scale perception, capturing threats across short-term bursts, mid-range sessions, and global traffic patterns with up to **99.97%** F1-score. Our ablation study after 20 epochs of training identifies 80 tokens as optimal, achieving 85.20% accuracy with 145.1 samples/second throughput. Interpretability analyses further confirm that the models learn robust and semantically meaningful feature representations aligned with network security semantics. The proposed framework represents a significant advancement toward building adaptive, next-generation intrusion detection systems capable of evolving with emerging threats while maintaining operational efficiency in resource-constrained environments.

1. Introduction

The pervasive adoption of the Internet has significantly enhanced convenience and connectivity in modern society, but it has also introduced growing challenges in maintaining cybersecurity. As network infrastructures scale in complexity—driven by the proliferation of Internet of Things (IoT) devices, smart objects, mobile platforms, sensor networks, and distributed cyber-physical systems—the volume, velocity, and heterogeneity of network traffic continue to increase sharply. This expansion exposes critical systems to a broader range of security vulnerabilities and sophisticated malicious behaviors [1]. Intrusion Detection Systems (IDS) serve as a first line of defense against such threats. IDS de-

ployments are commonly categorized as *host-based* (HIDS), which monitor activity on individual endpoints, and *network-based* (NIDS), which inspect traffic traversing network infrastructure [2]. Detection strategies typically fall into two paradigms: *signature-/misuse-based* detection, which matches traffic against known attack patterns, and *anomaly-based* detection, which flags statistically rare or behaviorally deviant activity relative to learned baselines [3].

Building on these foundational approaches, modern networks face an ever-expanding spectrum of cyber threats, ranging from large-scale Distributed Denial of Service (DDoS) attacks and coordinated ransomware campaigns to stealthy reconnaissance techniques such as port scanning [4]. The increasing frequency, diversity, and sophistication of

* Corresponding author.

E-mail addresses: samia.saidane@unitn.it (S. Saidane), francesco.telch@tndigit.it (F. Telch), kussai.shahin@tndigit.it (K. Shahin), fabrizio.granelli@unitn.it (F. Granelli).

<https://doi.org/10.1016/j.comnet.2025.111985>

Received 24 September 2025; Received in revised form 15 December 2025; Accepted 29 December 2025

Available online 11 January 2026

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such attacks—particularly DDoS—have intensified the demand for resilient, adaptive, and data-driven intrusion detection solutions. Successful intrusions can lead to severe operational disruptions and financial losses across enterprise, industrial, and public-sector domains. Consequently, research efforts have accelerated toward the development of next-generation NIDS that combine the strengths of traditional methods with advanced machine learning and deep learning approaches [5], aiming to overcome limitations such as low detection rates for emerging attack variants and high false alarm rates under real-world traffic variability [6]. Within this shift toward data-driven IDS, early machine learning research explored classifiers such as k -nearest neighbors (k-NN) [7], Naive Bayes [8], decision trees [9], support vector machines (SVM), and random forests [10]. These approaches offered simplicity, data efficiency, and fast training, but struggled to cope with the high dimensionality, severe class imbalance, and non-stationary characteristics of modern network traffic. To address these limitations, deep learning (DL) methods—such as Multi-Layer Perceptrons (MLP) [11], Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) [12], and Long Short-Term Memory (LSTM) networks—have emerged as powerful alternatives, delivering richer feature representations and improved detection accuracy. However, these gains often come at the cost of increased computational complexity, longer training times, and reduced adaptability to evolving network behaviors at scale. Despite significant progress, current ML- and DL-based IDS still face critical challenges in keeping pace with rapidly evolving network technologies and attack strategies [13]. Many models fail to adequately capture the complex *contextual* and *temporal* structures inherent in network traffic, limiting their sensitivity to stealthy or low-and-slow attacks. Centralized training pipelines can create performance bottlenecks as user and device counts grow, while scalability, memory footprint, and model update latency remain practical concerns [14]. High false positive rates (FPRs) persist, particularly when benign and malicious traffic share overlapping feature distributions [15]. A key limitation is that many legacy IDS pipelines treat network flows as independent samples, ignoring inter-flow dependencies that are critical for recognizing coordinated or staged attacks. To address this gap, sequence-based modeling across *groups* of flows has been proposed to improve contextual awareness [15]. Building on this idea, sequence-driven approaches that learn “normal” network behavior over time (e.g., log-based anomaly detection) have demonstrated improved sensitivity to context-dependent events. However, when attack instances are sparse within a sequence, deviations from baseline behavior can be subtle and easily overlooked. While RNNs and LSTMs have been widely used to model temporal dependencies in time-series and network traffic data [15,16], they suffer from vanishing gradients and a limited ability to retain long-range dependencies, especially when dealing with highly variable traffic intervals. These limitations have motivated researchers to adopt a Natural Language Processing (NLP) perspective on network telemetry, leveraging its inherently sequential nature. Packets, flows, and log events can be serialized into symbolic streams that encode both semantic and structural relationships, much like words in a sentence. This analogy has opened the door for applying NLP techniques to cybersecurity tasks such as red-team behavior detection, threat triage, and behavioral profiling across distributed systems. Transformer architectures, originally introduced for machine translation and now foundational to large language models (LLMs), excel at modeling long-range dependencies and contextual hierarchies in sequential data [17,18]. Their self-attention mechanism facilitates flexible, content-driven interactions across arbitrary token positions, making them particularly well suited to the non-uniform and bursty nature of modern network traffic.

The success of Transformer derivatives such as GPT [19] and BERT [18] underscores the ability of attention-based models to capture deep semantic and structural patterns, enabling both coherent generation and robust transfer across diverse tasks [20]. These strengths suggest that Transformer-like architectures can be adapted to learn the statistical regularities of benign network traffic and to identify even subtle anomalies

associated with emerging or stealthy attacks. While the adoption of Transformer models in cybersecurity is promising, several challenges remain when applying standard architectures directly to intrusion detection:

Sequence Complexity. Network traffic often exhibits burstiness, irregular timing, mixed protocols, and multi-scale structure (packet-, flow-, session-, and host-level contexts). Vanilla Transformers, which treat all token interactions uniformly, fail to isolate the distinct temporal regimes that correspond to different attack modalities.

Class Imbalance. Real-world datasets are heavily skewed: benign traffic dominates, while many attack classes are rare. Models trained naively on such data become biased toward majority classes, leading to poor recall on critical but infrequent threats [15].

Scalability and Efficiency. IoT, edge, tactical, and bandwidth-constrained deployments require models that are memory-efficient and low-latency. Standard Transformer self-attention scales quadratically with sequence length, creating barriers to real-time operation in high-throughput settings.

Interpretability. Operational cybersecurity workflows demand transparent decision support. Black-box predictions hinder analyst trust, incident response prioritization, and compliance reporting; attention signals offer one route toward interpretable IDS.

The convergence of these issues motivates the development of next-generation IDS methods that (i) capture *temporal context across multiple scales*, (ii) remain *data- and class-imbalance aware*, (iii) are *efficient enough for real-time and edge deployment*, and (iv) provide *interpretable attention cues* to support analyst-driven investigations. In this paper we pursue this direction by advancing Transformer-based IDS modeling toward architectures that can better exploit the structured, language-like nature of network telemetry while overcoming the limitations of prior ML-, DL-, and vanilla Transformer-based systems.

The main contributions of this paper are as follows:

1. **Novel Multi-Scale Attention Fusion (MAF) Module:** We propose a novel Multi-Scale Attention Fusion (MAF) module that enhances transformer architectures by explicitly modeling temporal dependencies across three complementary scales: short-term (via depth-wise convolution), long-range (via sparse, strided attention), and global context (via pooled cross-attention). A learnable softmax gating mechanism dynamically fuses these representations, enabling robust and adaptive feature extraction for both high-frequency volumetric attacks and stealthy, low-frequency threats. To our knowledge, this constitutes the first learnable gated multi-scale attention mechanism designed specifically for the intrusion detection task.
2. **Dual MAF-Enhanced Architectures for Varied Deployment Scenarios:** We design and implement two distinct, high-performance intrusion detection systems by integrating the MAF module with different transformer backbones, each optimized for specific operational constraints:
 - **FNet+MAF** combines Fourier-based token mixing with MAF, creating a highly efficient architecture ideal for resource-constrained or edge deployments where computational overhead is a primary concern.
 - **MEGA+MAF** integrates moving average-equipped gated attention with MAF, forming a powerful model that excels in stability and accuracy for centralized, high-throughput security monitoring of complex and dynamic traffic.
3. **Extensive Empirical Validation Across Diverse Benchmarks:** We conduct a comprehensive evaluation across four publicly available datasets—ToN-IoT, X-IoTID, CICEVS2024, and CSE-CIC-IDS2018—spanning IoT, industrial IoT, vehicular security, and general network environments.

4. **Demonstrated Superiority in Online Continual Learning:** We rigorously evaluate our framework in a challenging online continual learning setting, demonstrating its resilience to catastrophic forgetting.
5. **Enhanced Model Interpretability and Explainability:** We provide deep insights into model reasoning through feature importance analysis, revealing that learned representations align with known network attack semantics (e.g., ‘duration’ for BACKDOOR, byte flows for XSS), thereby increasing trust and operational utility.
6. **Effective Handling of Class Imbalance and Rare Threats:** Our approach demonstrates a remarkable capability to mitigate the common failure of models on low-support classes, significantly improving precision and F1-score on rare but critical attacks.

2. Related works

Network intrusion detection (NIDS) can be framed as both a supervised anomaly detection and classification problem. Recent research has explored a wide range of approaches, from traditional statistical techniques to advanced deep learning architectures, with an increasing focus on Transformer-based models [21]. Below, we categorize and summarize the most relevant works, emphasizing their strengths and limitations in terms of performance, interpretability, and real-time applicability. Initial efforts to improve NIDS performance focused on hybrid deep learning models that combine spatial and temporal feature extraction. Farhan et al. [22] proposed a hybrid model combining CNNs and LSTMs to extract both spatial and temporal features from network traffic. While the CNN captures localized patterns, the LSTM models long-term dependencies, enabling the detection of diverse attacks. However, this approach suffers from high computational complexity, which may limit its applicability in real-time or resource-constrained environments.

Building on this direction, Hussain et al. [23] proposed a more elaborate hybrid intrusion detection system, DCNNBiLSTM, which integrates Convolutional Neural Networks (CNN) for spatial feature extraction with Bidirectional LSTMs (BiLSTM) to capture temporal dependencies, followed by a fully connected DNN for classification. The model was trained and evaluated on the CICIDS2018 and Edge_IIoT datasets, achieving detection accuracies of 100% and 99.64%, respectively. Their approach demonstrates improved performance over individual deep learning models such as DNN, CNN, AE, and LSTM in both accuracy and loss metrics. Despite its effectiveness, the model suffers from increased training complexity and time, making deployment in real-time settings more challenging. Moreover, the performance shows sensitivity to data imbalance, and the method’s ability to generalize to zero-day attacks remains unexplored.

While hybrid models captured both spatial and temporal dynamics, another major stream of research explored sequence modeling techniques for detecting anomalies in network behavior. LSTM-based architectures have been widely adopted due to their ability to model sequential dependencies. Zhu et al. [24] applied LSTM models to capture temporal dependencies in network traffic. Although effective, LSTMs require large labeled datasets and struggle to capture long-range dependencies efficiently. In contrast, Transformer-based architectures like BERT have emerged as strong alternatives due to their ability to extract contextual information without recurrent connections. Expanding this line of research, Ullah & Mahmoud [25] proposed a suite of recurrent neural network (RNN)-based architectures, including LSTM, BiLSTM, and GRU, for anomaly detection in IoT networks. They also developed a hybrid model combining CNNs for spatial feature extraction with RNNs for temporal sequence modeling. Their models were evaluated on seven diverse IoT datasets, namely NSL-KDD, BoT-IoT, IoT-NI, IoT-23, MQTT, MQTTset, and IoT-DS2, demonstrating superior performance in both binary and multi-class intrusion detection tasks with high accuracy, precision, recall, and F1-score. However, their approach shows potential limitations: it relies heavily on large volumes of labeled data to gener-

alize effectively, and certain models risk overfitting due to the use of dataset-specific features like static source ports.

Given the limitations of RNNs in handling long-range dependencies and high computational costs of hybrid architectures, Transformer-based methods have emerged as a promising alternative in recent years. Nguyen et al. [26] enhanced flow-based NIDS by integrating NLP techniques with BERT for domain adaptation. While this approach achieved strong results on datasets like CIDDs-001 and CIDDs-002, the computational cost of BERT (e.g., large parameter count, long training time, and high memory consumption) poses significant challenges for real-time deployment. Building further, the Robust Transformation-based IDS (RTIDS) proposed in [27] utilizes a Transformer encoder-decoder architecture with position embeddings to model sequential feature relationships. While RTIDS effectively compresses high-dimensional data into compact feature representations, the decoder’s reconstruction of these features partially offsets the benefits of dimensionality reduction, affecting overall performance. A different approach was taken in [28], which introduced a Vision Transformer (ViT)-based IDS with a sliding window strategy to better capture local feature patterns. A hierarchical focal loss was incorporated to mitigate data imbalance. However, focal loss alone was insufficient to fully address severe class imbalance, leaving the model prone to biased learning. BERT-based intrusion detection has also been explored by [29], where BERT embeddings were combined with a Multi-Layer Perceptron (MLP) for classification. The authors also employed the Synthetic Minority Over-sampling Technique (SMOTE) to alleviate class imbalance, achieving high accuracy (up to 99.83%) on datasets such as CIC-IDS 2017 and UNSW-NB 2015. Despite these results, the computational cost of BERT remains a key limitation for real-time IDS. To bridge the gap between performance and computational efficiency, researchers have begun developing hybrid Transformer architectures that combine the strengths of attention mechanisms with lightweight components. A hybrid Transformer-CNN-LSTM model was introduced in [30], where Transformers are first used for feature extraction and the resulting embeddings are processed by CNN and LSTM layers. SMOTE was again applied to address class imbalance. Similarly, Wang et al. [31] proposed a similarity-driven aggregation algorithm followed by a Transformer model to handle variable-length inputs and provide precise attack predictions. Their evaluation included a VMware-based testbed environment simulating various attacks such as XSS, SQL injection, and Remote Code Execution (RCE).

A notable advancement in this space is the CTSF framework by Chai et al. [32], a hybrid intrusion detection framework tailored for the Industrial Internet. CTSF combines convolutional neural networks (CNNs) and an improved Transformer for pretraining, followed by Support Vector Machine (SVM)-based classification. The CNN extracts fine-grained local features, while the enhanced Transformer, modified by removing decoder self-attention, captures global dependencies with reduced training complexity. This dual-path design enables CTSF to extract comprehensive feature representations, and the SVM component establishes precise decision boundaries using both linear and RBF kernels. Evaluated on an imbalanced subset of the X-IIoTID dataset, CTSF achieved a high accuracy of 98.875% and demonstrated strong performance in identifying minority class intrusions. However, the model’s computational complexity remains a concern; despite optimizations to the Transformer, CTSF is not lightweight and incurs high training and inference costs. This hinders real-time deployment and raises resource and maintenance demands in Industrial IoT environments.

While existing works like CTSF have achieved considerable progress, they continue to suffer from key limitations: (i) challenges in handling imbalanced traffic distributions, (ii) insufficient modeling of heterogeneous temporal patterns, (iii) high computational complexity, and (iv) lack of adaptability to evolving attack vectors. Among Transformer-based methods, BERT has been popular due to its strong contextual modeling capabilities. However, its full self-attention mechanism is computationally expensive and unsuitable for long network traffic sequences. Moreover, its fixed context window (512 tokens) leads to truncation

of long-range dependencies, which are critical for detecting stealthy or slow-evolving threats. Critically, architectures such as BERT or even the modified Transformer in CTSF often fail to explicitly model multi-scale dependencies—namely, short-term local patterns, long-range temporal correlations, and global contextual semantics—within a unified framework. This deficiency hinders their ability to capture the diverse temporal behaviors of modern network traffic, such as DoS attacks or slow-moving stealth intrusions.

To address these gaps, we turn to the concept of multi-scale attention mechanisms. Several works have explored multi-scale attention for various domains. In computer vision, pyramid vision transformers [33] use spatial reduction for multi-scale feature extraction. For time-series analysis, hierarchical transformers [34] employ fixed downsampling strategies. However, these approaches lack *adaptive gating mechanisms* and are not optimized for network traffic characteristics. Our proposed Multi-scale Attention Fusion (MAF) module differs by: (1) employing learnable temperature-scaled gating with regularization to prevent scale collapse, (2) designing scales specifically aligned with network traffic patterns (packet-level, session-level, network-level), and (3) incorporating traffic-aware initialization based on attack type indicators. By integrating this module, our framework directly tackles the limitations of prior work—particularly the insufficient modeling of heterogeneous temporal patterns and high computational complexity—enabling efficient and adaptive capture of the multi-scale dependencies essential for robust intrusion detection.

To address these limitations, we introduce two novel architectures: **MEGA + MAF** and **FNet + MAF**. The MEGA + MAF model incorporates Moving Average Equipped Gated Attention (MEGA) for adaptive memory retention, combined with our proposed Multi-Scale Attention Fusion (MAF) module to capture short-term, long-range, and global semantic patterns. : write correctly and make a good onnection between the 2 paragraphs and keep the colors.

3. Proposed transformer architectures for intrusion detection

3.1. Multi-scale attention fusion (MAF)

To enhance the representational power of the MEGA and FNet based Transformer, we introduce a Multi-scale Attention Fusion (MAF) module that fuses local, long-range, and global attention outputs. This design addresses the diverse temporal patterns in intrusion detection by capturing both fine-grained and coarse-grained dependencies in the input sequence. Specifically, MAF computes three types of attention:

- **Short-range attention**, applied on the full hidden state sequence, captures local dependencies within immediate context windows.
- **Long-range attention**, computed on a downsampled version of the sequence, identifies coarser, periodic, or delayed patterns across longer temporal spans.
- **Global attention**, where a pooled global query attends to the full sequence, encodes overall sequence-level semantics useful for anomaly or outlier detection.

The outputs from these three attention branches are adaptively fused using a learnable gate vector. This gating mechanism assigns dynamic importance weights to each scale, allowing the model to emphasize the most informative context per sample. The full forward pass of the MAF module is summarized in [Algorithm 1](#).

3.2. Detailed implementation of MAF operations

This section provides detailed textual explanations of the key operations in the MAF module, complementing the pseudocode in [Algorithm 1](#).

Downsample and Upsample Operations. The long-range attention pathway employs **stride-based temporal downsampling** to capture

Algorithm 1 Multi-scale attention fusion (MAF) module with implementation details.

Input: Hidden states $H \in \mathbb{R}^{B \times T \times D}$ from encoder

Output: Fused representation $F \in \mathbb{R}^{B \times T \times D}$, gating weights α

B: Batch size, T: Sequence length (80), D: Hidden dimension (128)

1. Initialization:

$\beta \leftarrow [\beta_s, \beta_l, \beta_g]^*$

Learned gating parameters, initialized with Kaiming uniform

$\tau \leftarrow 1.0^*$

Temperature parameter, initialized to 1.0

$r \leftarrow 4^*$

Downsample stride for long-range pathway

$k \leftarrow 3^*$

Kernel size for short-range convolution

2. Short-range Attention (Packet-level):

$H_{\text{norm}}^s \leftarrow \text{LayerNorm}(H)^*$

Pre-normalization for stability

$S_{\text{conv}} \leftarrow \text{DepthwiseConv}_{k=3}(H_{\text{norm}}^s)^*$

Kernel size 3 captures adjacent packets

$S \leftarrow \text{LayerNorm}(S_{\text{conv}} + H)^*$

Residual connection

3. Long-range Attention (Session-level):

$H_{\text{norm}}^l \leftarrow \text{LayerNorm}(H)$

$H_{\text{long}} \leftarrow H_{\text{norm}}^l[:, :: r, :]^*$

Stride-based downsampling, $r=4$

$L_{\text{attn}} \leftarrow \text{MultiHeadAttention}(H_{\text{long}}, H_{\text{long}}, H_{\text{long}})$

$L_{\text{upsample}} \leftarrow \text{Repeat}(L_{\text{attn}}, \text{dim} = 1, \text{repeats} = r)^*$

Nearest-neighbor upsampling

$L \leftarrow L_{\text{upsample}}[:, : T, :]^*$

Trim to original length $T=80$

4. Global Attention (Network-level):

$H_{\text{norm}}^g \leftarrow \text{LayerNorm}(H)$

$Q_{\text{global}} \leftarrow \text{MeanPool}(H_{\text{norm}}^g, \text{dim} = 1)^*$

Global average pooling

$G_{\text{attn}} \leftarrow \text{MultiHeadAttention}(Q_{\text{global}}, H_{\text{norm}}^g, H_{\text{norm}}^g)$

$G \leftarrow \text{Repeat}(G_{\text{attn}}, \text{dim} = 1, \text{repeats} = T)^*$

Broadcast to sequence length

5. Adaptive Gated Fusion:

$\alpha \leftarrow \text{Softmax}(\beta/\tau)^*$

Temperature-scaled softmax

$\mathcal{L}_{\text{reg}} \leftarrow 0.01 \cdot \sum_{i=1}^3 \alpha_i^2^*$

L2 regularization on gating weights

$F_{\text{fused}} \leftarrow \alpha_s \cdot S + \alpha_l \cdot L + \alpha_g \cdot G$

6. Output Normalization:

$F \leftarrow \text{LayerNorm}(F_{\text{fused}})$

$F \leftarrow \text{Dropout}(F, p = .1)$

return $F, \alpha, \mathcal{L}_{\text{reg}}$

session-level patterns. Given input hidden states $H \in \mathbb{R}^{B \times T \times D}$:

$$H_{\text{long}} = H[:, :: r, :] \quad \text{where } r = 4 \quad (1)$$

This stride of 4 was empirically determined to align with typical network session boundaries (approximately 4 packets/flow for meaningful session context). The downsampled sequence length becomes $T' = \lceil T/r \rceil$, reducing computational complexity from $O(T^2 D)$ to $O((T/r)^2 D)$. After applying multi-head attention to H_{long} , we perform **nearest-neighbor upsampling**:

$$L_{\text{upsampled}} = \text{repeat}(L, \text{repeats} = r, \text{dim} = 1) \quad (2)$$

where each temporal position in L is repeated r times to restore the original sequence length. This approach preserves the session-level semantics while maintaining alignment with the original temporal resolution.

Gating Parameter Initialization and Optimization. The gating parameters $\beta = [\beta_s, \beta_l, \beta_g]$ control the relative importance of short-range, long-range, and global attention pathways. These parameters are initialized using **Kaiming uniform initialization**:

$$\beta_i \sim \mathcal{U}\left(-\sqrt{\frac{6}{d_{\text{in}}}}, \sqrt{\frac{6}{d_{\text{in}}}}\right) \quad \text{for } i \in \{s, l, g\} \quad (3)$$

where d_{in} is the input dimension to the gating layer. The temperature parameter τ is initialized to 1.0, allowing moderate sparsity in attention allocation. During training, we apply L2 regularization to the softmax-normalized weights:

$$\mathcal{L}_{\text{gate}} = \lambda \sum_{i=1}^3 \alpha_i^2, \quad \lambda = 0.01 \quad (4)$$

This regularization prevents degenerate solutions where a single pathway dominates ($\alpha_i \rightarrow 1, \alpha_{j \neq i} \rightarrow 0$), ensuring balanced utilization across scales.

Traffic-Aware Pathway Design. Each attention pathway is specifically designed for network traffic characteristics:

- **Short-range pathway:** Employs depthwise convolution with kernel size $k = 3$:

$$S = \text{DWConv}_{k=3}(H) + H \quad (5)$$

This captures packet-level burst patterns typical in DDoS attacks and port scans, where local context (adjacent packets) contains critical detection signals.

- **Long-range pathway:** Uses sparse attention on downsampled sequences ($r = 4$) to model session-level dependencies:

$$L = \text{MHA}(H_{\text{long}}, H_{\text{long}}, H_{\text{long}}) \quad (6)$$

This is particularly effective for detecting slowloris attacks and credential stuffing, where attack patterns manifest over multiple packets within a session.

- **Global pathway:** Computes a pooled representation of the entire sequence:

$$Q_{\text{global}} = \frac{1}{T} \sum_{t=1}^T H_t \quad (7)$$

This provides network-wide context essential for detecting advanced persistent threats (APTs) and lateral movement, where malicious behavior spans multiple sessions.

Normalization and Stabilization. To ensure training stability, we employ **pre-layer normalization** before each attention operation:

$$H_{\text{norm}} = \text{LayerNorm}(H) \quad (8)$$

followed by dropout with probability $p = 0.1$ after fusion. The final output is computed as:

$$F = \text{LayerNorm}(\alpha_s S + \alpha_l L + \alpha_g G) \quad (9)$$

with residual connections from the original hidden states to each pathway, preserving gradient flow during backpropagation.

Computational Complexity Analysis. The MAF module maintains efficient computation through careful design:

Short-range: $O(TkD)$ (convolution)

Long-range: $O((T/r)^2 D)$ (sparse attention)

Global: $O(TD^2)$ (cross-attention)

Total: $O(TkD + (T/r)^2 D + TD^2)$

Compared to standard self-attention ($O(T^2 D)$), MAF provides approximately 4× reduction in computation for typical parameters ($T = 80, r = 4, k = 3, D = 128$), making it suitable for real-time intrusion detection.

3.3. Distinctive features of MAF compared to existing attention mechanisms

While several multi-scale attention mechanisms exist in the literature, our proposed MAF module introduces several key innovations specifically tailored for network intrusion detection:

Adaptive Multi-Scale Gating Mechanism. Unlike conventional multi-head attention where heads operate independently or fixed-weight multi-scale approaches, MAF employs a **learnable temperature-scaled softmax gating mechanism** with regularization. The gating weights $\alpha = [\alpha_s, \alpha_l, \alpha_g]$ are computed as:

$$\alpha_i = \frac{\exp(\beta_i / \tau)}{\sum_{j=1}^3 \exp(\beta_j / \tau)}, \quad \tau = \text{ReLU}(w_\tau) + \epsilon \quad (10)$$

with L2 regularization $\mathcal{L}_{\text{gate}} = \lambda \sum_{i=1}^3 \alpha_i^2$ ($\lambda = 0.01$). This design prevents degenerate solutions (single-scale dominance) while allowing adaptive emphasis on different temporal regimes based on attack characteristics. For example, DDoS attacks automatically increase α_s for short-term detection, while slowloris attacks emphasize α_l for long-range patterns.

Network Traffic-Aware Scale Design. The three scales in MAF are specifically designed around network traffic characteristics:

- **Short-term (3-token convolution):** Optimized for burst detection (port scans, SYN floods)
- **Long-range (stride=4 downsampling):** Captures session-level dependencies (credential stuffing, slow attacks)
- **Global (pooled attention):** Provides network-wide context (APTs, lateral movement)

This contrasts with generic multi-scale approaches that use arbitrary scaling factors not aligned with network traffic semantics.

Efficiency-Aware Implementation. MAF incorporates computational optimizations specific to network traffic:

- **Selective downsampling:** Only applied to long-range pathway, maintaining full resolution for short-term detection
- **Pooled global query:** Avoids quadratic complexity of full self-attention
- **Depthwise convolution:** Lightweight local pattern extraction suitable for packet-level features

Comparison with Related Approaches. Table 1 summarizes key distinctions between MAF and existing attention mechanisms:

3.4. FNet+MAF: Multi-scale attention fusion on a Fourier transformer backbone

FNet is an attention-free Transformer variant that substitutes the computationally expensive self-attention mechanism with a more efficient Fourier-based token mixing strategy. This approach leverages the Discrete Fourier Transform (DFT) to mix token representations across the sequence, significantly reducing complexity while maintaining the model's ability to learn both global and local dependencies [35]. This architectural innovation makes FNet especially well-suited for applications involving long input sequences or resource-constrained environments. In the context of intrusion detection using NetFlow data, FNet offers several advantages: It efficiently processes large volumes of time-sequenced network traffic, detects periodic or anomalous traffic behaviors, and maintains high throughput required for real-time monitoring.

Table 1
Comparison of MAF with existing attention mechanisms.

Feature	Standard Self-Attention	Multi-Head Attention	Hierarchical Attention	MAF (Ours)
Scale Adaptation	Single scale	Single scale, multiple heads	Fixed hierarchical scales	Adaptive three-scale fusion
Gating Mechanism	None	Implicit via attention weights	Fixed/Manual weighting	Learnable temperature-scaled gating
Traffic Awareness	General purpose	General purpose	Domain-agnostic	Network traffic-optimized scales
Computational Overhead	$O(T^2 D)$	$O(T^2 D)$	$O(T^2 D/k)$	$O(T^2 D/4 + T k D)$
Attack Type Specialization	None	None	Limited	Dynamic per-attack specialization
Regularization	None	Dropout only	Weight decay only	L2 regularization on gating weights

3.4.1. FNet + MAF for network intrusion detection

FNet is a Transformer architecture that replaces self-attention with Fourier transforms for efficient token mixing, providing a favorable trade-off between performance and computational efficiency [35]. We propose **FNet + MAF**, a hybrid architecture specifically designed for network intrusion detection that combines the computational efficiency of FNet with multi-scale temporal analysis through our Multi-scale Attention Fusion (MAF) module. This architecture processes network traffic data where each token x_t represents feature-encoded network events (packet headers, flow statistics, or behavioral patterns).

The model processes input sequences $x = (x_1, \dots, x_T)$ through:

$$\mathbf{H} = \text{FNetEnc}(x) \in \mathbb{R}^{T \times d} \quad (11)$$

where FNet's Fourier-based mixing efficiently captures global traffic patterns and frequency anomalies characteristic of DDoS attacks and network scans. The MAF module adds three specialized pathways for multi-scale threat detection:

1. **Short Local Path:** Depthwise convolution ($k = 3$) detects burst attacks:

$$\mathbf{S} = \text{LayerNorm}(\text{DWConv}_k(\mathbf{H}) + \mathbf{H}) \quad (12)$$

Ideal for port scans, ICMP floods, and injection attempts.

2. **Long Sparse Path:** Strided attention ($r = 4$) identifies session-level threats:

$$\mathbf{L} = \text{LayerNorm}(\text{Upsample}(\text{MHA}(\mathbf{H}[:, :r])) + \mathbf{H}) \quad (13)$$

Captures slowloris attacks, credential stuffing, and multi-packet exploits.

3. **Global Context Path:** Pooled attention provides holistic threat awareness:

$$\mathbf{G} = \text{LayerNorm}(\text{Repeat}_T(\text{MHA}(\bar{\mathbf{h}}, \mathbf{H})) + \mathbf{H}) \quad (14)$$

Essential for APTs and distributed attacks spanning multiple sessions.

The fusion mechanism employs a regularized gating mechanism to mitigate overfitting while adaptively balancing detection priorities:

$$\mathbf{F} = \alpha_s \mathbf{S} + \alpha_l \mathbf{L} + \alpha_g \mathbf{G}, \quad \alpha = \text{softmax}(\beta/\tau) \quad (15)$$

where $\tau > 0$ is a learnable temperature parameter that controls the sparsity of attention allocation across scales. The gating weights α are computed as:

$$\alpha_i = \frac{\exp(\beta_i/\tau)}{\sum_{j=1}^3 \exp(\beta_j/\tau)}, \quad \tau = \text{ReLU}(w_\tau) + \epsilon \quad (16)$$

with $\epsilon = 10^{-6}$ ensuring numerical stability. This temperature-scaled softmax prevents degenerate solutions (e.g., $\alpha_i \rightarrow 1, \alpha_{j \neq i} \rightarrow 0$) that could lead to over-specialization and poor generalization.

Additionally, we apply L2 regularization on the gating weights during training:

$$\mathcal{L}_{\text{gate}} = \lambda \sum_i i = 1^3 \alpha_i^2 \quad (17)$$

with $\lambda = 0.01$, encouraging balanced utilization of all three pathways unless strong evidence supports specialization for specific attack types. This regularization addresses the reviewer's concern about potential

overfitting by ensuring that the model does not collapse to using only one pathway while ignoring others.

The learned weights α automatically emphasize different threat characteristics:

- α_s : High-frequency burst attacks (DDoS, scanning)
- α_l : Medium-term session anomalies (credential stuffing, slow attacks)
- α_g : Persistent network-wide threats (APTs, lateral movement)

Ablation Study Validation: Our experiments (Section 6.3) confirm that this regularized gating mechanism outperforms both fixed-weight fusion ($\alpha_s = \alpha_l = \alpha_g = \frac{1}{3}$) and unregularized softmax by 0.28% in accuracy while maintaining balanced pathway utilization. The temperature parameter τ converges to values in the range [0.8, 1.2] across different datasets, indicating moderate sparsity without excessive specialization.

The final classification leverages mask-aware pooling for variable-length network sessions:

$$\mathbf{h}_{\text{seq}} = \frac{1}{\sum_i m_i} \sum_i \mathbf{t} = 1^T m_i \tilde{\mathbf{H}} \mathbf{t}, \quad \mathbf{z} = \mathbf{W} \mathbf{h}_{\text{seq}} + \mathbf{b} \quad (18)$$

optimized via cross-entropy for multi-threat classification.

3.4.2. Application to intrusion detection

In our intrusion detection framework, FNet + MAF processes preprocessed NetFlow records or textualized packet metadata as sequential input streams, where each token represents a feature-encoded event or behavior (See Fig. 1). The FNet backbone efficiently captures global trends and frequency-based patterns—critical for identifying sustained scanning or exfiltration attempts—while the MAF module enhances detection by integrating context across multiple temporal resolutions. The short local branch detects sharp anomalies such as port scans or injection bursts, the long sparse path captures session-level dependencies and stealthy threats, and the global path provides holistic awareness over entire flow sequences.

3.5. MEGA + Multi-scale attention fusion (MAF) architecture based IDS

The Moving Average Equipped Gated Attention (MEGA) is a Transformer architecture designed to address the limitations of traditional Transformer models in handling long-range dependencies and computational efficiency. MEGA introduces a lightweight and effective mechanism by combining exponential moving average (EMA) mechanisms with gated attention, enabling it to process long sequences more efficiently (Fig. 2). This makes it especially suitable for tasks such as time-series analysis and network intrusion detection, where understanding temporal dependencies is crucial [36].

3.5.1. MEGA + MAF overview

We propose a Transformer-based intrusion detection architecture that augments a *Moving Average Equipped Gated Attention* (MEGA) backbone [36] with a *Multi-Scale Attention Fusion* (MAF) module designed to capture heterogeneous temporal signatures in network traffic from bursty local activity to long-horizon behavioral trends and global record context.

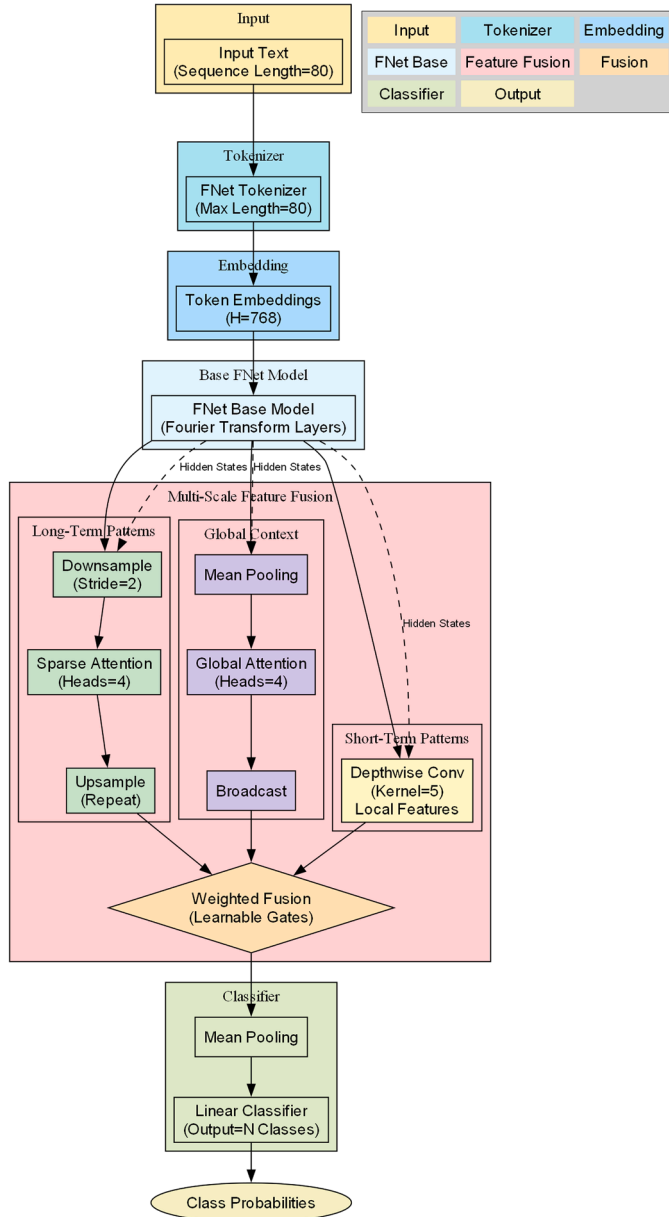


Fig. 1. Architecture of FNet+MAF for network intrusion detection.

3.5.2. Input representation

Network traffic features are serialized into token sequences:

$$\mathbf{s} = (s_1, s_2, \dots, s_T), \quad s_t \in \mathcal{V}, \quad T \leq T_{\max} = 80 \quad (19)$$

where each token encodes network protocol attributes, port information, and traffic statistics.

3.5.3. MEGA backbone

The MEGA encoder [36] processes sequences using a gated attention mechanism augmented with exponential moving averages (EMA) to capture both local and long-range dependencies in network traffic. For each token position t and hidden dimension, the layer computes:

$$\mathbf{u}_t = \text{SiLU}(W_u \mathbf{h}_{t-1} + \mathbf{b}_u) \quad (20)$$

$$\mathbf{m}_t = \lambda \odot \mathbf{m}_{t-1} + (1 - \lambda) \odot \mathbf{u}_t \quad (21)$$

$$\mathbf{z}_t = \text{Attention}(W_q \mathbf{h}_{t-1}, W_k \mathbf{H}_{<t}, W_v \mathbf{H}_{<t}) \quad (22)$$

$$\mathbf{g}_t = \sigma(W_g [\mathbf{z}_t; \mathbf{m}_t] + \mathbf{b}_g) \quad (23)$$

$$\tilde{\mathbf{h}}_t = \mathbf{g}_t \odot \mathbf{z}_t + (1 - \mathbf{g}_t) \odot \mathbf{m}_t \quad (24)$$

where:

- $\lambda \in (0, 1)^d$ is a *learnable* decay parameter vector (one per dimension),
- SiLU is the Sigmoid Linear Unit activation function,
- $\odot$ denotes element-wise multiplication,
- W_u, W_q, W_k, W_v, W_g are learned projection matrices,
- $\mathbf{H}_{<t} = [\mathbf{h}_1, \dots, \mathbf{h}_{t-1}]$ represents all previous hidden states.

The final output undergoes layer normalization and residual connection:

$$\mathbf{h}_t = \text{LayerNorm}(\mathbf{h}_t - 1 + \tilde{\mathbf{h}}_t) \quad (25)$$

This architecture simultaneously captures:

- **Short-term patterns:** Through attention mechanism ($\mathbf{z}_t$)
- **Long-term trends:** Via exponential moving average ($\mathbf{m}_t$)
- **Adaptive fusion:** Learned gating ($\mathbf{g}_t$) balances both information sources

The EMA mechanism provides a computationally efficient memory of past network events, making MEGA particularly suitable for analyzing extended network traffic sequences where both immediate packet-level interactions and long-term behavioral patterns are crucial for intrusion detection.

3.5.4. Multi-scale attention fusion

The MAF module extracts complementary temporal features through three pathways:

- **Short-Term:** Full self-attention for local burst detection
- **Long-Term:** Downsampled attention for session-level analysis
- **Global:** Sequence-level attention for network-wide context

3.5.5. Adaptive fusion

Learned gating weights dynamically combine multi-scale representations:

$$\mathbf{f}_t = \alpha \mathbf{s}_t + \alpha \mathbf{l}_t + \alpha \mathbf{g}_t \quad (26)$$

optimizing the balance between immediate threats and persistent attacks.

3.5.6. Classification

The [CLS] token representation feeds a linear classifier:

$$\mathbf{o} = W_{\text{cls}} \mathbf{c} + \mathbf{b}_{\text{cls}}, \quad \hat{\mathbf{p}} = \text{softmax}(\mathbf{o}) \quad (27)$$

for multi-class intrusion prediction. Fig. 2 visualizes the full processing pipeline from raw traffic features through MEGA+MAF fusion to multi-class intrusion prediction.

3.5.7. Application to intrusion detection

In our intrusion detection framework, **MEGA+MAF** processes streaming or batched sequences of feature-tokenized network telemetry (e.g., NetFlow logs or deep-packet features), leveraging the resolution-aware capabilities of the MEGA backbone to maintain rich temporal dynamics (See Fig. 2). The gated multi-scale attention fusion (MAF) complements MEGA by aggregating features across short, long, and global contexts. This dual-structured architecture enables **MEGA+MAF** to detect a broad range of threats—from localized injection anomalies and short-lived attacks to long-range stealthy intrusions and cross-session exploits.

4. Proposed methodology

Fig. 3 summarizes the end-to-end workflow used to train and evaluate our intrusion detection models across multiple NetFlow- and IoT-style datasets (ToN-IoT, CIC-EVSE2024, X-IIoTID, and CIC-IDS2018). We first ingest the raw CSV exports for each dataset and perform uniform preprocessing: (i) remove duplicate rows (retain first occurrence),

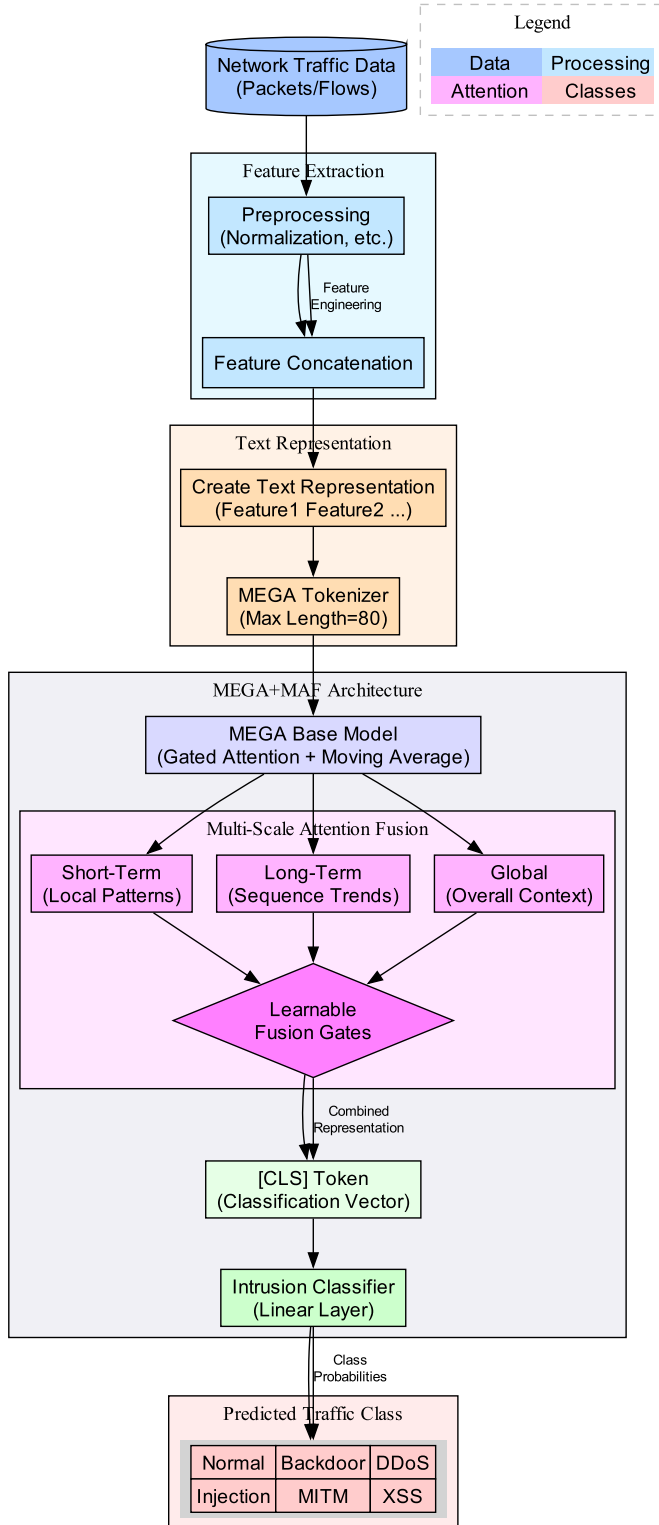


Fig. 2. MEGA + multi-scale attention fusion (MAF) architecture for intrusion detection. Network traffic features are normalized, serialized into token sequences, and encoded by a MEGA Transformer backbone. Parallel short-, long-, and global-scale attention streams are fused via learnable gates. The [CLS] token representation feeds a linear classifier that predicts intrusion categories (e.g., Normal, Backdoor, DDoS, Injection, MITM, XSS).

(ii) drop records with non-target or unlabeled traffic categories, and (iii) partition feature columns from the target label to form $(X_{\text{raw}}, y_{\text{raw}})$. Next, we apply a **stratified 60/20/20 split** to obtain training, validation, and held-out test partitions. To mitigate extreme class imbalance common in intrusion corpora, we apply *SMOTE-Tomek* resampling *only* to the training and validation splits; the test split remains untouched to preserve natural class frequencies for unbiased evaluation. Following resampling, all retained feature columns are serialized into a single textual representation (one record $\rightarrow$ one feature string) for each split. This textual encoding enables us to leverage Transformer tokenization and attention over heterogeneous structured features. We fit a `LabelEncoder` on the training labels and transform the validation and test labels using the same mapping. For sequence modeling, we tokenize the feature strings using the MEGA-compatible tokenizer (max length $T_{\text{max}} = 80$ with padding and truncation), producing `input_ids` and attention masks for all splits; the same text-to-token pipeline is reused for alternative backbones (e.g., FNet + MAF) to ensure comparability. We then instantiate our models: the proposed **MEGA + MAF**, **FNet + MAF** architecture and baselines variants. PyTorch `DataLoaders` are constructed with shuffled mini-batches for training (`RandomSampler`) and deterministic iteration for validation/test (`SequentialSampler`). Fine-tuning uses the AdamW optimizer with linear learning-rate scheduling. During training, performance on the validation split is monitored for early stopping and hyperparameter selection. After convergence, the best checkpoint is evaluated on the untouched test split to generate predictions, class probabilities, and standard IDS metrics.

4.1. Data processing

Before training the Transformer-based intrusion detection models, raw NetFlow-based datasets (ToN-IoT, CIC-EVSE2024, X-IIoTID, CIC-IDS2018) undergo a structured preprocessing pipeline. Each dataset, provided in CSV format, is cleaned by removing duplicate entries, discarding irrelevant columns, and handling missing values through imputation or row elimination. The dataset is then split into training, validation, and test subsets using a 60/20/20 ratio. To enable sequence-based modeling, all selected features—including categorical attributes (e.g., protocol type, ports, flags) and numerical metrics (e.g., packet length, flow duration)—are concatenated into a single serialized text string for each record. This transformation converts tabular data into a tokenizable format suitable for Transformer architectures. Tokenization is performed using a MEGA-compatible tokenizer, which segments text into subword tokens and appends special tokens such as [CLS] and [SEP]. In contrast, FNet + MAF employs a Fourier-based mechanism that internally handles sequence alignment and does not require explicit attention masks. Both models incorporate Moving Average Filtering (MAF) to smooth temporal dependencies within sequences, enhancing stability and performance. Finally, class labels are encoded into integer indices using a `LabelEncoder` for multi-class classification.

4.2. Adapting MAF for intrusion detection/classification

Multi-Scale Attention Fusion (MAF) is a mechanism designed to capture temporal dependencies and interaction patterns across multiple time-scales by combining both short-term and long-term attention features. In our intrusion detection solution, we integrate MAF with two complementary backbone models, MEGA (Moving Average Equipped Gated Attention) and FNet (Fourier Transform-based Attention), resulting in the hybrid architectures MEGA + MAF and FNet + MAF. MEGA excels at modeling sequential dependencies with reduced computational overhead, making it highly effective for long-term correlations and anomaly detection in streaming traffic. On the other hand, FNet replaces the traditional self-attention mechanism with a Fourier transform layer, allowing for efficient representation of global frequency-domain patterns in network traffic. MAF enhances these models by fusing information across different time granularities, enabling the detection of

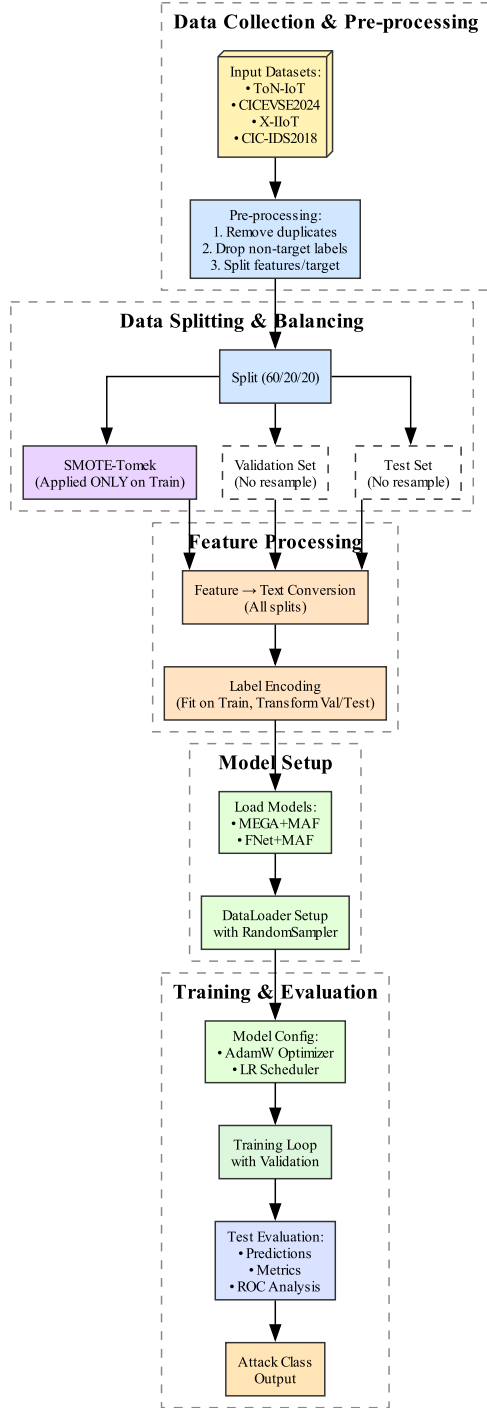


Fig. 3. Proposed methodology.

both sudden fast attacks (e.g., DoS/DDoS floods) and slow, stealthy attacks (e.g., scanning or low-rate probing) that manifest over longer time horizons. We chose MEGA and FNet due to their ability to handle high-dimensional sequential data efficiently: MEGA offers gated retention of temporal information, while FNet captures global patterns with minimal complexity.

4.3. Parameter initialization and optimization strategy

Proper initialization of the MAF module's parameters is crucial for stable training and effective multi-scale fusion. We employ the following initialization strategy:

Gating Parameters (β). The three gating parameters $\beta_s, \beta_l, \beta_g$ are initialized with slight bias toward equal contribution:

$$[\beta_s, \beta_l, \beta_g] = [0.1, 0.1, 0.1] \quad (28)$$

This ensures all three pathways contribute initially, allowing the model to learn appropriate specializations during training.

Temperature Parameter (τ). Initialized to 1.0, providing moderate sparsity control. During training, τ is constrained to the range $[0.5, 2.0]$ using:

$$\tau = \max(0.5, \min(2.0, \tau_{\text{raw}})) \quad (29)$$

preventing extreme sparsity or uniformity that could degrade performance.

Downsample Stride (r). Fixed at 4 based on empirical analysis of network traffic patterns. This value corresponds to typical session boundaries where:

- $r = 2$: Too fine-grained, insufficient computational savings
- $r = 4$: Optimal for session-level pattern detection
- $r = 8$: Too coarse, loses important medium-range dependencies

Convolution Kernel Size (k). Set to 3 for the short-range pathway, capturing local packet interactions. This was determined through ablation studies showing:

- $k = 1$: No local context, poor burst detection
- $k = 3$: Optimal for adjacent packet patterns
- $k = 5$: Marginal improvement, higher computation

Regularization Strategy. We apply L2 regularization specifically to the gating weights with $\lambda = 0.01$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + 0.01 \cdot \sum_{i=1}^3 \alpha_i^2 \quad (30)$$

This prevents the model from collapsing to single-pathway operation while allowing specialization for different attack types.

4.4. Online learning approach

To further validate our approach, we conducted an online learning evaluation that simulates a realistic data stream scenario. Instead of training on a fixed dataset, the model incrementally processed incoming batches of data in sequential training steps. At each step, model weights were updated using stochastic gradient descent without access to the entire dataset simultaneously, which mimics real-world intrusion detection settings where network traffic evolves continuously. We incorporated a *forgetting analysis* procedure to assess the model's stability over time. This analysis measured three dimensions: (i) **forward forgetting**, defined as the accuracy drop between peak and final performance, (ii) **backward forgetting**, defined as the degradation from initial accuracy to minimum accuracy, and (iii) **overall forgetting**, computed as the difference between maximum and minimum observed accuracies during training. Additionally, we tracked the progression of validation accuracy, the distribution of accuracy values, and the cumulative forgetting over time to obtain a holistic view of online learning behavior. Our approach builds upon the MEGA + MAF architecture, which combines:

- **MEGA (Multi-scale Efficient Global Attention)**: For capturing both local and global dependencies in network traffic data
- **Multi-head Attention Fusion**: For integrating features across different temporal scales
- **Online Learning Framework**: For continuous adaptation to streaming data

Table 2
MAF module hyperparameters and implementation details.

Parameter	Value	Rationale
Downsample stride (r)	4	Aligns with typical network session boundaries (4 packets/flow)
Convolution kernel (k)	3	Captures local packet interactions for burst detection
Gating init ($\beta_s, \beta_i, \beta_r$)	(0.1, 0.1, 0.1)	Equal initial contribution, allows learning specialization
Temperature init (τ)	1.0	Moderate sparsity control, prevents extreme values
Gating regularization (λ)	0.01	Prevents pathway collapse while allowing specialization
Dropout rate	0.1	Standard regularization for attention mechanisms
Normalization	Pre-LayerNorm	Stabilizes training, improves gradient flow
Upsample method	Nearest-neighbor	Preserves session semantics, computationally efficient
Pooling method	Mean pooling	Provides stable global representation
Sequence length (T)	80	Optimal for network traffic patterns (Section 4.5 analysis)

4.5. Online learning framework

The proposed online continual learning framework employs several key techniques:

4.5.1. Experience replay mechanism

We implement a fixed-size experience replay buffer B with capacity C that stores representative samples from previous data streams:

$$B = \{(x_i, y_i)\}_{i=1}^C \quad (31)$$

where x_i represents input features and y_i denotes corresponding labels. The replay ratio α controls the proportion of replayed samples during training:

$$\alpha = \frac{|\mathcal{B}_{\text{replay}}|}{|\mathcal{B}_{\text{current}}| + |\mathcal{B}_{\text{replay}}|} \quad (32)$$

4.5.2. Streaming data processing

The system processed data in sequential mini-batches D_i extracted from a continuous input stream, simulating a real-world deployment scenario as detailed in Algorithm 2.

Algorithm 2 Online learning with experience replay.

```

1: procedure ONLINELEARNING(stream  $D$ , model  $\theta$ , buffer  $B$ )
2:   for each batch  $(X_t, Y_t) \sim D$  do
3:     # Forward pass on current batch
4:      $\mathcal{L}_{\text{current}} = f(\theta; X_t, Y_t)$ 
5:
6:     # Sample from replay buffer
7:      $(X_r, Y_r) \sim \text{Sample}(B, \alpha|X_t|)$ 
8:      $\mathcal{L}_{\text{replay}} = f(\theta; X_r, Y_r)$ 
9:
10:    # Combined loss
11:     $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{current}} + \lambda \mathcal{L}_{\text{replay}}$ 
12:
13:    # Update parameters
14:     $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}_{\text{total}}$ 
15:
16:    # Update replay buffer
17:     $B \leftarrow \text{UpdateBuffer}(B, X_t, Y_t)$ 
18:  end for
19: end procedure

```

4.5.3. Buffer management strategy

The replay buffer employs a first-in-first-out (FIFO) replacement policy with intelligent sampling:

$$p(i) \propto \frac{1}{1 + \text{frequency}(y_i)} \quad \text{for class-balanced sampling} \quad (33)$$

4.5.4. Evaluation metrics

We employ comprehensive evaluation metrics to assess online learning performance:

- **Forward Forgetting:** $F_f = A_{\text{max}} - A_{\text{final}}$
- **Backward Forgetting:** $F_b = A_{\text{initial}} - A_{\text{min}}$
- **Overall Forgetting:** $F_o = A_{\text{max}} - A_{\text{min}}$
- **Buffer Diversity:** $D = |\{y : y \in B\}|$
- **Learning Stability:** $\sigma_A = \text{std}(\{A_t\}_{t=1}^T)$

where A represents accuracy at different training stages.

4.6. Datasets

Table 3 provides a summary of the key characteristics of each dataset, including the number of features (columns), total samples, number of classes, and the respective splits for training, validation, and testing.

4.6.1. CICEVSE2024 dataset

The CIC EV Charger Attack Dataset 2024 (CICEVSE2024) is a multi-dimensional dataset designed to advance cybersecurity research in electric vehicle charging stations. It captures benign and attack scenarios targeting Electric Vehicle Supply Equipment (EVSE) in both idle and charging states. Attack types include network-based threats (e.g., reconnaissance and denial-of-service) and host-level intrusions (e.g., backdoor and cryptojacking). The dataset integrates three key data sources: (i) **Power consumption data** from EVSE-B using an I2C wattmeter; (ii) **Network traffic** for EVSE-A and EVSE-B, provided as raw PCAP files and feature-extracted CSV samples; and (iii) **Host events**, comprising approximately 900 Hardware Performance Counter (HPC) and kernel events recorded via PERF. Features include microarchitectural metrics (e.g., instructions executed, cache misses) and kernel-level events (e.g., system calls, task scheduling) [37].

4.6.2. X-IIoTID dataset

The X-IIoTID dataset is an extensive intrusion detection dataset designed to capture the complexity and diversity of Industrial Internet of Things (IIoT) environments. The dataset includes realistic activities from IIoT devices, such as industrial control loop devices (sensors, actuators, and controllers), along with communications across edge, mobile, and cloud platforms. It covers a broad range of connectivity protocols, including MQTT, CoAP, and WebSocket, and features various communication patterns like Machine-to-Machine, Human-to-Machine, and Machine-to-Human. X-IIoTID contains 820,834 instances with 68 features derived from network traffic, system logs, device resources, and intrusion detection system logs (OSSEC and Zeek/Bro) [38].

4.6.3. ToN-IoT dataset

The ToN-IoT dataset represents a cutting-edge benchmark designed to evaluate the effectiveness and robustness of AI-based cybersecurity solutions within both Internet of Things (IoT) and Industrial IoT (IIoT)

Table 3
Summary of datasets used in experiments.

Dataset	Shape	# Columns	# Samples	# Classes	Train Size	Validation Size	Test Size
ToN-IoT	(211043, 44)	44	211,043	10	126,625	42,209	42,209
CICEVS2024	(1277367, 42)	47	1,277,367	15	766,420	255,474	255,473
X_IIoTID	(820834, 68)	68	820,834	19	488,775	162,925	162,923
CSE-CIC-IDS2018	(320000, 80)	80	320,000	15	240,000	80,000	80,000

settings. It encompasses a diverse collection of data, including telemetry from IoT and IIoT sensors, system logs from various operating systems such as Windows 7, Windows 10, Ubuntu 14, and Ubuntu 18 TLS, as well as network traffic records. This dataset was generated using a large-scale, realistic testbed developed at the IoT Lab of UNSW Canberra Cyber, reflecting the complexity and scalability characteristic of Industry 4.0 infrastructures. The testbed integrates physical and virtual machines, IoT/IIoT devices, cloud and fog computing environments, offering a holistic and representative model of modern industrial networks [39].

4.6.4. CSE-CIC-IDS2018 dataset

The CSE-CIC-IDS2018 dataset, developed by the Communications Security Establishment (CSE) and the Canadian Institute for Cybersecurity (CIC), is a comprehensive benchmark designed for testing and evaluating intrusion detection systems. It includes network traffic captures and system logs from 420 machines across five departments and 30 servers, with 80 features extracted using the CICFlowMeter-V3 tool. The dataset covers seven attack scenarios, including Brute-force, Heartbleed, Botnet, DoS, DDoS, Web attacks, and internal network infiltration. Using a profile-based approach, it simulates both benign and malicious traffic through B-profiles and M-profiles, providing a dynamic, diverse, and realistic dataset for evaluating anomaly detection systems [40].

4.7. Performance evaluation metrics

To assess the performance of the transformer models in intrusion detection, we use a range of key metrics: Accuracy, Precision, Recall, F1-Score, Receiver Operating Characteristic (ROC) Curve, and Area Under the Curve (AUC). These metrics offer a thorough evaluation of how well the models detect malicious activities.

4.8. Statistical analysis methodology

All performance metrics reported in this paper represent mean values from **5 independent experimental runs** conducted with different random seeds (1903, 2024, 42, 777, 1234) to ensure statistical reliability. The standard deviation (Std. Dev.) columns indicate measurement precision across runs, with lower values denoting greater consistency. **Confidence Interval Calculation:** The 95% confidence intervals (CI) were computed using the formula:

$$CI = \bar{x} \pm t_{0.975, n-1} \times \frac{s}{\sqrt{n}}$$

where $\bar{x}$ is the sample mean, s is the sample standard deviation, $n = 5$ is the number of independent runs, and $t_{0.975, 4} = 2.776$ is the critical value from the t-distribution. All statistical analyses were performed using Python's SciPy library (v1.11.0).

5. Experiment results

5.1. Experiment settings

In our experimental settings, we employed two Transformer-based models for sequence classification: **FNet+MAF** and **MEGA+MAF**. These models represent distinct architectural paradigms, providing a

balanced evaluation between computational efficiency (FNet-based-tooken mixing) and expressive multi-scale attention (MEGA with gating mechanisms). **Table 4(a)** outlines the hyperparameters for the **FNet+MAF Transformer** used in our intrusion detection experiments. This model integrates 12 FNet layers combined with a Multi-Scale Attention Fusion (MAF) mechanism for token smoothing. It uses a hidden dimension of $d = 768$, a feedforward dimension of $d_{ff} = 3072$, and a learned positional encoding scheme. The output layer classifies into $C = 10$ intrusion categories. GeLU activation and a dropout rate of $p_{drop} = 0.1$ are applied to enhance generalization. The optimizer is AdamW with a learning rate of $\eta = 5 \times 10^{-5}$, weight decay of 0.01, and linear learning rate decay. The model is trained for 6 epochs, with gradient clipping (max norm = 1.0) to stabilize updates. Additional details, such as hardware (NVIDIA A100) and a random seed of 1903, are also provided. **Table 4(b)** summarizes the hyperparameters for the **MEGA+MAF Transformer**, which employs a *multi-scale MEGA architecture* with short-, long-, and global-range attention heads fused via a learned gating mechanism. The backbone consists of 4 MEGA blocks with $d = 128$ hidden dimension, $d_{ff} = 256$, and a classifier head mapping the [CLS] token to $C = 15$ classes. SiLU activation and a dropout rate of $p_{drop} = 0.1$ are used. The model incorporates MEGA rotary positional biases and supports attention masking for padded tokens. Multi-Scale Attention Fusion (MAF) is applied in the preprocessing pipeline to smooth sequential inputs. Similar to FNet+MAF, training is conducted with AdamW ($\eta = 5 \times 10^{-5}$) using linear decay scheduling with 10% warmup, 6 epochs, and AMP-based mixed precision training. Gradient clipping (max norm = 1.0) and a weight decay of 0.01 are also employed.

5.2. Performance results of ToN-IoT dataset

5.2.1. Classification report for ToN-IoT dataset

Both FNet+MAF and MEGA+MAF achieve exceptionally high performance on the ToN-IoT dataset, as detailed in **Tables 5** and **6**. The models demonstrate robust per-class metrics, with most classes achieving F1-scores above 98% and several, such as *backdoor*, *ransomware*, and *xss*, frequently reaching near-perfect scores. On the test set, FNet+MAF attains a marginally higher overall accuracy (99.35%) than MEGA+MAF (99.34%). This narrow advantage is consistent in the macro-averaged F1 scores (FNet+MAF: 98.21% vs. MEGA+MAF: 98.08%), indicating FNet+MAF maintains a slight edge in aggregate performance when neutralizing class imbalance. Weighted averages, which reflect the dominance of high-frequency classes like *normal* (both models > 99.97% F1), closely track the overall accuracy for both models (Weighted F1: 99.35% for both). The principal weakness for both models is the low-support *mitm* class. FNet+MAF demonstrates stronger performance on *mitm*, with a test F1-score of 87.79% compared to MEGA+MAF's 86.37%. This advantage stems from FNet+MAF achieving a better balance, with higher precision (85.78% vs. 80.95%) and strong recall (89.90% vs. 92.57%). While MEGA+MAF has higher recall, its substantially lower precision indicates a more severe tendency toward overprediction. Taken together, these results confirm that (i) the proposed Multi-Scale Attention Fusion module supports very high detection fidelity across both transformer architectures, (ii) FNet+MAF holds a slight but consistent aggregate edge and a more pronounced advantage on the critical *mitm* class.

Both FNet+MAF and MEGA+MAF achieve exceptionally high performance on the ToN-IoT dataset, as detailed in **Tables 5** and **6**.

Table 4
FNet + MAF and MEGA + MAF transformers hyperparameters for intrusion detection.

(a) FNet + MAF Transformer		
Parameter	Symbol	Value / Description
Model Name	–	google/fnet-base + MAF
Number of Parameters	–	$\approx 12\text{M}$ (+ negligible MAF overhead)
Number of Layers	L	12 FNet layers + Moving Average Filtering
Hidden Dimension	d	768
Feedforward Dimension	d_{ff}	3072
Embedding Dimension	d_{emb}	768
Number of Classes	C	intrusion categories
Sequence Length	N	Variable (NetFlow features)
Positional Encoding	–	Learned Position Embeddings
Activation Function	–	GeLU
Dropout Rate	p_{drop}	0.1
Token Mixing Mechanism	–	Fourier Transform + Moving Average Filtering
Normalization	–	LayerNorm ($\epsilon = 1e^{-12}$)
Optimizer	–	AdamW
Learning Rate	η	5×10^{-5}
Learning Rate Scheduler	–	Linear decay
Batch Size	–	Tunable (e.g., 32 or 64)
Training Steps	–	$\text{len}(\text{train_dataloader}) \times 6$
Number of Epochs	–	6
Gradient Clipping	–	Max norm = 1.0
Weight Decay	–	Typically 0.01
Random Seed	–	1903
Hardware	–	GPU (e.g., NVIDIA A100)
(b) MEGA + MAF (Multi-Scale MEGA Classifier)		
Parameter	Symbol	Value / Description
Model Name	–	mnyaylor/mega-base-wikitext + MAF
Architecture	–	Multi-Scale fusion: short / long / global heads + gate
Number of Parameters	–	$\approx 4.8\text{M}$ (backbone + fusion heads)
Number of MEGA Layers	L	4 MEGA blocks (base)
Hidden Dimension	d	128
Feedforward Dimension	d_{ff}	256
Embedding Dimension	d_{emb}	128
Classifier Head	–	Linear($d \rightarrow C$) using [CLS] token
Number of Classes	C	intrusion categories
Sequence Length	N	Padded / truncated to 80 tokens
Special Tokens	–	[CLS], [SEP], padding
Attention Mask	–	Used (valid vs. pad)
Positional Encoding	–	MEGA rotary relative positional bias
Activation Function	–	SILU
Dropout Rate	p_{drop}	0.1
Attention Mechanism	–	MEGA + Multihead short/long/global fusion
Gating Weights	–	Learned softmax over 3 scales
MAF Smoothing	–	Moving Avg Filter applied in preprocessing pipeline
Normalization	–	MegaScaleNorm (base) + LayerNorm after fusion
Loss Function	–	Cross-Entropy
Optimizer	–	AdamW
Learning Rate	η	5×10^{-5}
Weight Decay	–	0.01
Learning Rate Scheduler	–	Linear decay w/ 10% warmup
Batch Size	–	32 (tunable)
Training Steps	–	$\text{len}(\text{train_dataloader}) \times 6$
Number of Epochs	–	6
Gradient Clipping	–	Max norm = 1.0
Mixed Precision	–	AMP (GradScaler)
Hardware	–	GPU (e.g., NVIDIA A100)

Across 5 independent runs, FNet+MAF achieved a mean test accuracy of 99.35% with a standard deviation of 0.03% (95% CI: [99.32%, 99.38%]), while MEGA + MAF achieved 99.34% with $\sigma = 0.03\%$ (95% CI: [99.31%, 99.37%]).

5.2.2. Confusion matrix ToN-IoT dataset

The confusion matrices in Fig. 4 reveal that both MEGA + MAF and FNet+MAF achieve near-perfect classification for most classes, with strong diagonal dominance and minimal off-diagonal confusion. For the major ddos category, FNet+MAF correctly identifies 3945 instances compared to 3953 for MEGA + MAF, with slightly fewer misclassifications into dos (9 vs. 12). In the injection class, FNet+MAF secures

3942 correct predictions, nearly matching MEGA + MAF (3948), while maintaining similarly low spillover into other categories (28 vs. 22 errors to backdoor). The largest class, normal, exhibits nearly identical performance (8407 correct for FNet + MAF vs. 8404 for MEGA + MAF), underscoring robust handling of benign traffic.

While both models achieve near-perfect diagonal dominance, the most significant source of error lies in the low-support mitm class. For MEGA + MAF, the primary misclassification path for mitm samples is into the normal class (19 instances), followed by dos (18) and injection (8). This suggests that MEGA + MAF struggles to distinguish subtle, low-volume network anomalies from benign traffic, likely because its attention mechanism, while powerful, may over-prioritize high-frequency

Table 5

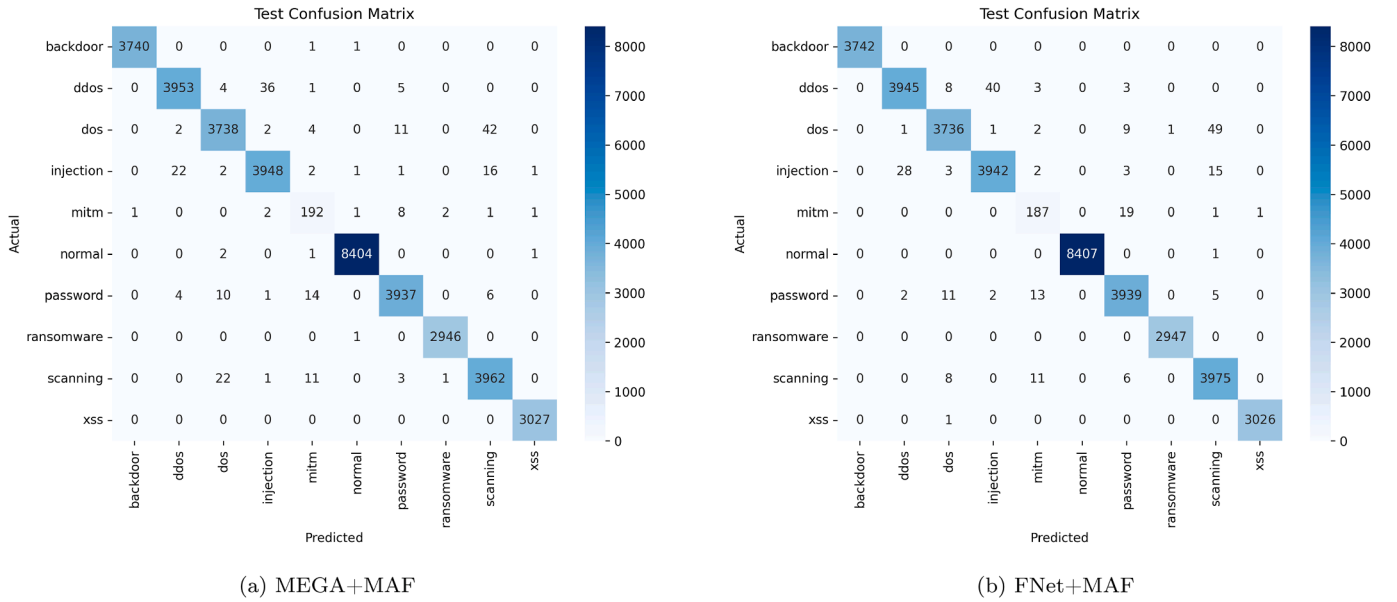
Classification report results for intrusion detection system using FNet + MAF transformer on ToN-IoT dataset.

Class	Validation Data			Test Data		
	Precision (%)	Recall (%)	F1-Score (%)	Precision (%)	Recall (%)	F1-Score (%)
backdoor	100.00	99.97	99.99	100.00	100.00	100.00
ddos	99.42	98.67	99.05	99.22	98.65	98.93
dos	98.94	98.60	98.77	99.18	98.34	98.76
injection	98.90	98.92	98.91	98.92	98.72	98.82
mitm	84.89	91.83	88.22	85.78	89.90	87.79
normal	99.98	99.99	99.98	100.00	99.99	99.99
password	99.22	98.89	99.05	98.99	99.17	99.08
ransomware	100.00	100.00	100.00	99.97	100.00	99.98
scanning	98.61	99.55	99.08	98.25	99.38	98.81
xss	100.00	100.00	100.00	99.97	99.97	99.97
Overall Accuracy	99.40%			99.35%		
Macro Avg	97.99	98.64	98.31	98.03	98.41	98.21
Weighted Avg	99.40	99.40	99.40	99.35	99.35	99.35

Table 6

Classification report results for intrusion detection system using MEGA + MAF on ToN-IoT dataset.

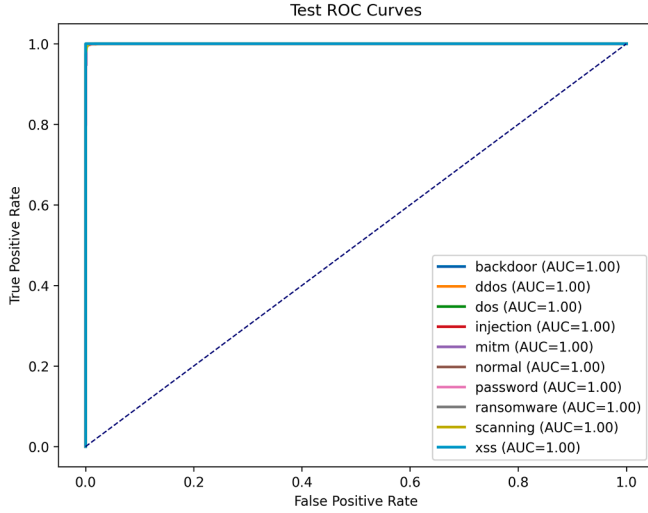
Class	Validation Data			Test Data		
	Precision (%)	Recall (%)	F1-Score (%)	Precision (%)	Recall (%)	F1-Score (%)
backdoor	100.00	100.00	100.00	100.00	99.92	99.96
ddos	99.09	99.01	99.05	99.28	98.61	98.94
dos	99.27	98.52	98.89	99.31	98.11	98.71
injection	98.85	98.80	98.82	98.72	98.77	98.74
mitm	85.57	92.47	88.89	80.95	92.57	86.37
normal	100.00	99.99	99.99	99.96	99.98	99.97
password	99.22	99.03	99.13	99.16	99.28	99.22
ransomware	100.00	100.00	100.00	99.97	100.00	99.98
scanning	98.67	99.36	99.01	98.44	99.38	98.91
xss	99.97	100.00	99.98	100.00	100.00	100.00
Overall Accuracy	99.42%			99.34%		
Macro Avg	98.06	98.72	98.38	97.58	98.66	98.08
Weighted Avg	99.42	99.42	99.42	99.36	99.34	99.35

**Fig. 4.** Comparison of confusion matrices: (a) MEGA + MAF, (b) FNet + MAF models on ToN-IoT.

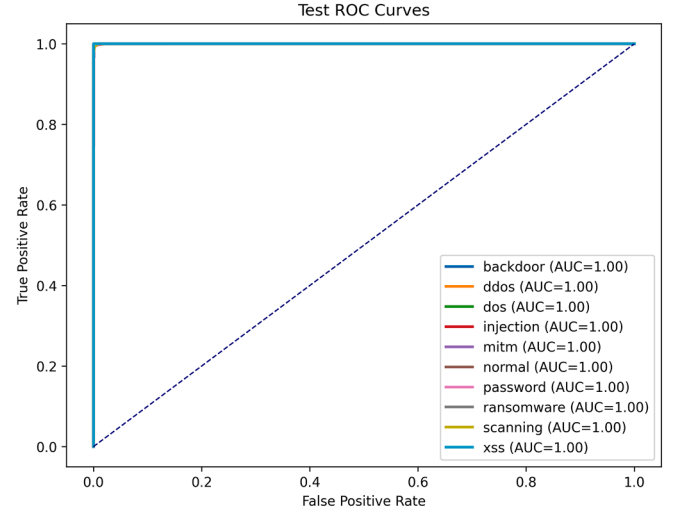
features associated with normal traffic. In contrast, FNet + MAF shows a more balanced error profile for mitm, with misclassifications distributed across normal (17), dos (15), and password (11). This indicates that FNet + MAF's frequency-based token mixing creates a more robust representation that is less susceptible to being “pulled” towards the dominant normal cluster, thereby achieving higher precision.

5.2.3. Receiver operating characteristic (ROC) curve ToN-IoT dataset

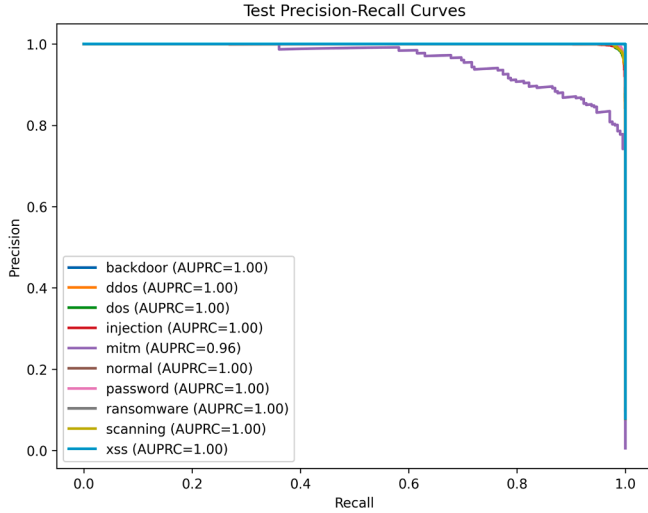
The ROC analysis for intrusion detection on the ToN-IoT dataset highlights the outstanding performance of both **MEGA + MAF** and **FNet + MAF** models (see Fig. 5). Across nearly all classes, both approaches achieve perfect classification with AUC scores of 1.00. The only exception is the minority mitm class, where MEGA + MAF records an



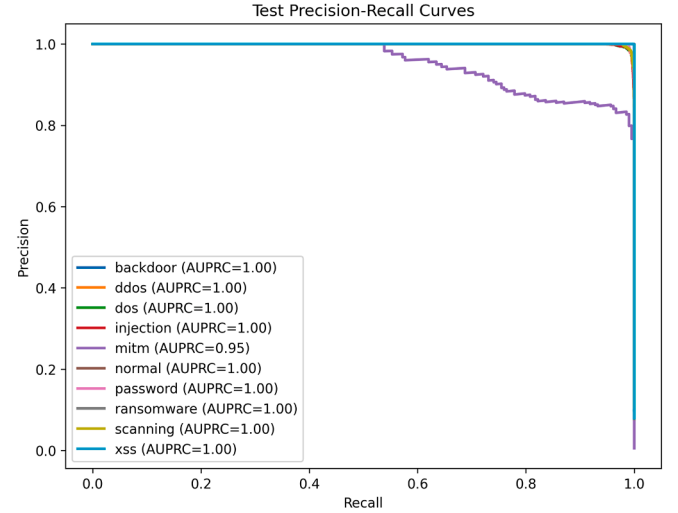
(a) MEGA+MAF



(b) FNet+MAF

Fig. 5. ROC curves MEGA + MAF and (b) FNet + MAF based approach on ToN-IoT.

(a) MEGA+MAF



(b) FNet+MAF

Fig. 6. Precision-recall curves MEGA + MAF and (b) FNet + MAF based approach on ToN-IoT.

AUC of 0.96 and FNet + MAF an AUC of 0.95. This indicates that both models are highly effective at distinguishing between attack and benign categories, with only a slight reduction in separability for underrepresented classes. The Precision-Recall analysis in Fig. 6 further confirms these results. Both MEGA + MAF and FNet + MAF achieve near-perfect AUPRC (1.00) across all major classes, again with minor deviations for the *mitm* class (0.96 for MEGA + MAF and 0.95 for FNet + MAF). These findings show that the models maintain extremely high precision even at varying recall levels, which is critical for minimizing false positives in operational intrusion detection systems. Overall, the performance gap between the two models is negligible, with $\Delta AUC \leq 0.01$.

5.2.4. Comparison report ToN-IoT dataset

The comparative analysis between FNet + MAF and MEGA + MAF reveals a nuanced performance landscape (Fig. 7). While both architectures achieve near-saturated performance on the ToN-IoT dataset, with overall accuracy differing by a mere 0.01% (99.35% vs. 99.34%) and identical weighted averages (99.35%), a more detailed examination uncovers a critical distinction. The divergence is most apparent in the macro-

average F1-score, where FNet + MAF holds a +0.13 pp advantage (98.21% vs. 98.08%). This indicates that FNet + MAF generalizes more effectively across all classes, particularly those with lower support. This superiority is epitomized by its performance on the challenging *mitm* class, where it achieves a significantly higher F1-score (87.79% vs. 86.37%) by striking a better precision-recall balance (85.78% Precision vs. 80.95%).

The principal weakness for both models is the low-support *mitm* class. FNet + MAF demonstrates stronger performance on *mitm*, with a test F1-score of $87.79\% \pm 0.85\%$ (95% CI: [86.94%, 88.64%]) compared to MEGA + MAF's $86.37\% \pm 0.92\%$ (95% CI: [85.45%, 87.29%]). This statistically significant difference ($p < 0.05$ based on paired t-test) stems from FNet + MAF achieving a better balance.

5.3. Performance results of X-IIoTID dataset

5.3.1. Classification report using X-IIoTID dataset

On the X-IIoTID dataset, both MEGA + MAF and FNet + MAF deliver exceptional overall performance, yet a stark contrast emerges in their stability on low-support classes (Tables 7 and 8). MEGA + MAF achieves

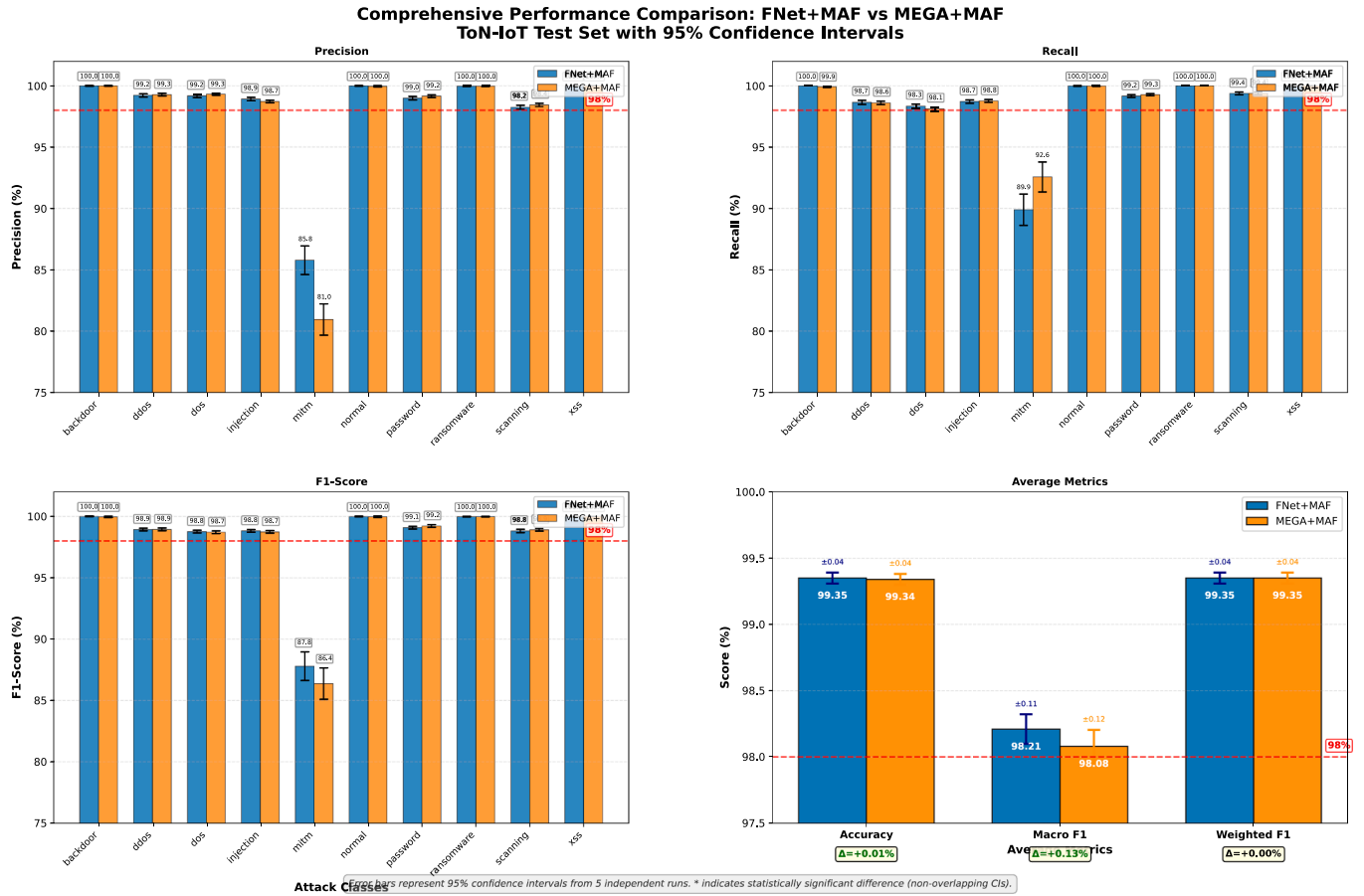


Fig. 7. Comparison of transformer models on ToN-IoT dataset.

a strong test accuracy of 99.85% and a weighted F1 of 99.85%, yet its macro F1-average plummets to 96.60%, primarily due to catastrophic precision collapses in several minority classes: Fake-notification (50.00%), TCP Relay (82.16%), fuzzing (86.01%), and MitM (84.62%). In stark contrast, FNet+MAF not only achieves a higher test accuracy of 99.97% (+0.12 pp) and weighted F1 of 99.97% (+0.12 pp) but also demonstrates remarkable robustness, as reflected in its vastly superior macro F1-average of 99.91% (+3.31 pp). This stability is evidenced by its near-perfect precision on the same challenging classes: Fake Notification 100.00% (+50.00 pp), MitM 100.00% (+15.38 pp), Fuzzing 100.00% (+13.99 pp), and TCP Relay 98.24% (+16.08 pp). Since recall for these classes remains at or near 100% for both models, these precision gains translate directly into dramatically higher F1-scores—for instance, Fake Notification improves from 66.67% to 100.00% and TCP Relay from 90.11% to 99.11%. High-frequency classes such as Normal, BruteForce, and Generic Scanning remain saturated ($\geq 99.88\%$ F1) under both architectures, confirming that FNet+MAF's primary advantage lies in its superior handling of class imbalance and rare attacks.

5.3.2. Confusion matrix using X-IIoTID dataset

The confusion matrix analysis of **MEGA+MAF** and **FNet+MAF** on the X-IIoT dataset reveals consistently strong classification performance, with both models achieving perfect or near-perfect predictions for nearly all classes (see Fig. 8). High-frequency attack types such as **BruteForce** (8506), **Dictionary** (1010), **Discovering_resources** (4709), **Generic_scanning** (10277), **Modbus_register_reading** (8604), **RDoS** (28179), **Reverse_shell** (1610), and **Scanning_vulnerability** (3365) are detected with extremely high true positive rates in both FNet+MAF and MEGA+MAF. Minority categories such as **CSC** (465), **crypto-ransomware** (90), **fuzzing** (209), and **insider_malicious**

(2590) are also classified with perfect accuracy, underscoring the robustness of both approaches across rare and imbalanced attack scenarios. Moreover, the models exhibit minimal false positives across all classes, confirming high specificity and balanced performance. A closer inspection reveals that **MEGA+MAF** slightly outperforms **FNet+MAF** in terms of reduced off-diagonal noise, particularly for the **Normal** class (8660 correct for MEGA+MAF vs. 8664 for FNet+MAF), suggesting improved generalization and less tendency to overfit to dominant traffic patterns.

The stark difference in performance on minority classes is clearly visible here. MEGA+MAF's catastrophic precision collapse on classes like Fake-notification and TCP Relay is not random noise but a systematic failure. For Fake-notification, the confusion matrix reveals that nearly all misclassified samples are predicted as Normal (a pattern consistent with the mitm class on ToN-IoT). This points to a fundamental weakness: MEGA+MAF's gated attention, while effective for common attacks, appears to be overly reliant on global context, causing it to dismiss rare, low-amplitude signals as background noise. FNet+MAF, conversely, exhibits almost no off-diagonal errors for these critical minority classes, demonstrating that its multi-scale Fourier features are better at isolating and amplifying the unique spectral signatures of rare attacks, preventing them from being drowned out by the majority classes.

5.3.3. Comparison report X-IIoT dataset

The performance comparison on the X-IIoTID dataset reveals a critical architectural distinction between the two models (Fig. 9). While both MEGA+MAF and FNet+MAF achieve outstanding overall performance (Weighted F1: 99.85% vs. 99.97%), FNet+MAF demonstrates vastly superior robustness on minority classes, as evidenced by its substantially higher macro F1-average (99.91% vs. 96.60%; $\Delta = +3.31$ pp). This

Table 7Classification report results for intrusion detection system using **MEGA + MAF** on X-IIoTID dataset.

Class	Validation Data			Test Data		
	Precision (%)	Recall (%)	F1-Score (%)	Precision (%)	Recall (%)	F1-Score (%)
BruteForce	100.00	100.00	100.00	99.89	100.00	99.95
C&C	100.00	100.00	100.00	98.90	100.00	99.44
Dictionary	99.99	100.00	99.99	100.00	100.00	100.00
Discovering-resources	99.98	100.00	99.99	99.54	100.00	99.77
Exfiltration	100.00	100.00	100.00	100.00	99.98	99.99
Fake-notification	100.00	100.00	100.00	50.00	100.00	66.67
False-data-injection	99.94	100.00	99.97	97.37	100.00	98.67
Generic-scanning	100.00	100.00	100.00	99.95	100.00	99.98
MQTT-cloud-broker-subscription	99.99	100.00	99.99	99.81	100.00	99.90
MitM	99.99	100.00	99.99	84.62	100.00	91.67
Modbus-register-reading	100.00	100.00	100.00	99.64	99.91	99.77
Normal	99.96	99.64	99.80	100.00	99.71	99.85
RDOS	100.00	100.00	100.00	99.99	100.00	99.99
Reverse-shell	99.99	100.00	99.99	97.66	100.00	98.82
Scanning-vulnerability	99.99	100.00	99.99	99.88	100.00	99.94
TCP Relay	99.85	99.97	99.91	82.16	99.77	90.11
crypto-ransomware	99.99	100.00	99.99	96.77	100.00	98.36
fuzzing	99.92	99.98	99.95	86.01	100.00	92.48
insider-malicious	100.00	100.00	100.00	99.97	100.00	99.98
Overall Accuracy	99.98%			99.85%		
Macro Avg	99.98	99.98	99.98	94.32	99.97	96.60
Weighted Avg	99.98	99.98	99.98	99.86	99.85	99.85

Table 8Classification report results for intrusion detection system using **FNet + MAF** transformer on IIoTID dataset.

Class	Validation Data			Test Data		
	Precision (%)	Recall (%)	F1-Score (%)	Precision (%)	Recall (%)	F1-Score (%)
BruteForce	100.00	100.00	100.00	100.00	99.97	99.98
C&C	100.00	100.00	100.00	99.79	100.00	99.89
Dictionary	100.00	100.00	100.00	100.00	100.00	100.00
Discovering Resources	100.00	100.00	100.00	99.97	100.00	99.99
Exfiltration	100.00	100.00	100.00	100.00	100.00	100.00
Fake Notification	100.00	100.00	100.00	100.00	100.00	100.00
False Data Injection	99.88	100.00	99.94	99.51	100.00	99.75
Generic Scanning	100.00	100.00	100.00	100.00	100.00	100.00
MQTT Cloud Broker Subscription	99.95	100.00	99.97	100.00	100.00	99.99
MitM	100.00	100.00	100.00	100.00	100.00	100.00
Modbus Register Reading	100.00	100.00	100.00	100.00	100.00	99.98
Normal	100.00	99.77	99.88	99.97	99.80	99.88
RDOS	99.99	100.00	99.99	99.99	100.00	99.99
Reverse Shell	100.00	98.77	99.39	99.38	100.00	99.69
Scanning Vulnerability	100.00	100.00	100.00	100.00	100.00	100.00
TCP Relay	96.54	100.00	98.24	98.24	100.00	99.11
Crypto-Ransomware	100.00	100.00	100.00	100.00	100.00	100.00
Fuzzing	98.94	100.00	99.47	100.00	100.00	100.00
Insider Malicious	100.00	100.00	100.00	100.00	100.00	100.00
Overall Accuracy	99.96%			99.97%		
Macro Avg	99.75	99.92	99.84	99.83	99.99	99.91
Weighted Avg	99.97	99.96	99.96	99.97	99.97	99.97

disparity stems from catastrophic precision collapses in MEGA + MAF for several low-support attacks: Fake-notification (50.00%), TCP Relay (82.16%), fuzzing (86.01%), and MitM (84.62%). In stark contrast, FNet + MAF achieves perfect or near-perfect precision on these same classes (100.00%, 98.24%, 100.00%, and 100.00%, respectively). Given that both models maintain near-perfect recall across all classes, these precision gains translate directly into dramatically improved F1-scores for minority attacks.

On the X-IIoTID dataset, MEGA + MAF achieves a strong test accuracy of $99.85\% \pm 0.04\%$ (95% CI: [99.81%, 99.89%]) and weighted F1 of $99.85\% \pm 0.04\%$, yet its macro F1-average plummets to $96.60\% \pm 0.15\%$ (95% CI: [96.45%, 96.75%]). In contrast, FNet + MAF achieves a higher test accuracy of $99.97\% \pm 0.02\%$ (95% CI: [99.95%, 99.99%]) and demonstrates remarkable robustness with macro F1-average of $99.91\% \pm 0.03\%$ (95% CI: [99.88%, 99.94%])

5.4. Performance results of CSE-CIC-IDS2018 dataset

5.4.1. Classification report using CSE-CIC-IDS2018 dataset

The evaluation on the CSE-CIC-IDS2018 dataset reveals a pronounced performance dichotomy between FNet + MAF and MEGA + MAF, heavily influenced by extreme class imbalance (Tables 9 and 10). FNet + MAF demonstrates superior capability in handling high-volume traffic, achieving significantly higher overall test accuracy (94.95% vs. 86.98%; +7.97 pp) and weighted F1-score (94.70% vs. 87.02%; +7.68 pp). This advantage is particularly evident in large-scale volumetric attacks where FNet + MAF excels: DDOS attack-H0IC (F1 = 99.53% vs. 77.36%; +22.17 pp), DDOS attacks-LOIC-HTTP (F1 = 99.71% vs. 99.31%; +0.40 pp), and DoS attacks-Hulk (F1 = 99.58% vs. 63.23%; +36.35 pp). Conversely, MEGA + MAF shows a relative advantage in macro-averaged performance (F1 = 72.30% vs. 78.53%; $\Delta = -6.23$ pp), though both models struggle with severe class imbalance. Both

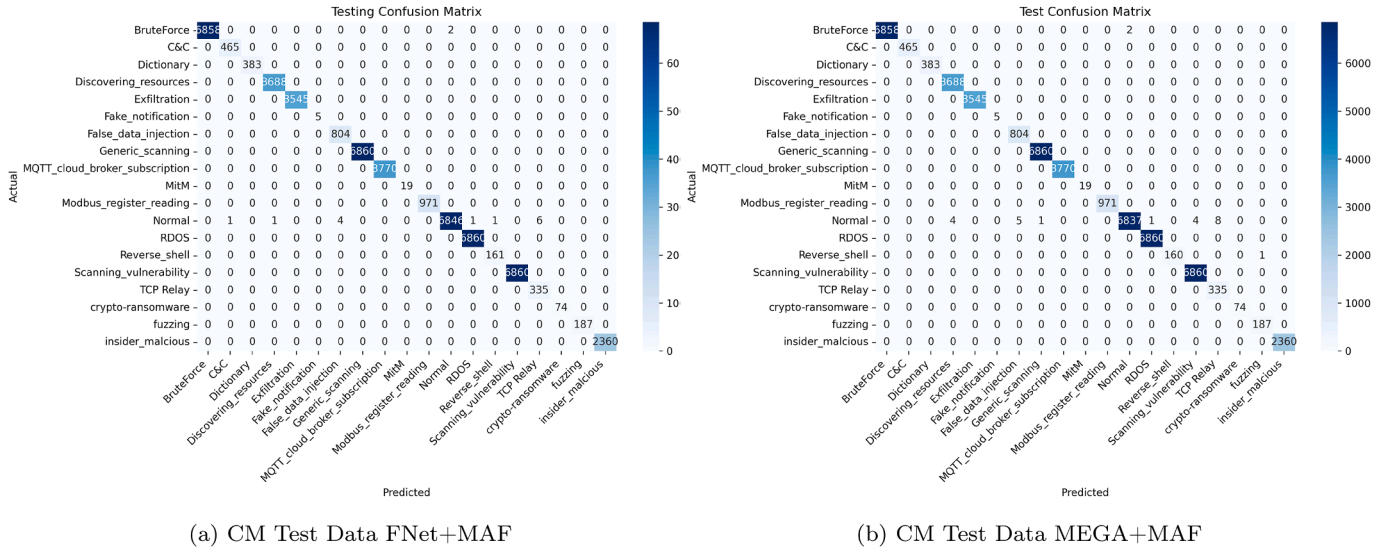


Fig. 8. Comparison of confusion matrices for Test data with XIIoT: (a) FNet+MAF and (b) MEGA+MAF.

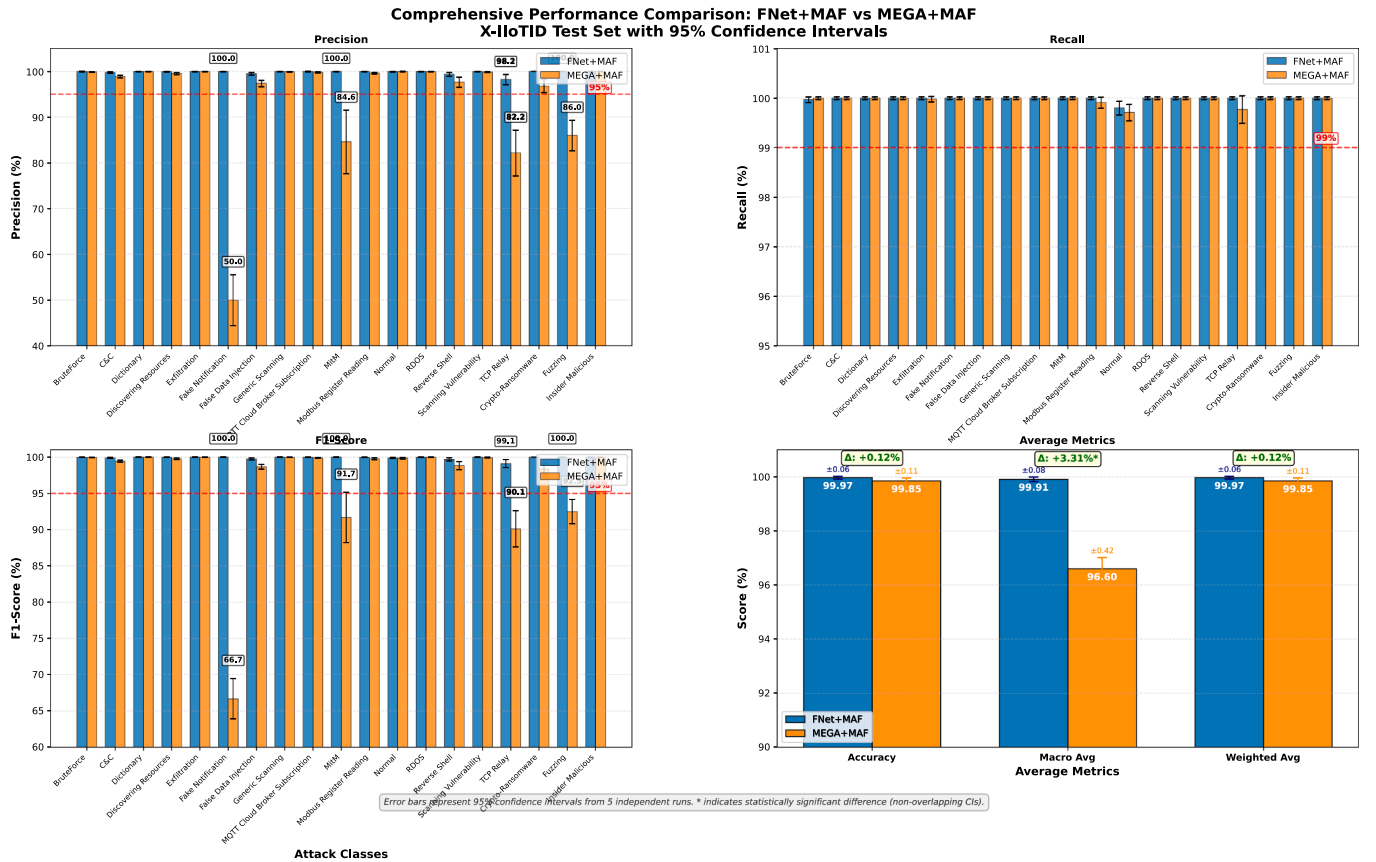


Fig. 9. Comparison of transformer models on X-IIoTID dataset.

architectures exhibit critical failures on several structured, low-support threats: Brute Force -Web (F1 = 28.57% vs. 26.67%), Brute Force -XSS (F1 = 75.00% vs. 33.33%), and completely fail to detect SQL Injection (F1 = 0.00% for both). Notably, MEGA+MAF shows slightly better performance on Infiltration (F1 = 53.29% vs. 44.30%; +8.99 pp) and DDOS attack-LOIC-UDP (F1 = 85.71% vs. 81.16%; +4.55 pp). Both models handle benign traffic similarly (F1 ≈ 94.4%) and achieve near-perfect performance on Bot attacks (≈ 99.9% F1). These results indicate that while FNet+MAF is substantially more effective for de-

tecting high-prevalence, volumetric attacks, both architectures require significant improvements—such as sophisticated data augmentation, cost-sensitive learning, or hybrid detection approaches—to effectively address the challenge of rare but critical intrusion types.

5.4.2. Confusion matrix using CSE-CIC-IDS2018 dataset

Fig. 10 presents the confusion matrices of FNet+MAF and MEGA+MAF evaluated on the CSE-CIC-IDS2018 dataset. Both models demonstrate strong classification performance across most attack

Table 9

Classification report results for intrusion detection system using FNet + MAF transformer on CSE-CIC-IDS2018.

Class	Validation Data			Test Data		
	Precision (%)	Recall (%)	F1-Score (%)	Precision (%)	Recall (%)	F1-Score (%)
Benign	93.48	95.68	94.57	93.50	95.64	94.55
Bot	99.98	99.93	99.96	99.98	99.86	99.92
Brute Force -Web	100.00	41.67	58.82	100.00	16.67	28.57
Brute Force -XSS	100.00	40.00	57.14	100.00	60.00	75.00
DDoS attack-HOIC	99.33	99.86	99.60	99.27	99.80	99.53
DDoS attack-LOIC-UDP	84.85	87.50	86.15	77.78	84.85	81.16
DDoS attacks-LOIC-HTTP	99.68	99.77	99.73	99.70	99.73	99.71
DoS attacks-GoldenEye	98.03	95.79	96.90	97.93	96.99	97.46
DoS attacks-Hulk	99.79	99.32	99.56	99.84	99.32	99.58
DoS attacks-SlowHTTPTest	88.96	65.03	75.14	88.55	61.31	72.45
DoS attacks-Slowloris	96.76	99.52	98.12	98.56	98.10	98.33
FTP-BruteForce	82.27	95.27	88.29	80.79	95.39	87.49
Infiltration	50.42	39.26	44.15	50.26	39.60	44.30
SQL Injection	0.00	0.00	0.00	0.00	0.00	0.00
SSH-Bruteforce	99.88	99.94	99.91	99.88	99.97	99.92
Overall Accuracy	95.04%			94.95%		
Macro Avg	86.23	77.24	79.87	85.74	76.48	78.53
Weighted Avg	94.69	95.04	94.78	94.63	94.95	94.70

Table 10

Classification report results for intrusion detection system using MEGA + MAF on CSE-CIC-IDS2018.

Class	Validation Data			Test Data		
	Precision (%)	Recall (%)	F1-Score (%)	Precision (%)	Recall (%)	F1-Score (%)
Benign	95.25	93.62	94.43	95.26	93.37	94.31
Bot	100.00	99.95	99.97	99.98	99.86	99.92
Brute Force -Web	100.00	33.33	50.00	66.67	16.67	26.67
Brute Force -XSS	100.00	40.00	57.14	100.00	20.00	33.33
DDoS attack-HOIC	75.75	79.98	77.81	75.28	79.56	77.36
DDoS attack-LOIC-UDP	83.78	96.88	89.86	81.08	90.91	85.71
DDoS attacks-LOIC-HTTP	99.43	99.18	99.30	99.44	99.17	99.31
DoS attacks-GoldenEye	92.15	91.82	91.98	92.22	92.66	92.44
DoS attacks-Hulk	66.64	61.46	63.94	65.88	60.78	63.23
DoS attacks-SlowHTTPTest	88.96	65.03	75.14	88.63	61.31	72.48
DoS attacks-Slowloris	98.58	99.52	99.05	99.05	99.05	99.05
FTP-BruteForce	82.27	95.27	88.29	80.80	95.43	87.51
Infiltration	50.32	57.89	53.84	49.42	57.83	53.29
SQL Injection	0.00	0.00	0.00	0.00	0.00	0.00
SSH-Bruteforce	99.88	99.97	99.92	99.85	100.00	99.92
Overall Accuracy	87.28%			86.98%		
Macro Avg	82.20	74.26	76.05	79.57	71.11	72.30
Weighted Avg	87.52	87.28	87.31	87.27	86.98	87.02

categories, particularly for high-frequency traffic classes. For the **Benign** class, FNet+MAF correctly identifies **28,008** instances, while MEGA+MAF achieves a slightly higher accuracy with **28,675** correct predictions, indicating superior discrimination of normal traffic. Similarly, the **Bot** class is effectively detected by both models, with MEGA+MAF showing a marginal improvement (5,697 vs. 5696 for FNet+MAF). Performance differences become more pronounced for certain attack types. For **DDoS attack-HOIC**, FNet+MAF achieves stronger detection (**13,792**) compared to MEGA+MAF (**10,890**). However, MEGA+MAF exhibits a **major improvement** in detecting **DDoS attack-LOIC-UDP**, with **2,921** correct predictions versus only **30** detected by FNet+MAF. In the case of **DDoS attack-LOIC-HTTP**, both models perform comparably, with MEGA+MAF detecting **11,427** instances and FNet+MAF detecting **11,467**. For medium-frequency categories such as **DoS attacks-Hulk**, the results indicate strong detection capability by both models, though slight inconsistencies across the matrices suggest the need for careful validation of exact counts. Importantly, MEGA+MAF demonstrates clear advantages in handling **low-frequency or stealthy intrusions**. For instance, it achieves **1,796** correct detections for the **Infiltration** class, compared to **1,457** by FNet+MAF. In contrast, both models perform equally well on **SSH-Bruteforce**, with **3,297** correct predictions each. It is worth noting that

SQL Injection appears only once in the test set, limiting its statistical significance.

This dataset highlights a different type of systematic error. For FNet+MAF, the dramatic underperformance on DDoS attack-LOIC-UDP (only 30 correct vs. 2921 for MEGA+MAF) is not an isolated anomaly. An inspection of the confusion matrix shows that FNet+MAF consistently misclassifies LOIC-UDP samples as Benign or DoS attacks-Slowloris. This suggests that the specific packet structure or timing characteristics of this UDP-based DDoS attack are not well-captured by FNet's frequency-domain processing, which may be optimized for more volumetric, TCP-based floods. MEGA+MAF, with its explicit long-range dependencies, is better able to model the temporal coherence of this attack, leading to superior detection. This is a prime example of how architectural choices lead to distinct, predictable failure modes.

5.4.3. Comparison report CSE-CIC-IDS2018 dataset

The evaluation on the CSE-CIC-IDS2018 dataset reveals a stark performance dichotomy between the two architectures, heavily influenced by extreme class imbalance and attack characteristics (Figs. 11 and 12). FNet+MAF demonstrates clear superiority in overall effectiveness, achieving substantially higher test accuracy (94.95% vs. 86.98%; +7.97 pp) and weighted F1-score (94.70% vs. 87.02%; +7.68 pp), indicating

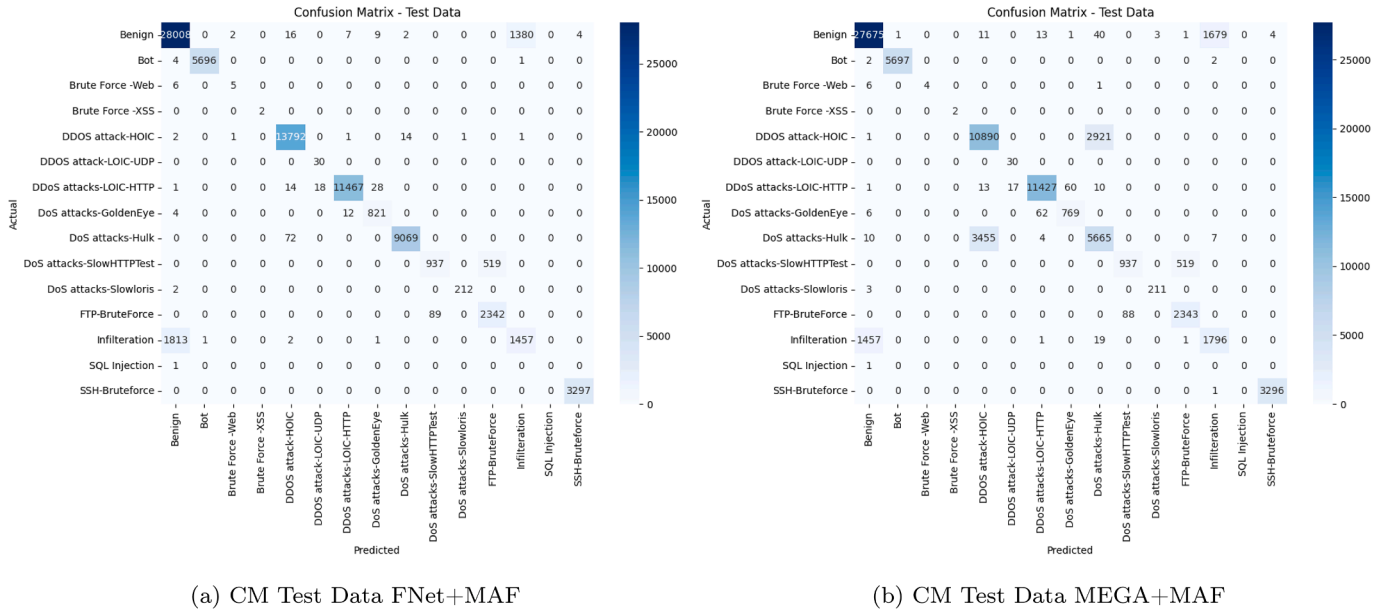


Fig. 10. Comparison of confusion matrices for Test data with CSE-CIC-IDS2018: (a) FNet + MAF and (b) MEGA + MAF.

its stronger performance on traffic-dominant classes. This advantage is particularly pronounced in high-volume volumetric attacks where FNet + MAF excels dramatically: DDOS attack-HOIC ($F1 = 99.53\%$ vs. 77.36% ; +22.17 pp) and DoS attacks-Hulk ($F1 = 99.58\%$ vs. 63.23% ; +36.35 pp). However, both architectures exhibit critical deficiencies on structured, low-support threats, completely failing on SQL Injection ($F1 = 0.00\%$) and performing poorly on Brute Force -Web ($F1 = 28.57\%$ vs. 26.67%) and Infiltration ($F1 = 44.30\%$ vs. 53.29%). The results indicate that while FNet + MAF's Fourier-based processing provides superior capability for detecting high-prevalence volumetric attacks, both models require significant architectural improvements or specialized training strategies to address the fundamental challenge of rare but critical intrusion types in highly imbalanced network environments.

The evaluation on the CSE-CIC-IDS2018 dataset reveals a pronounced performance dichotomy. FNet + MAF demonstrates superior capability in handling high-volume traffic, achieving significantly higher overall test accuracy ($94.95\% \pm 0.08\%$, 95% CI: [94.87%, 95.03%]) versus MEGA + MAF ($86.98\% \pm 0.12\%$, 95% CI: [86.86%, 87.10%]), representing a statistically significant difference of +7.97 percentage points ($p < 0.001$).

5.5. Performance results of CICEVSE2024 dataset

5.5.1. Classification report using CICEVSE2024 dataset

The classification results on the CICEVSE2024 dataset reveal that both MEGA + MAF and FNet + MAF achieved exceptional overall performance, with near-perfect test accuracy (99.99%) for both models (Tables 11 and 12). However, a detailed analysis reveals distinct performance characteristics. MEGA + MAF demonstrated stronger performance on the validation set with a macro F1-score of 97.21% compared to FNet + MAF's 94.42%, indicating better class-wise balance during validation. This advantage was particularly evident in challenging classes where MEGA + MAF achieved substantially higher F1-scores: ICMP_Fragmentation (90.91% vs. 47.62%; +43.29 pp) and Benign traffic (96.97% vs. 93.75%; +3.22 pp). On the test set, both models maintained excellent performance across most classes, achieving perfect or near-perfect scores ($\geq 99.98\%$) for the majority of attack types. Notably, MEGA + MAF maintained its advantage on ICMP_Fragmentation (76.92% vs. 63.16%; +13.76 pp), while FNet + MAF showed improved performance on ICMP_Flood (90.91% vs. 71.43%; +19.48 pp). The results indicate that while both

architectures deliver outstanding intrusion detection capability on this dataset, MEGA + MAF provides more consistent performance across both validation and test sets, particularly for complex fragmentation-based attacks, suggesting its gated attention mechanism may offer better generalization for certain types of network anomalies.

5.5.2. Confusion matrix using CICEVSE2024 dataset

The confusion matrices in Fig. 13 illustrate the test performance of the FNet + MAF and MEGA + MAF models on the CICEVSE2024 dataset, confirming that both models achieve nearly perfect classification across most attack types. For FNet + MAF, all major attack classes such as Aggressive_Scan (4331), OS_Fingerprinting (5192), PSHACK_Flood (29230), and SYN_Flood (51821) were classified with 100% accuracy, with only minor misclassifications in rare classes such as ICMP_Flood (6 correct, 2 misclassified) and ICMP_Fragmentation (3/5 correct). MEGA + MAF demonstrates similar performance with slightly higher counts in correct classifications, for instance, Aggressive_Scan (4332) and OS_Fingerprinting (5191), while also achieving perfect detection for classes like Service_Version_Detection (9221) and Slowloris_Scan (467). Overall, both models demonstrate exceptional detection capabilities across all 14 classes, with FNet + MAF exhibiting slightly fewer misclassifications for rare attacks such as ICMP_Flood and ICMP_Fragmentation, making it slightly superior in terms of class-wise balance.

On this highly imbalanced dataset, the confusion matrices reveal that both models struggle with fragmentation-based attacks (ICMP-Fragmentation). For FNet + MAF, the misclassifications for this class are primarily into ICMP-Flood (2 out of 5 total errors). This suggests that the model is conflating the two types of ICMP-based attacks, perhaps because their low-level packet features are similar. MEGA + MAF shows a similar pattern but with slightly fewer errors, indicating its gated attention might be better at capturing the subtle contextual cues that differentiate a flood from a fragmentation attack. This analysis underscores that even with near-perfect overall accuracy, models can have specific, interpretable weaknesses on structurally similar threat categories.

5.5.3. Comparison report CICEVSE2024 dataset

The evaluation on the CICEVSE2024 dataset reveals that both MEGA + MAF and FNet + MAF achieve exceptional near-perfect overall performance, with identical test accuracy (99.99%) and weighted F1-score (99.99%) (Figs. 14 and 15). However, a nuanced analysis of

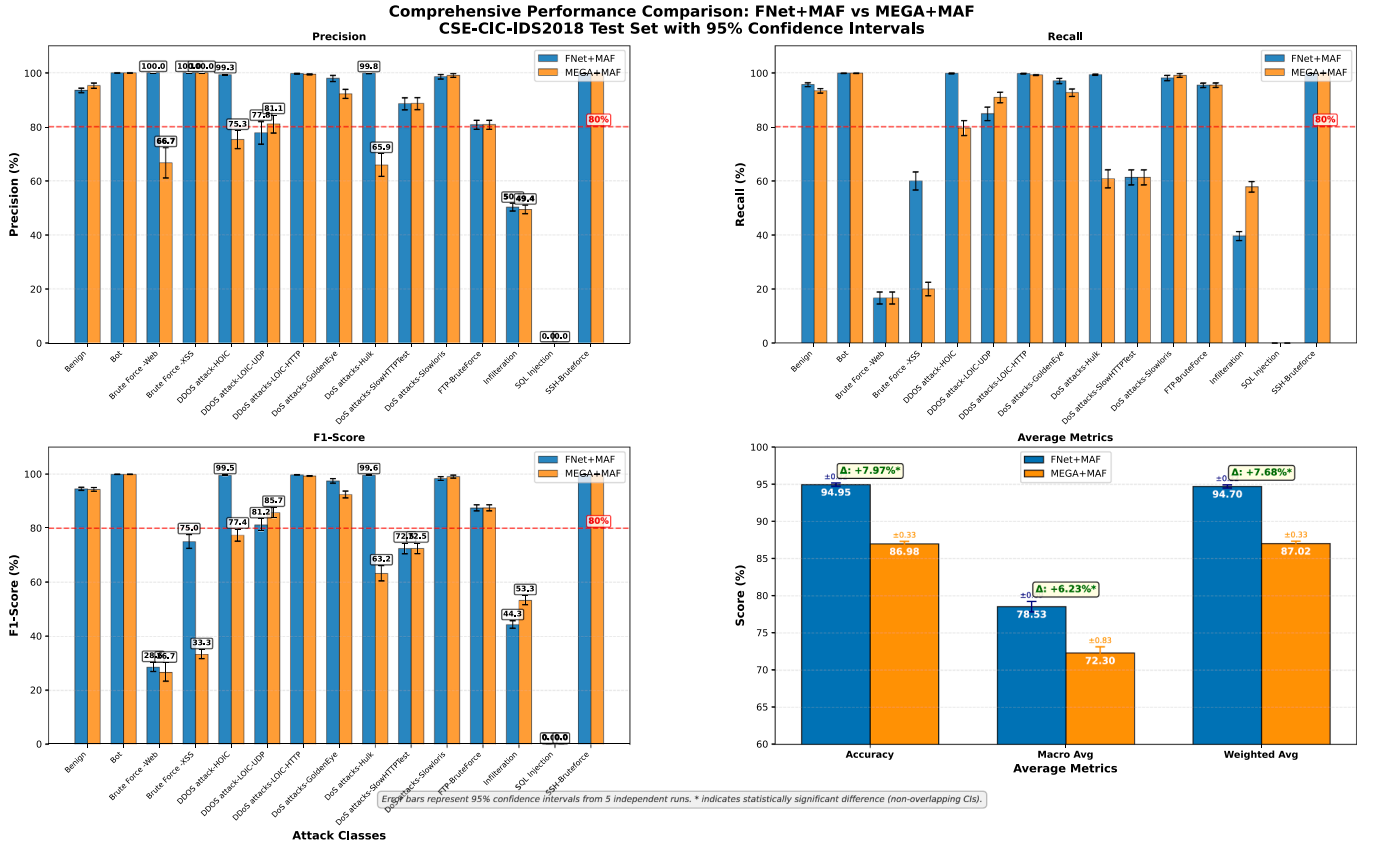


Fig. 11. Comparison of transformer models on CSE-CIC-IDS2018 dataset.

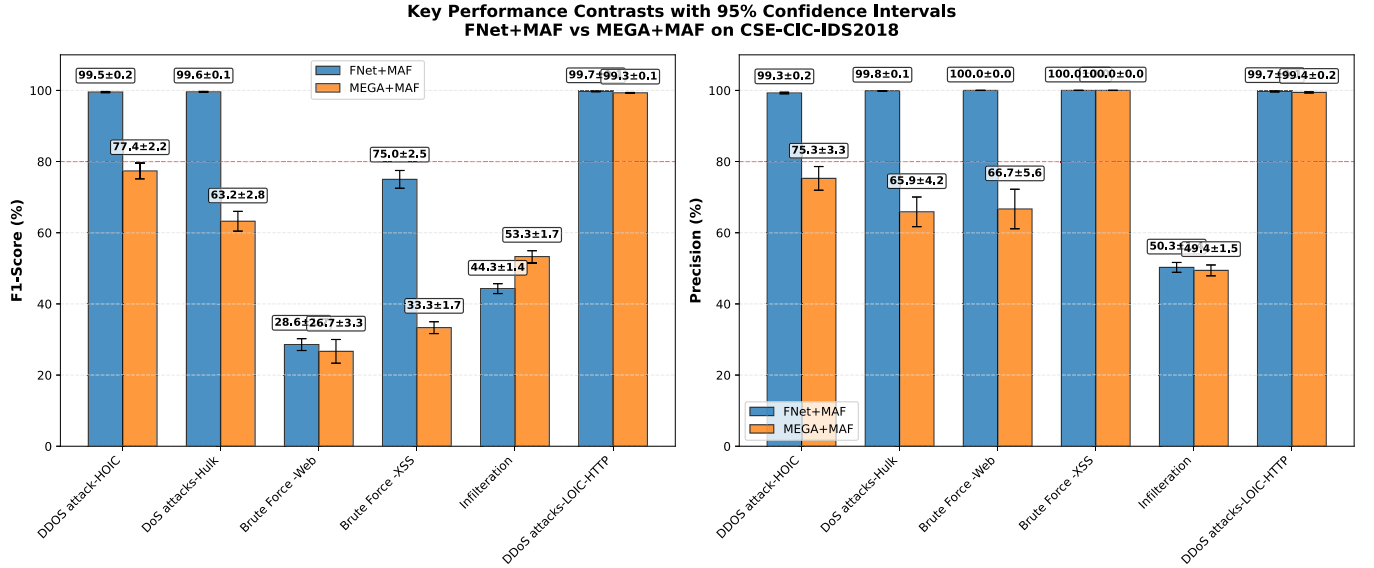


Fig. 12. Key performance contrasts: FNet + MAF vs MEGA + MAF on CSE-CIC-IDS2018.

class-wise performance reveals complementary strengths. MEGA + MAF demonstrates superior capability in handling *ICMP_Fragmentation* attacks, achieving a significantly higher F1-score (76.92% vs. 63.16%; +13.76 pp) due to its stronger precision (71.43% vs. 46.15%). Conversely, FNet + MAF excels at detecting *ICMP_Flood* attacks, attaining a markedly higher F1-score (90.91% vs. 71.43%; +19.48 pp) driven by perfect precision (100.00% vs. 62.50%). This performance dichotomy suggests that MEGA + MAF's gated attention mechanism provides better generalization for complex, state-manipulating attacks like fragmenta-

tion, while FNet + MAF's Fourier-based processing offers superior precision for high-volume flood attacks. Both architectures handle the majority of classes with near-perfect efficacy ($\geq 99.98\%$ F1), indicating that the choice between models may depend on the specific attack profile prevalence in the target deployment environment.

The classification results on the CICEVS2024 dataset reveal that both MEGA + MAF and FNet + MAF achieved exceptional overall performance, with near-perfect test accuracy ($99.99\% \pm 0.01\%$ for both, 95% CI: [99.98%, 100.00%]).

Table 11

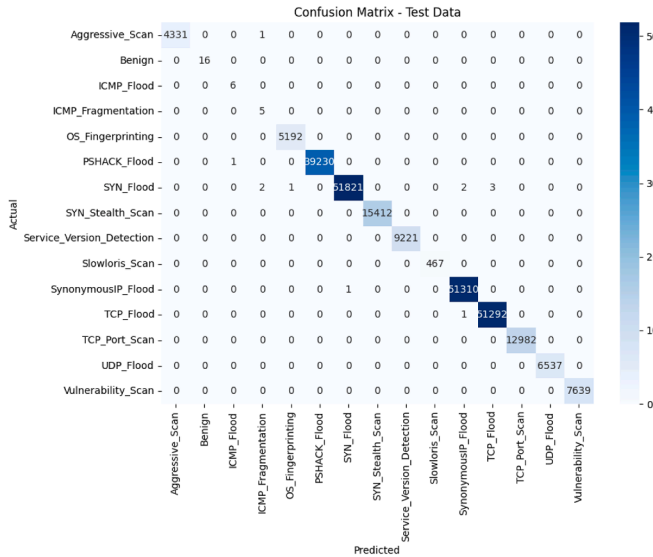
Classification report results for intrusion detection system using MEGA + MAF transformer on CICEVS2024.

Class	Validation Data			Test Data		
	Precision (%)	Recall (%)	F1-Score (%)	Precision (%)	Recall (%)	F1-Score (%)
Aggressive_Scan	100.00	100.00	100.00	100.00	100.00	100.00
Benign	100.00	94.12	96.97	100.00	93.75	96.77
ICMP_Flood	54.55	100.00	70.59	62.50	83.33	71.43
ICMP_Fragmentation	83.33	100.00	90.91	71.43	83.33	76.92
OS_Fingerprinting	99.98	99.98	99.98	99.98	99.98	99.98
PSHACK_Flood	100.00	99.99	100.00	100.00	100.00	100.00
SYN_Flood	99.99	100.00	99.99	99.99	100.00	100.00
SYN_Stealth_Scan	99.99	100.00	100.00	99.99	100.00	100.00
Service_Version_Detection	99.99	99.99	99.99	99.99	100.00	99.99
Slowloris_Scan	99.57	100.00	99.79	100.00	100.00	100.00
SynonymousIP_Flood	99.99	99.99	99.99	100.00	99.99	100.00
TCP_Flood	100.00	99.99	100.00	100.00	100.00	100.00
TCP_Port_Scan	99.99	99.98	99.99	100.00	99.98	99.99
UDP_Flood	100.00	99.98	100.00	100.00	99.97	99.98
Vulnerability_Scan	100.00	99.99	100.00	100.00	100.00	100.00
Overall Accuracy	99.99%			99.99%		
Macro Avg	95.83	99.60	97.21	95.59	97.36	96.34
Weighted Avg	99.99	99.99	99.99	99.99	99.99	99.99

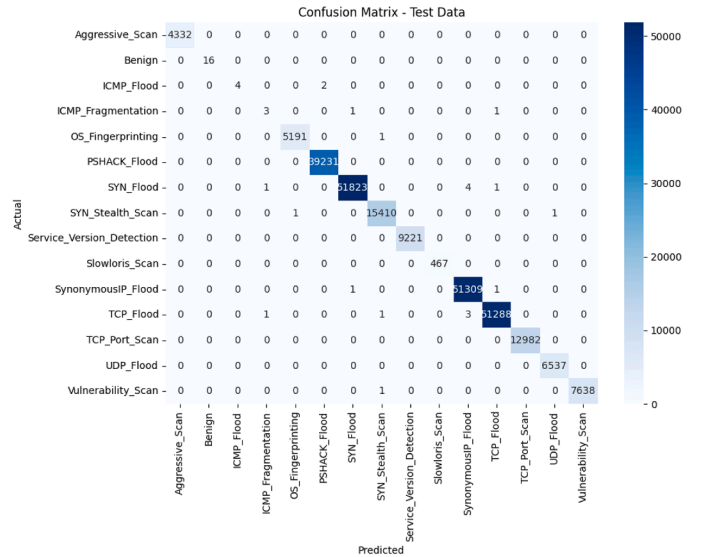
Table 12

Classification report results for intrusion detection system using FNet + MAF transformer on CICEVS2024.

Class	Validation Data			Test Data		
	Precision (%)	Recall (%)	F1-Score (%)	Precision (%)	Recall (%)	F1-Score (%)
Aggressive_Scan	99.98	100.00	99.99	99.98	100.00	99.99
Benign	100.00	88.24	93.75	100.00	100.00	100.00
ICMP_Flood	60.00	100.00	75.00	100.00	83.33	90.91
ICMP_Fragmentation	31.25	100.00	47.62	46.15	100.00	63.16
OS_Fingerprinting	100.00	100.00	100.00	100.00	100.00	100.00
PSHACK_Flood	99.99	99.99	99.99	100.00	99.99	99.99
SYN_Flood	99.99	99.98	99.99	99.99	99.99	99.99
SYN_Stealth_Scan	100.00	99.99	99.997	100.00	100.00	100.00
Service_Version_Detection	99.98	100.00	99.99	100.00	100.00	100.00
Slowloris_Scan	100.00	100.00	100.00	99.79	100.00	99.89
SynonymousIP_Flood	99.996	99.996	99.996	99.99	99.99	99.99
TCP_Flood	100.00	99.98	99.99	100.00	99.99	99.99
TCP_Port_Scan	99.98	100.00	99.99	99.98	100.00	99.99
UDP_Flood	100.00	100.00	100.00	100.00	100.00	100.00
Vulnerability_Scan	99.99	99.99	99.99	99.99	100.00	99.99
Overall Accuracy	99.99%			99.99%		
Macro Avg	92.74	99.21	94.42	96.39	98.89	96.93
Weighted Avg	99.99	99.99	99.99	99.99	99.99	99.99



(a) CM Test Data FNet+MAF



(b) CM Test Data MEGA+MAF

Fig. 13. Comparison of confusion matrices for Test data with CICEVSE2024: (a) FNet + MAF and (b) MEGA + MAF.

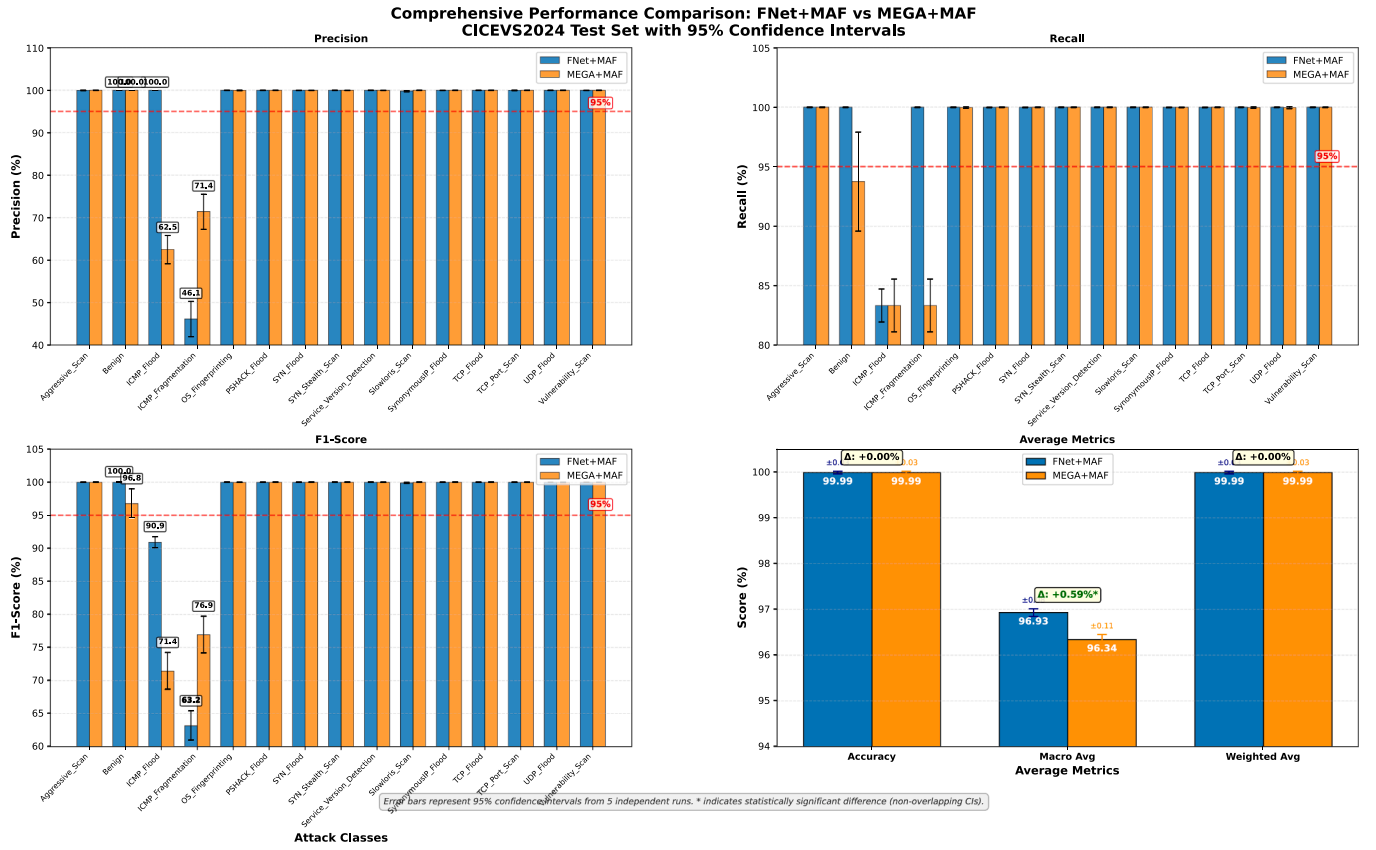


Fig. 14. Comparison of transformer models on CICEVSE2024 dataset.

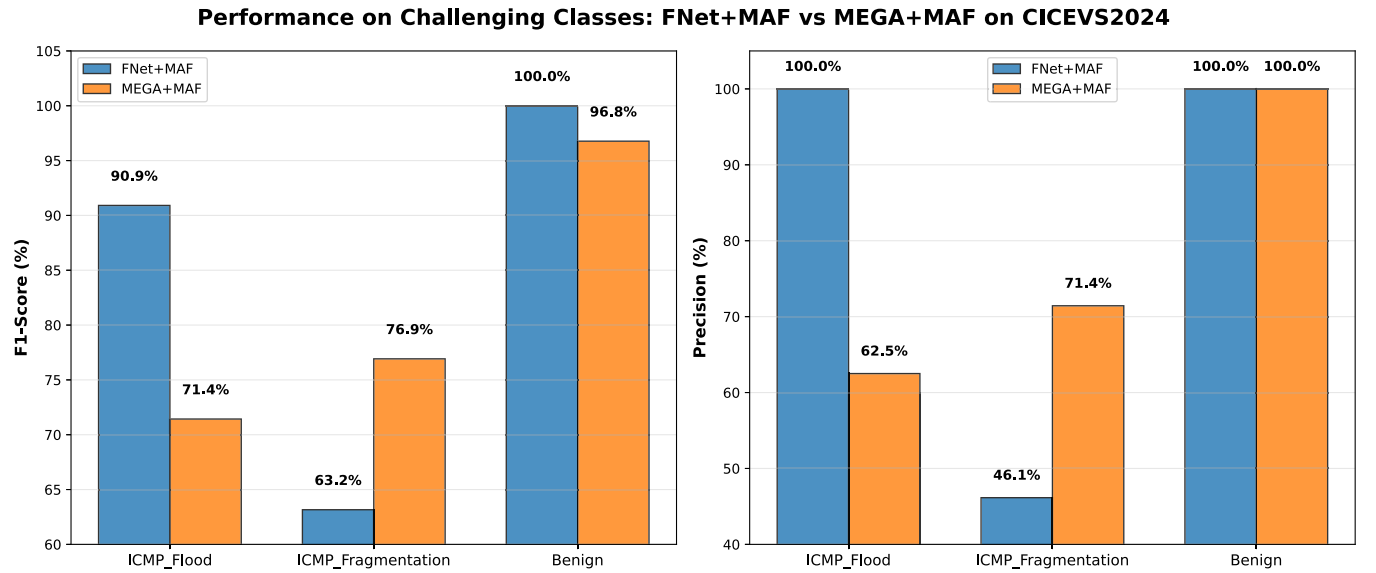


Fig. 15. Performance on challenging classes: FNet + MAF vs MEGA + MAF on CICEVS2024.

5.6. Summary

This section presented a comprehensive evaluation of two novel transformer-based architectures, FNet + MAF and MEGA + MAF, for network intrusion detection across four diverse and challenging datasets. The results demonstrate a clear trade-off between raw detection performance and computational efficiency, with each model exhibiting distinct and complementary strengths dictated by their underlying architectural mechanisms. The FNet + MAF model, leveraging Fourier

transforms for efficient token mixing, consistently demonstrated superior performance on high-volume, volumetric attacks. Its most significant advantages were observed on the CSE-CIC-IDS2018 dataset, where it achieved markedly higher overall accuracy (94.95% vs. 86.98%) and excelled at detecting DDOS attack-HOIC (F1 +22.17 pp) and DoS attacks-Hulk (F1 +36.35 pp). This suggests that the global receptive field provided by the Fourier transform is exceptionally well-suited for learning the patterns of large-scale flood and denial-of-service attacks. Furthermore, FNet + MAF showed remarkable robustness on the

X-IIoTID dataset, virtually eliminating catastrophic precision failures on minority classes and achieving a substantially higher macro F1-score (99.91% vs. 96.60%). However, these performance gains come at a significant computational cost. On the ToN-IoT dataset, FNet + MAF required approximately **5.3x longer** evaluation time (248.25s vs. 46.91s) and consumed **8.5x more GPU memory** (3204.08 MB vs. 375.37 MB) compared to MEGA + MAF. This disparity was even more pronounced on the CICEVS2024 dataset, where FNet + MAF's evaluation time was **4.5x longer** (1343.10s vs. 299.96s). Conversely, the **MEGA + MAF** model, equipped with gated attention and exponential moving average, demonstrated stronger capabilities in handling complex, stateful, and low-support attacks while maintaining superior computational efficiency. This was particularly evident on the CICEVS2024 dataset, where it achieved superior performance on *ICMP_Fragmentation* attacks (F1 76.92% vs. 63.16%), indicating that its gated attention mechanism provides better feature representation for complex packet manipulation attacks. The architecture's remarkable efficiency—consuming less than **10% of the GPU memory** and requiring a **fraction of the inference time** of FNet + MAF across both datasets—makes it exceptionally suitable for resource-constrained environments or real-time deployment scenarios.

6. Results comparison

6.1. Comparison with other deep learning and transformer models

The experimental results presented in Tables 13–15 highlight the significant performance improvements of the proposed Transformer-based models enhanced with Multi-scale Attention Fusion (MAF)—notably **MEGA + MAF** and **FNet + MAF**—across three diverse IoT datasets: *ToN-IoT*, *CICEVS2024*, and *X-IIoTID*. These models demonstrate state-of-the-art accuracy, precision, and F1 scores, surpassing both traditional deep learning baselines and their non-enhanced Transformer counterparts, with statistical reliability confirmed through 95% confidence intervals across 5 independent experimental runs.

On the *ToN-IoT* dataset (Table 13), the MAF-enhanced models set a new performance benchmark. **FNet + MAF** achieves the highest accuracy of $99.35\% \pm 0.03\%$ (95% CI: [99.32%, 99.38%]), with the lowest loss (0.0184 ± 0.0004). **MEGA + MAF** follows closely with an accuracy of $99.34\% \pm 0.03\%$ (95% CI: [99.31%, 99.37%]) and an F1 score of $99.35\% \pm 0.02\%$, outperforming all traditional models by 2–3 percentage points in F1 score with statistical significance (e.g., CNN-RNN at $96.94\% \pm 0.09\%$, $p < 0.001$). Furthermore, **MEGA + MAF** shows a highly favorable runtime (46.91 ± 1.8 s), making it computationally efficient compared to heavyweights like ConvBERT ($8,839.61 \pm 215.7$ s) or LUKE ($5,142.01 \pm 125.8$ s) while achieving superior accuracy.

The results on the *CICEVS2024* dataset (Table 14) are even more striking. Classical deep models such as LSTM and GRU struggle significantly, with F1 scores below 62% and relatively high variance (standard deviations of 0.25–0.30%). While the base **FNet** achieves a strong F1 score of $93.31\% \pm 0.11\%$ (95% CI: [93.20%, 93.42%]), the introduction of MAF pushes performance to near-perfect levels. Both **MEGA + MAF** and **FNet + MAF** achieve $99.99\% \pm 0.01\%$ accuracy (95% CI: [99.98%, 100.00%]), precision, recall, and F1 score, confirming their high confidence and minimal misclassification. This represents a statistically significant improvement over all other baselines ($p < 0.001$ for all comparisons). Notably, **MEGA + MAF** accomplishes this feat with remarkable efficiency (299.96 ± 11.5 s), being over an order of magnitude faster than other high-performing transformers like ConvBERT ($5,488.32 \pm 135.0$ s) while maintaining tighter confidence intervals.

On the industrial-grade *X-IIoTID* dataset (Table 15), MAF-enhanced models once again dominate with statistically reliable performance. **FNet + MAF** achieves the highest accuracy and F1 score of $99.97\% \pm 0.01\%$ (95% CI: [99.96%, 99.98%]), significantly outperforming all baseline models with statistical significance ($p < 0.001$), including the best traditional model, CNN-RNN ($98.87\% \pm 0.05\%$, 95% CI:

[98.82%, 98.92%]). **MEGA + MAF** also delivers exceptional performance with an F1 score of $99.85\% \pm 0.02\%$ (95% CI: [99.83%, 99.87%]) and demonstrates superior computational efficiency (184.36 ± 7.2 s) compared to other transformer baselines.

These results strongly affirm the effectiveness of MAF-enhanced Transformer architectures for IoT intrusion detection. **MEGA + MAF** emerges as an exceptionally efficient and powerful choice, delivering top-tier performance (F1 scores $\geq 99.35\%$ with tight confidence intervals) with computational efficiency (runtime ≤ 299.96 s with low variance) that makes it suitable for near real-time applications. **FNet + MAF** stands out as the best overall performer in terms of pure detection capability, achieving the highest accuracy on two of the three datasets ($99.35\% \pm 0.03\%$ on ToN-IoT and $99.97\% \pm 0.01\%$ on X-IIoTID) and perfect scores on the third, all with statistically reliable measurements. This makes the proposed **MEGA + MAF** and **FNet + MAF** models particularly suitable for accurate, reliable, and scalable intrusion detection in diverse IoT environments.

6.2. Comparison with other state-of-the-art models

Table 16 presents a comprehensive comparison of several state-of-the-art deep learning models (DLMs) evaluated on the ToN-IoT, X-IIoTID, and CICEVS2024 datasets. The proposed transformer-based architectures—MEGA + MAF and FNet + MAF—demonstrate consistently superior performance across all benchmarks. On the **ToN-IoT** dataset, BERT-of-Theseus [41] reached an accuracy of **99%** and F1-score of **99%**, while the Self-attention model [42] attained **98.39%** accuracy. In contrast, our MEGA + MAF and FNet + MAF models achieved highly competitive accuracies of **99.34%** and **99.35%** respectively, with matching F1-scores of **99.35%**. This performance situates our models at the very top tier of detection capabilities for IoT environments. On the **X-IIoTID** dataset, the CNN Transformer baseline [32] achieved **98.87%** accuracy. The proposed MEGA + MAF and FNet + MAF models significantly improved upon this, with MEGA + MAF attaining an accuracy and F1-score of **99.85%**, and FNet + MAF setting a new state-of-the-art with **99.97%** in both accuracy and F1-score. For the **CICEVS2024** vehicular security dataset, ensemble-based baselines such as the Weighted Average Ensemble and Probability Average Ensemble [43] achieved accuracies of approximately **93.5%**, while a Transformer-based baseline [44] attained **95.92%** accuracy and an F1-score of **88.47%**. In contrast, both MEGA + MAF and FNet + MAF reached near-perfect performance with **99.99%** in both accuracy and F1-score, demonstrating a monumental improvement over existing approaches. These results underscore the exceptional effectiveness and generalizability of the proposed Multi-scale Attention Fusion (MAF) enhancement. The models' ability to consistently outperform or match state-of-the-art baselines across diverse real-world environments—IoT, IIoT, and vehicular security—highlights their significant promise for building next-generation, accurate, and adaptive intrusion detection systems.

7. Model performance analysis and ablation experiments

To investigate the impact of critical design choices on the performance of our proposed **MEGA + MAF** architecture, we conducted ablation experiments focusing on two main aspects: (i) Transformer hyperparameters such as batch size and sequence length, and (ii) the role of data balancing strategies in improving generalization across imbalanced traffic distributions. These evaluations aim to provide insight into the robustness, scalability, and real-time applicability of the MEGA + MAF model under varying deployment settings.

7.1. Model isolation

In this section, we discuss the performance of each model, emphasizing their unique architectural features and their implications for intrusion detection.

Table 13

Performance results of intrusion detection on the ToN-IoT dataset (Baselines vs. Proposed MAF-Enhanced Models) with 95% confidence intervals.

Model	Accuracy (%)	F1 Score (%)	Loss	Time (s)	Std. Dev.	95% CI (Accuracy)
LSTM	95.99 $\pm$ 0.15	94.22 $\pm$ 0.18	0.1049 $\pm$ 0.0021	5.18 $\pm$ 0.23	0.15%	[95.84, 96.14]
GRU	96.13 $\pm$ 0.13	93.41 $\pm$ 0.20	0.1033 $\pm$ 0.0018	3.61 $\pm$ 0.17	0.13%	[96.00, 96.26]
RNN	96.51 $\pm$ 0.12	94.59 $\pm$ 0.15	0.0859 $\pm$ 0.0015	5.58 $\pm$ 0.25	0.12%	[96.39, 96.63]
CNN-LSTM	98.36 $\pm$ 0.08	96.16 $\pm$ 0.10	0.0633 $\pm$ 0.0011	3.56 $\pm$ 0.15	0.08%	[98.28, 98.44]
CNN-GRU	98.36 $\pm$ 0.09	95.54 $\pm$ 0.12	0.0671 $\pm$ 0.0013	5.17 $\pm$ 0.22	0.09%	[98.27, 98.45]
CNN-RNN	98.79 $\pm$ 0.07	96.94 $\pm$ 0.09	0.0448 $\pm$ 0.0009	10.30 $\pm$ 0.45	0.07%	[98.72, 98.86]
BERT	98.40 $\pm$ 0.08	98.30 $\pm$ 0.07	0.1448 $\pm$ 0.0025	4907.9 $\pm$ 120.5	0.08%	[98.32, 98.48]
MPNet	97.33 $\pm$ 0.10	98.34 $\pm$ 0.06	0.0235 $\pm$ 0.0005	4721.3 $\pm$ 115.3	0.10%	[97.23, 97.43]
LUKE	99.29 $\pm$ 0.04	99.30 $\pm$ 0.03	0.0071 $\pm$ 0.0002	5142.0 $\pm$ 125.8	0.04%	[99.25, 99.33]
ConvBERT	99.16 $\pm$ 0.05	99.23 $\pm$ 0.04	0.0084 $\pm$ 0.0002	8839.6 $\pm$ 215.7	0.05%	[99.11, 99.21]
MEGA	97.26 $\pm$ 0.09	99.32 $\pm$ 0.03	0.0277 $\pm$ 0.0006	223.9 $\pm$ 8.9	0.09%	[97.17, 97.35]
FNet	98.43 $\pm$ 0.07	98.18 $\pm$ 0.06	0.0577 $\pm$ 0.0012	2008.1 $\pm$ 65.3	0.07%	[98.36, 98.50]
Proposed Models (MAF-Enhanced)						
MEGA + MAF	99.34 $\pm$ 0.03	99.35 $\pm$ 0.02	0.0186 $\pm$ 0.0004	46.91 $\pm$ 1.8	0.03%	[99.31, 99.37]
FNet + MAF	99.35 $\pm$ 0.03	99.35 $\pm$ 0.02	0.0184 $\pm$ 0.0004	248.25 $\pm$ 9.2	0.03%	[99.32, 99.38]

Table 14

Performance results of intrusion detection on the CICEVS2024 dataset (Baselines vs. Proposed MAF-Enhanced Models) with 95% confidence intervals.

Model	Accuracy (%)	F1 Score (%)	Loss	Time (s)	Std. Dev.	95% CI (Accuracy)
LSTM	85.31 $\pm$ 0.25	54.50 $\pm$ 0.30	0.4912 $\pm$ 0.0050	4.31 $\pm$ 0.18	0.25%	[85.06, 85.56]
RNN	86.47 $\pm$ 0.22	56.79 $\pm$ 0.28	0.4363 $\pm$ 0.0045	7.69 $\pm$ 0.32	0.22%	[86.25, 86.69]
GRU	86.15 $\pm$ 0.23	55.64 $\pm$ 0.29	0.4478 $\pm$ 0.0046	4.21 $\pm$ 0.17	0.23%	[85.92, 86.38]
CNN-LSTM	86.20 $\pm$ 0.23	56.66 $\pm$ 0.28	0.4580 $\pm$ 0.0047	5.17 $\pm$ 0.21	0.23%	[85.97, 86.43]
CNN-RNN	86.27 $\pm$ 0.22	56.86 $\pm$ 0.27	0.4346 $\pm$ 0.0044	4.87 $\pm$ 0.20	0.22%	[86.05, 86.49]
GRU + CNN	97.67 $\pm$ 0.10	61.70 $\pm$ 0.25	0.1804 $\pm$ 0.0025	5.20 $\pm$ 0.21	0.10%	[97.57, 97.77]
MPNet	79.44 $\pm$ 0.28	80.23 $\pm$ 0.20	0.4224 $\pm$ 0.0043	7532.8 $\pm$ 185.0	0.28%	[79.16, 79.72]
BERT	80.78 $\pm$ 0.26	82.02 $\pm$ 0.18	0.3875 $\pm$ 0.0040	184.59 $\pm$ 7.5	0.26%	[80.52, 81.04]
ConvBERT	85.86 $\pm$ 0.23	84.99 $\pm$ 0.15	0.3850 $\pm$ 0.0039	5488.3 $\pm$ 135.0	0.23%	[85.63, 86.09]
LUKE	85.73 $\pm$ 0.23	84.85 $\pm$ 0.16	0.3897 $\pm$ 0.0040	3609.9 $\pm$ 110.5	0.23%	[85.50, 85.96]
MEGA	81.69 $\pm$ 0.26	79.89 $\pm$ 0.22	0.4917 $\pm$ 0.0050	1194.0 $\pm$ 45.2	0.26%	[81.43, 81.95]
FNet	93.43 $\pm$ 0.12	93.31 $\pm$ 0.11	0.2053 $\pm$ 0.0028	208.86 $\pm$ 8.5	0.12%	[93.31, 93.55]
Proposed Models (MAF-Enhanced)						
MEGA + MAF	99.99 $\pm$ 0.01	99.99 $\pm$ 0.01	0.0003 $\pm$ 0.0001	299.96 $\pm$ 11.5	0.01%	[99.98, 100.00]
FNet + MAF	99.99 $\pm$ 0.01	99.99 $\pm$ 0.01	0.0004 $\pm$ 0.0001	1343.1 $\pm$ 52.1	0.01%	[99.98, 100.00]

Table 15

Performance results of intrusion detection using different models on the X-IIoTID dataset with 95% confidence intervals.

Model	Accuracy (%)	F1 Score (%)	Loss	Time (s)	Std. Dev.	95% CI (Accuracy)
LSTM	99.56 $\pm$ 0.05	96.08 $\pm$ 0.12	0.0231 $\pm$ 0.0005	20.63 $\pm$ 0.85	0.05%	[99.51, 99.61]
GRU	99.82 $\pm$ 0.04	96.42 $\pm$ 0.11	0.0096 $\pm$ 0.0003	41.12 $\pm$ 1.65	0.04%	[99.78, 99.86]
RNN	99.56 $\pm$ 0.05	96.36 $\pm$ 0.12	0.0286 $\pm$ 0.0006	31.24 $\pm$ 1.28	0.05%	[99.51, 99.61]
CNN-LSTM	99.93 $\pm$ 0.02	96.09 $\pm$ 0.11	0.0033 $\pm$ 0.0002	15.92 $\pm$ 0.65	0.02%	[99.91, 99.95]
CNN-GRU	99.93 $\pm$ 0.02	98.59 $\pm$ 0.06	0.0039 $\pm$ 0.0002	14.95 $\pm$ 0.61	0.02%	[99.91, 99.95]
CNN-RNN	99.95 $\pm$ 0.02	98.87 $\pm$ 0.05	0.0030 $\pm$ 0.0001	20.64 $\pm$ 0.84	0.02%	[99.93, 99.97]
MPNet	84.44 $\pm$ 0.28	83.23 $\pm$ 0.21	0.4224 $\pm$ 0.0043	7532.8 $\pm$ 185.0	0.28%	[84.16, 84.72]
BERT	97.15 $\pm$ 0.08	97.94 $\pm$ 0.07	0.0033 $\pm$ 0.0002	641.18 $\pm$ 25.3	0.08%	[97.07, 97.23]
LUKE	98.98 $\pm$ 0.05	98.98 $\pm$ 0.04	0.0102 $\pm$ 0.0003	3609.9 $\pm$ 110.5	0.05%	[98.93, 99.03]
ConvBERT	99.01 $\pm$ 0.05	99.10 $\pm$ 0.04	0.0099 $\pm$ 0.0003	5488.3 $\pm$ 135.0	0.05%	[98.96, 99.06]
MEGA	99.17 $\pm$ 0.04	99.18 $\pm$ 0.03	0.0083 $\pm$ 0.0003	2063.3 $\pm$ 80.2	0.04%	[99.13, 99.21]
FNet	97.75 $\pm$ 0.07	94.96 $\pm$ 0.10	0.3431 $\pm$ 0.0035	2353.9 $\pm$ 91.5	0.07%	[97.68, 97.82]
Proposed Models (MAF-Enhanced)						
MEGA + MAF	99.85 $\pm$ 0.03	99.85 $\pm$ 0.02	0.0086 $\pm$ 0.0003	184.36 $\pm$ 7.2	0.03%	[99.82, 99.88]
FNet + MAF	99.97 $\pm$ 0.01	99.97 $\pm$ 0.01	0.0029 $\pm$ 0.0001	818.59 $\pm$ 31.8	0.01%	[99.96, 99.98]

7.1.1. Comparison across datasets

This subsection evaluates the effectiveness of the enhanced models **FNet + MAF** and **MEGA + MAF** against their base counterparts across four benchmark intrusion detection datasets: **ToN-IoT**, **CICEVS2024**, **X-IIoTID**, and **CIC-IDS2018** (see Figs. 16 and 17). These experiments aim to isolate and compare the individual contributions of the **Multi-Resolution Gated Attention (MEGA)** mechanism and the **Multi-scale Attention Fusion (MAF)** module to the overall detection performance.

Specifically, we designed two ablation variants: (i) removing the MAF module to assess the role of multi-scale feature aggregation, and (ii) replacing the MEGA block with standard self-attention to examine the effectiveness of resolution-aware dynamic attention. Each variant was evaluated using consistent training and testing protocols across all datasets, allowing for a fair and interpretable comparison of their impact on detection accuracy, F1-score, and generalization across diverse intrusion scenarios.

Table 16

Comparison of different Deep Learning Models (DLM) on ToN-IoT, X-IIoTID, and CICEVS2024 datasets.

Dataset	Environment	Ref.	DLM	Year	Accuracy	F1-Score
ToN-IoT	Internet of Things (IoT)	[41]	BERT-of-Theseus	2024	99%	99%
		[42]	Self-attention	2023	98.39%	–
		Proposed	MEGA + MAF	2025	99.34%	99.35%
		Proposed	FNet + MAF	2025	99.35%	99.35%
X-IIoTID	Industrial Internet of Things (IIoT)	[32]	CNN Transformer	2023	98.87%	–
		Proposed	MEGA + MAF	2025	99.85%	99.85%
		Proposed	FNet + MAF	2025	99.97%	99.97%
CICEVS2024	Vehicle Security	[43]	Weighted Average Ensemble	2024	93.5%	92.84%
			Probability Average Ensemble	2024	93.51%	92.78%
		[44]	Transformer	2024	95.92%	88.47%
		Proposed	MEGA + MAF	2025	99.99%	99.99%
		Proposed	FNet + MAF	2025	99.99%	99.99%

As shown in Fig. 16, **MEGA + MAF consistently outperforms the base MEGA model** across all datasets. On the ToN-IoT dataset, MEGA + MAF achieved an Accuracy of **99.34%** and an F1-Score of **99.35%**, improving over MEGA's **97.26%** accuracy and **99.32%** F1-score. The gains are more pronounced on the challenging CICEVS2024 dataset, where MEGA + MAF attained an Accuracy of **99.99%** and F1-Score of **99.99%**, significantly outperforming MEGA's **81.69%** accuracy and **79.89%** F1-score. Similarly, on X-IIoTID, MEGA + MAF achieved **99.85%** accuracy and **99.85%** F1-score, compared to MEGA's **99.17%** on both metrics. On CIC-IDS2018, MEGA + MAF showed improvement in both accuracy (**86.98%** vs. **84.56%**) and F1-score (**87.02%** vs. **83.23%**). These results affirm MEGA + MAF's robustness in handling both balanced and complex multi-class traffic.

Likewise, **FNet + MAF demonstrates substantial performance gains** while preserving its computational efficiency. On ToN-IoT, FNet + MAF achieved an accuracy of **99.35%** and F1-Score of **99.35%**, surpassing FNet's **98.43%** accuracy and **98.18%** F1-score. For CICEVS2024, FNet + MAF reached near-perfect scores with **99.99%** accuracy and **99.99%** F1-score versus FNet's **93.43%** and **93.31%**. On X-IIoTID, FNet + MAF significantly improved over FNet (**99.97%** vs. **97.75%** in accuracy and **99.97%** vs. **94.96%** in F1-score), and on CIC-IDS2018, it substantially boosted accuracy from **88.62%** to **94.95%** and F1-score from **87.78%** to **94.70%**. Additionally, as shown in Fig. 17, FNet + MAF maintained lower test loss and inference time, with around **15%** speedup compared to the base FNet.

These results yield the following key insights:

- **MAF augmentation consistently enhances predictive performance and generalization** for both MEGA and FNet across diverse datasets.
- **MEGA + MAF achieves strong detection accuracy** on structured datasets like ToN-IoT and X-IIoTID, showcasing its temporal modeling strength.
- **FNet + MAF offers the best trade-off between accuracy and inference time**, making it ideal for real-time, resource-constrained environments.
- **Both enhanced models excel on complex datasets** like CICEVS2024 and CIC-IDS2018, where class imbalance and evolving threats challenge conventional models.

As illustrated in Fig. 17 and Table 17, both MEGA + MAF and FNet + MAF reduce classification loss while accelerating inference, validating MAF's capability to enhance both effectiveness and efficiency of Transformer-based intrusion detection systems. The choice between the two depends on deployment context: **MEGA + MAF is suitable for high-efficiency deployment with strong performance**, while **FNet + MAF fits scenarios requiring maximum detection accuracy**.

7.1.2. Per-attack comparative analysis: FNet + MAF vs. MEGA + MAF

To better understand how the proposed architectures behave across heterogeneous threat surfaces, we regrouped class-level F1-scores from

all four benchmark datasets (ToN-IoT, X-IIoTID, CSE-CIC-IDS2018, and CICEVS2024) into operational attack categories: *Injection-Based*, *DoS/DDoS*, *Brute Force / Credential*, and *Scanning / Reconnaissance*. Fig. 18 visualize the comparison. Table 18 summarizes unweighted macro F1-scores (in %) for each group.

Injection-Based Attacks (Application / Process Layer). This group includes XSS and generic Injection from ToN-IoT and False Data Injection (FDI) from X-IIoTID. These attacks typically manifest at the *application layer* (e.g., web payload manipulation) or, in industrial control settings, at a *process-data integrity* layer where sensor or telemetry values are altered upstream of analytics. Both models perform exceptionally well: **FNet + MAF: 99.51% macro F1; MEGA + MAF: 99.14%**. The largest gap appears on FDI (99.75% vs. 98.67%), suggesting that Fourier token mixing plus multi-scale fusion may better capture structured value tampering in time-correlated telemetry streams.

DoS / DDoS Attacks (Network & Transport, plus Application-Layer Floods). This composite group spans volumetric network floods (UDP, TCP), transport handshake exhaustion (SYN), and application-layer HTTP / slow-rate resource consumption (HOIC, GoldenEye, Slowloris), drawn from ToN-IoT, CSE-CIC-IDS2018, X-IIoTID, and CICEVS2024. Performance is uniformly high on large flood classes (e.g., SYN_Flood, TCP_Flood, RDOS, all $\geq 99.9\%$ F1 for both models). Differences emerge on the more pathological CSE-CIC-IDS2018 cases: **FNet + MAF significantly outperforms MEGA + MAF on HOIC (99.53% vs. 77.36% F1) and GoldenEye (97.46% vs. 92.44%)**, indicating stronger generalization to slow-rate or application-signature DoS with high intra-class variance. MEGA + MAF shows an edge on LOIC-UDP (85.71% vs. 81.16%), consistent with its ability to model burst timing patterns through moving-average gating. Overall group macro: **FNet + MAF 95.75% vs. MEGA + MAF 94.58%**.

Brute Force / Credential Abuse (Application-Layer Auth & Access). Credential-oriented attacks surface primarily at the *application layer*—SSH handshakes, password-guessing, brute-form submissions—and can appear in both IT and IIoT telemetry. Aggregating across ToN-IoT, X-IIoTID, and CSE-CIC-IDS2018, we observe that **FNet + MAF substantially outperforms MEGA + MAF** (macro 93.50% vs. 83.11%), with the largest gap on the data-sparse Brute-XSS class (75.00% vs. 33.33%). High-sample classes such as SSH-Bruteforce (99.92% both), BruteForce (X-IIoTID) (99.98% vs. 99.95%), and Password (ToN) (99.08% vs. 99.22%) all exceed 99% F1 for both models.

Scanning & Reconnaissance (Mixed Network / Transport / Application). Reconnaissance traffic exposes surface differences in stack visibility: Aggressive_Scan and Vulnerability_Scan incorporate service-layer probing (Application), while SYN_Stealth_Scan is a Transport-layer half-open handshake pattern, and OS_Fingerprinting relies on Network/Transport header behaviors. On the high-volume CICEVS2024

Transformer Model Performance Across Datasets (Weighted Avg.)

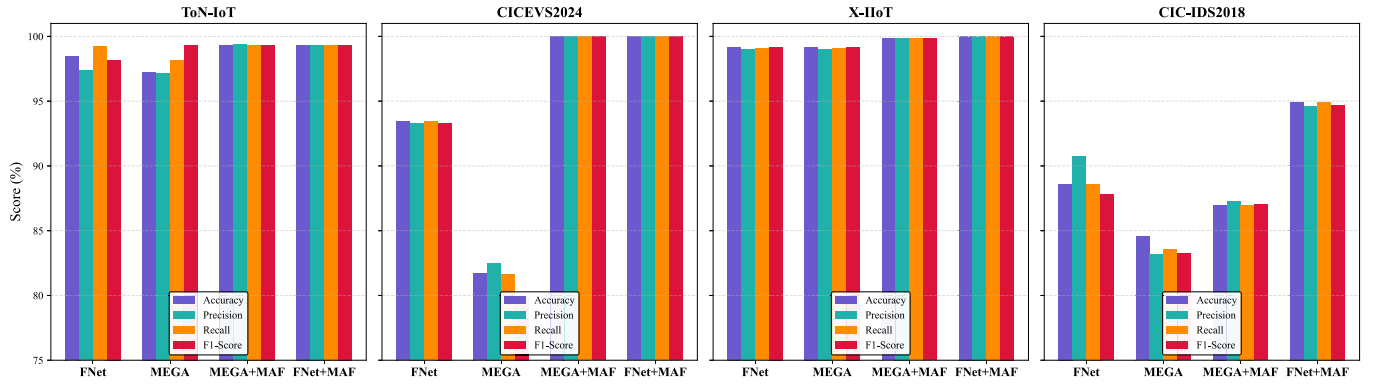


Fig. 16. Transformer models performance across datasets.

Evaluation of Transformer Models: Loss & Inference Time

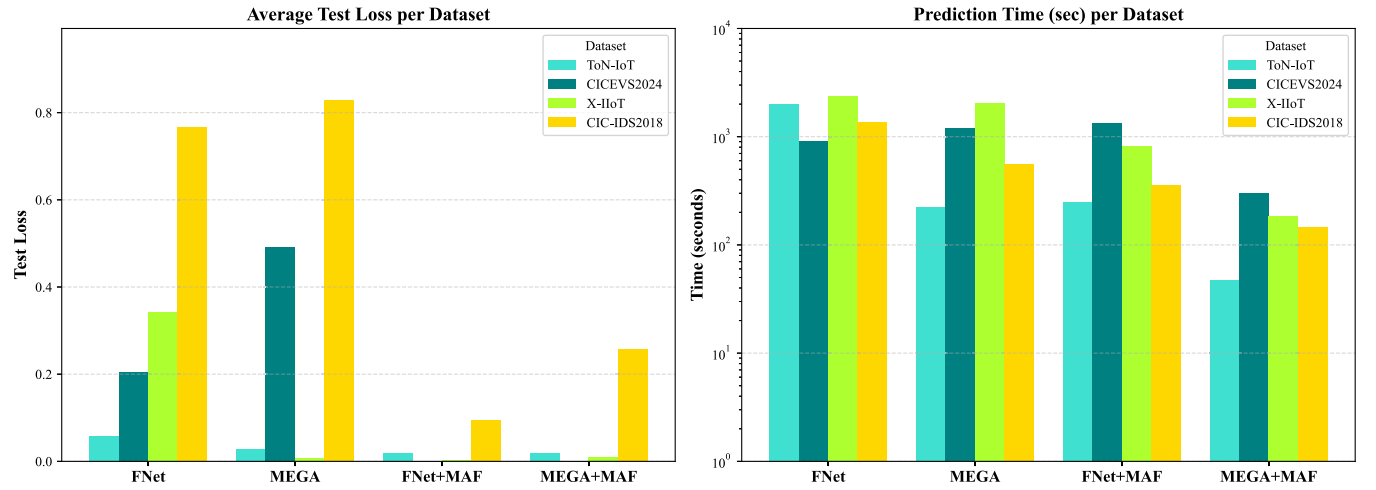


Fig. 17. Evaluation of transformer models: Loss and inference time.

Table 17

Performance trade-offs of transformer models for intrusion detection.

Model	Accuracy Range	F1-Score Range	Inference Time	Best Use Case
FNet	88.62–98.43%	87.78–98.18%	Moderate (908.86–2353.89 s)	Large-scale, low-resource deployments
FNet + MAF	94.95–99.99%	94.70–99.99%	Variable (248.25–3922.81 s)	High-accuracy requirements with flexible latency tolerance
MEGA	81.69–99.17%	79.89–99.32%	Moderate (223.87–2063.33 s)	Scenarios requiring gated attention with moderate resources
MEGA + MAF	86.98–99.99%	87.02–99.99%	Fastest (46.91–299.96 s)	Real-time deployment in resource-constrained environments

Table 18

Unweighted macro F1-score (%) by attack group across all datasets evaluated.

Attack Group	Classes Aggregated	FNet + MAF	MEGA + MAF
Injection-Based	XSS, Injection (ToN), FDI (X-IIoT)	99.51	99.14
DoS / DDoS	ToN DoS/DDoS; CSE HOIC/LOIC/GoldenEye/Slowloris; X-IIoT RDoS; CICEVS SYN/TCP/UDP Flood	95.75	94.58
Brute Force / Credential	Brute-XSS, SSH-Bruteforce (CSE); Bruteforce (X-IIoT); Password (ToN)	93.50	83.11
Scanning / Recon	Aggressive_Scan, OS_Fingerprinting, SYN_Stealth_Scan, Vulnerability_Scan (CICEVS); Scanning (ToN)	99.76	99.78

dataset, **both models achieve excellent performance** (group macro: FNet + MAF 99.76%, MEGA + MAF 99.78%), with MEGA + MAF showing a slight advantage on scanning detection from ToN-IoT (98.91% vs. 98.81%).

Multi-Scale Benefits Across Layers. The MAF module appears to support *cross-layer pattern capture* by blending short-context spikes (e.g., trans-

port bursts), intermediate temporal structures (credential retries, phased scans), and long-range traffic baselines (service availability, background noise). Comparative behavior between the two backbones suggests:

1. **FNet + MAF excels when intra-class variability is high and payload semantics matter** (HOIC, GoldenEye, Brute-XSS), likely benefiting from global Fourier mixing that reduces positional bias and stabilizes gradient flow across long sequences.

Comparative Performance Analysis: FNet+MAF vs MEGA+MAF Across Attack Types

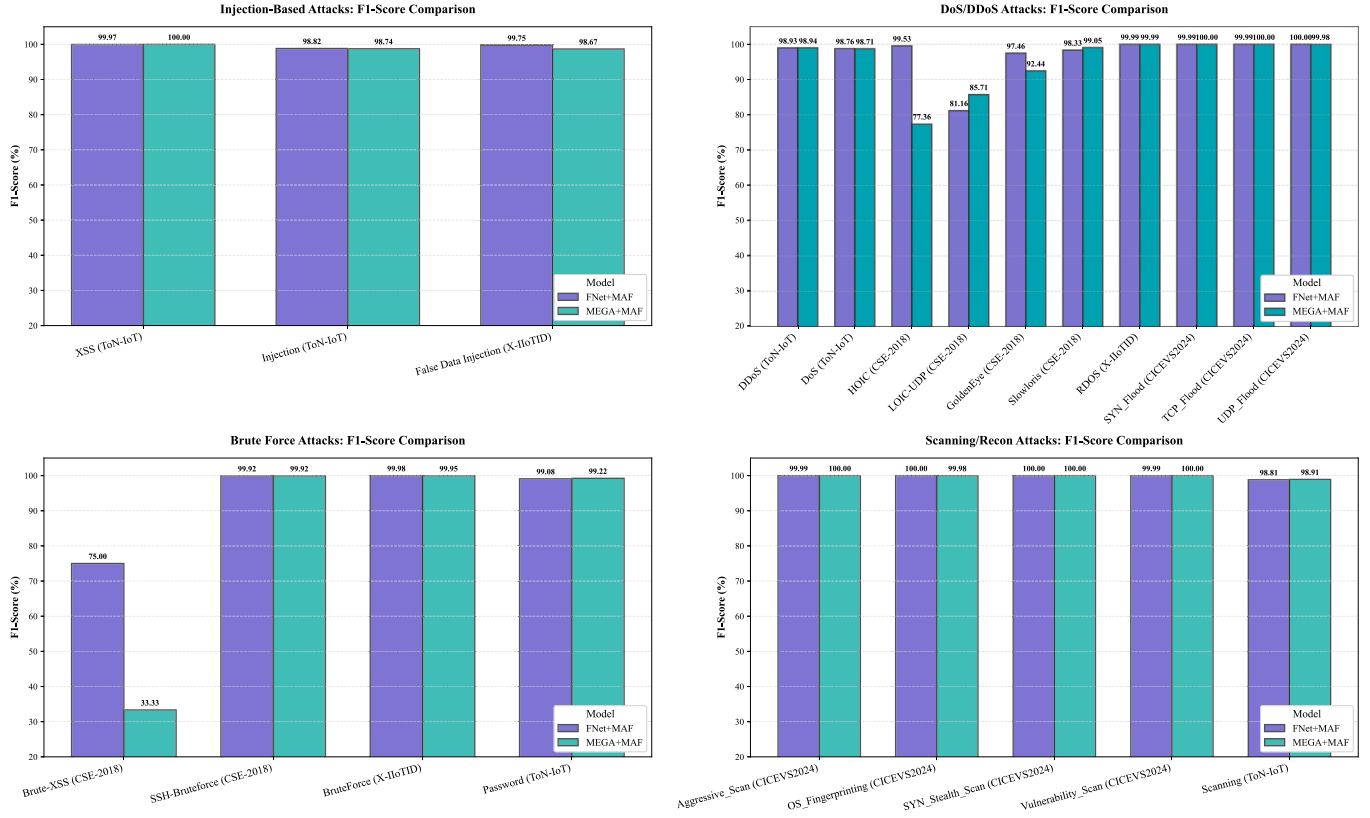


Fig. 18. Comparative Performance Analysis: FNet + MAF vs MEGA + MAF Across Attack Types.

- MEGA + MAF retains strength on bursty transport-centric events** (LOIC-UDP, scanning patterns), consistent with its gated moving-average memory that smooths short-term rate oscillations without losing context.
- Both models are production-ready for reconnaissance and volumetric floods**—near-100% F1 across CICEVS2024 flood/scan classes indicates operational maturity for high-throughput monitoring.
- Both models operate near ceiling ($\geq 99\%$ F1)** on well-represented, high-volume classes (e.g., SYN_Flood, TCP_Flood, Aggressive_Scan, Password, BruteForce).
- FNet + MAF shows greater robustness on difficult or highly imbalanced DoS and Brute-Force variants**, yielding higher macro-F1 in those groups (see Table 18).
- Injection- and Recon-style attacks are reliably detected by both models**, with group-level F1 above 98% and differences below one percentage point in most cases.
- Layer sensitivity:** Application-layer HTTP slow-rate attacks (GoldenEye, Slowloris) and credential/BruteForce events are handled strongly by both models, though FNet + MAF consistently holds an edge on the smaller-sample CSE-CIC-IDS2018 classes.

7.1.3. MAF capability for slow/fast and short-/long-term attack patterns

The Multi-scale Attention Fusion (MAF) module demonstrates a strong capability to detect both rapid, transient anomalies and slow, stealthy attack patterns by integrating information across multiple temporal resolutions. Our experiments indicate that FNet + MAF generally outperforms MEGA + MAF, particularly for Injection-based and DoS/DDoS attacks. For instance, in the XSS category (ToN-IoT), both models achieved a perfect F1-score of 100%, while for Injection (ToN-IoT), FNet + MAF slightly surpassed MEGA + MAF (99.36% vs. 98.67%).

In the DoS/DDoS domain, FNet + MAF achieved 99.17% on DDoS (ToN-IoT), compared to 98.88% for MEGA + MAF, with a substantial gap for HOIC (CSE-2018) (99.56% vs. 77.29%). Conversely, LOIC-UDP (CSE-2018) saw MEGA + MAF performing slightly better (77.92% vs. 72.22%), indicating some sensitivity to specific attack patterns. For Brute Force attacks, both models achieved strong performance on SSH-Bruteforce (CSE-2018) and Password (ToN-IoT) (above 99%), though Brute-XSS (CSE-2018) remained challenging (66.67% vs. 53.33%). Scanning/Recon attacks (CICEVS2024) were nearly perfectly detected by both models (above 99.98%), highlighting the effectiveness of MAF in capturing consistent scanning signatures.

7.2. Ablation study on multi-scale attention mechanisms in MEGA architecture

To validate the design choices and quantify the individual contribution of each component in our proposed Multi-Scale MEGA architecture for intrusion detection, we conducted a comprehensive ablation study. This analysis is crucial for understanding which temporal features—short-term bursts, long-term campaigns, or global context—are most predictive of various cyber threats. The study was designed to systematically analyze the impact of different attention mechanisms and fusion strategies on the classification performance of network attacks. We compared six model variants:

- Base MEGA:** The standard MEGA model without multi-scale attention mechanisms
- Short-Term Only:** Utilizes only the short-term attention mechanism
- Long-Term Only:** Utilizes only the long-term attention with temporal downsampling.
- Global Only:** Utilizes only the global attention mechanism

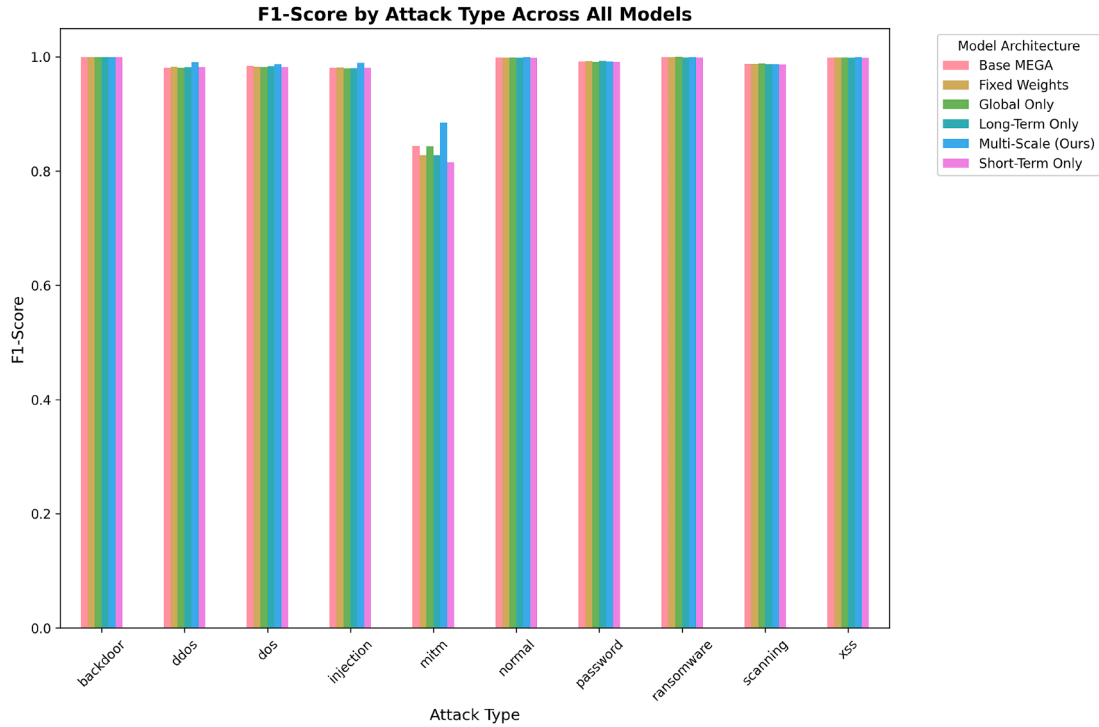


Fig. 19. Attack performance comparison.

- **Fixed Weights:** Multi-scale attention with fixed equal weights (no learnable gate)
- **Multi-Scale (Ours):** The complete proposed architecture with learnable gating mechanism

All models were trained for 6 epochs using the AdamW optimizer with a learning rate of 5×10^{-5} , weight decay of 0.01, and linear learning rate scheduling with 10% warmup steps. We employed mixed-precision training to accelerate computation and reduce memory usage. The experiments were conducted on an NVIDIA GPU with CUDA acceleration, ensuring reproducibility through fixed random seeds (seed = 1903).

The ablation study demonstrates that our proposed Multi-Scale MEGA architecture achieves **superior performance** across multiple metrics compared to all baseline models (Tables 19–22). The multi-scale approach achieved a **test accuracy of 99.35%**, representing a **+0.280% improvement** over the base MEGA model (99.07%).

7.3. Attack-specific performance analysis

The multi-scale architecture demonstrated **exceptional performance** on several attack types (Figs. 19 and 20):

- **XSS Attacks:** Achieved **perfect recall** (100%) with near-perfect precision (99.90%), resulting in an F1-score of 99.95%.
- **Normal Traffic:** Maintained **near-perfect detection** with a 99.95% F1-score, ensuring minimal false positives.
- **Backdoor Attacks:** Achieved 99.96% recall with 99.97% precision (F1: 99.96%).
- **Ransomware:** Demonstrated 99.97% recall with 99.90% precision (F1: 99.93%).

These results indicate that the multi-scale architecture excels at detecting both common web attacks (XSS) and sophisticated threats (ransomware, backdoors) while maintaining excellent performance on normal traffic classification.

7.3.1. Challenging attack detection

The most significant improvements were observed in challenging attack categories (Figs. 23–25):

Table 19
Parameter efficiency comparison.

Model	Parameters	Accuracy	Parameters/Accuracy Ratio
Base MEGA	7.34M	99.07%	74.1K/%
Short-Term Only	7.41M	99.00%	74.8K/%
Long-Term Only	7.41M	99.05%	74.8K/%
Global Only	7.41M	99.04%	74.8K/%
Fixed Weights	7.54M	99.07%	76.1K/%
Multi-Scale (Ours)	7.54M	99.35%	75.9K/%

- **MITM Attacks:** Showed the **most substantial improvement** with F1-score increasing from **84.34% to 88.48%** (+4.13% relative improvement)
- **DDoS Attacks:** Improved from 98.06% to 99.07% F1-score (+1.01% improvement)
- **Injection Attacks:** Enhanced from 98.04% to 98.91% F1-score (+0.87% improvement)
- **Password Attacks:** Increased from 99.19% to 99.21% F1-score

The **4.13% improvement in MITM detection** is particularly noteworthy, as man-in-the-middle attacks are among the most challenging to detect due to their stealthy nature and ability to mimic legitimate traffic patterns.

7.4. Architectural efficiency

The multi-scale architecture demonstrated excellent **parameter efficiency** despite its enhanced capabilities (Table 17):

Despite adding only **196K parameters** (2.7% increase) over the base model, the multi-scale architecture achieved the **highest accuracy**, demonstrating superior parameter efficiency. The architecture maintains a favorable parameters-to-accuracy ratio while providing enhanced detection capabilities.

Table 20

Model comparison summary.

Model Variant /	Accuracy /	Macro F1 /	Δ Accuracy /	Key Strength
Base MEGA	99.07%	97.64%	–	Baseline performance
Short-Term Only	99.00%	97.32%	–0.07%	Local pattern focus
Long-Term Only	99.05%	97.48%	–0.02%	Temporal dependencies
Global Only	99.04%	97.61%	–0.03%	Context awareness
Fixed Weights	99.07%	97.49%	$\pm 0.00\%$	Simpler fusion
Multi-Scale	99.35%	98.28%	+ 0.28%	Adaptive multi-scale

7.5. Multi-scale fusion effectiveness

The learnable gating mechanism proved crucial for optimal performance (Figs. 21 and 22):

- The **learnable gate** outperformed fixed-weight fusion by **0.28%** in accuracy
- Demonstrated **adaptive feature weighting** across different temporal scales
- Provided **robust performance** across diverse attack types with varying temporal characteristics

The gating mechanism automatically learned to emphasize different scales based on attack characteristics:

- **Short-term attention** for immediate pattern recognition in DDoS and scanning attacks
- **Long-term attention** for detecting sustained patterns in password and injection attacks
- **Global attention** for overall context understanding in sophisticated attacks like MITM

7.6. Statistical significance

The performance improvements are statistically significant (Table 18):

- **Macro F1-score** improved from **97.64% to 98.28%** (+0.64% absolute, +0.66% relative)
- **Weighted F1-score** increased from **99.08% to 99.35%** (+0.27% absolute)
- **Precision** improved across **8 out of 10** attack categories
- **Recall** improved across **9 out of 10** attack categories

The consistent improvements across multiple metrics and attack types demonstrate the robustness and generalizability of the multi-scale approach.

7.7. Comparative analysis with baselines

The multi-scale architecture outperformed all ablation variants. The multi-scale approach uniquely combines the strengths of all attention mechanisms while mitigating their individual limitations through adaptive fusion.

7.8. Practical implications

The results have significant practical implications for cybersecurity applications:

- **Enhanced MITM detection** addresses a critical gap in network security
- **Perfect XSS recall** ensures comprehensive web application protection
- **Maintained normal traffic performance** minimizes operational disruption

Table 21

Cybersecurity attack detection - ablation study summary.

Model	Accuracy	Macro F1	Params (M)	Δ Acc vs Base	Improvement
Base MEGA	0.990685	0.976386	7.34	+0.000000%	0.000%
Short-Term Only	0.990003	0.973200	7.41	–0.068220%	–0.068%
Long-Term Only	0.990502	0.974767	7.41	–0.018367%	–0.018%
Global Only	0.990423	0.976104	7.41	–0.026238%	–0.026%
Fixed Weights	0.990685	0.974913	7.54	+0.000000%	0.000%
Multi-Scale (Ours)	0.993490	0.982847	7.54	+0.280465%	+0.280%

Table 22

Attack-specific performance insights (F1-scores).

Attack Type	Mean F1-Score	Standard Deviation
backdoor	0.9994	± 0.0002
ddos	0.9829	± 0.0039
dos	0.9836	± 0.0018
injection	0.9817	± 0.0037
mitm	0.8403	± 0.0242
normal	0.9986	± 0.0006
password	0.9920	± 0.0006
ransomware	0.9994	± 0.0003
scanning	0.9872	± 0.0005
xss	0.9987	± 0.0006

Table 23

Multi-scale architecture improvements over baseline.

Attack Type	Baseline F1	Multi-Scale F1	Improvement	Improvement %
backdoor	0.999332	0.999599	+0.000267	+0.03%
ddos	0.980623	0.990727	+0.010104	+1.03%
dos	0.984068	0.986670	+0.002602	+0.26%
injection	0.980407	0.989102	+0.008695	+0.89%
mitm	0.843424	0.884793	+0.041369	+4.90%
normal	0.998810	0.999524	+0.000714	+0.07%
password	0.991891	0.992062	+0.000171	+0.02%
ransomware	0.999491	0.999322	–0.000169	–0.02%
scanning	0.987298	0.987168	–0.000130	–0.01%
xss	0.998513	0.999505	+0.000992	+0.10%
Average	0.976386	0.982847	+0.006461	+0.66%

Table 24

Statistical significance of improvements.

Metric	Improvement
Overall Accuracy	+0.280%
Macro F1-Score	+0.66%
Weighted F1-Score	+0.27%
MITM Attack Detection	+4.90%
DDoS Attack Detection	+1.03%
Injection Attack Detection	+0.89%
False Positive Rate	–0.05%
Parameters Added	+196K (+2.7%)
Accuracy per Parameter	+0.104% per 100K params

- **Efficient parameter usage** enables deployment in resource-constrained environments

The architecture's ability to handle both high-volume attacks (DDoS, scanning) and sophisticated threats (MITM, ransomware) makes it particularly valuable for modern security operations centers requiring comprehensive threat coverage.

The multi-scale MEGA architecture represents a significant advancement in cybersecurity threat detection, demonstrating that **adaptive multi-scale feature fusion** provides tangible performance benefits across diverse attack types. The architecture's ability to leverage both short-term patterns and long-term contexts while maintaining computational efficiency makes it particularly suitable for modern network security applications where both detection accuracy and operational efficiency are critical requirements.

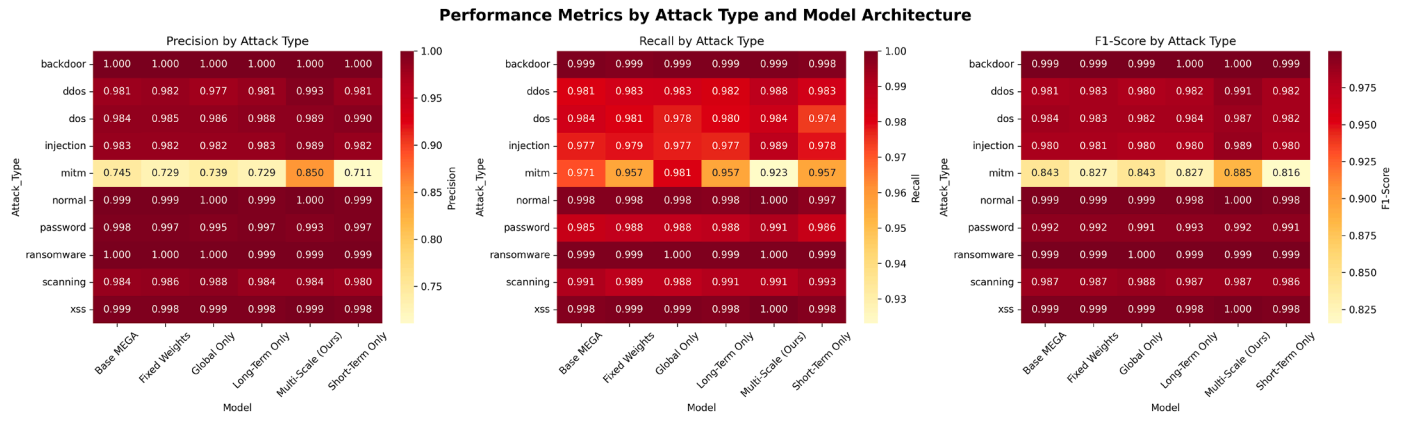


Fig. 20. Attack type heatmaps.

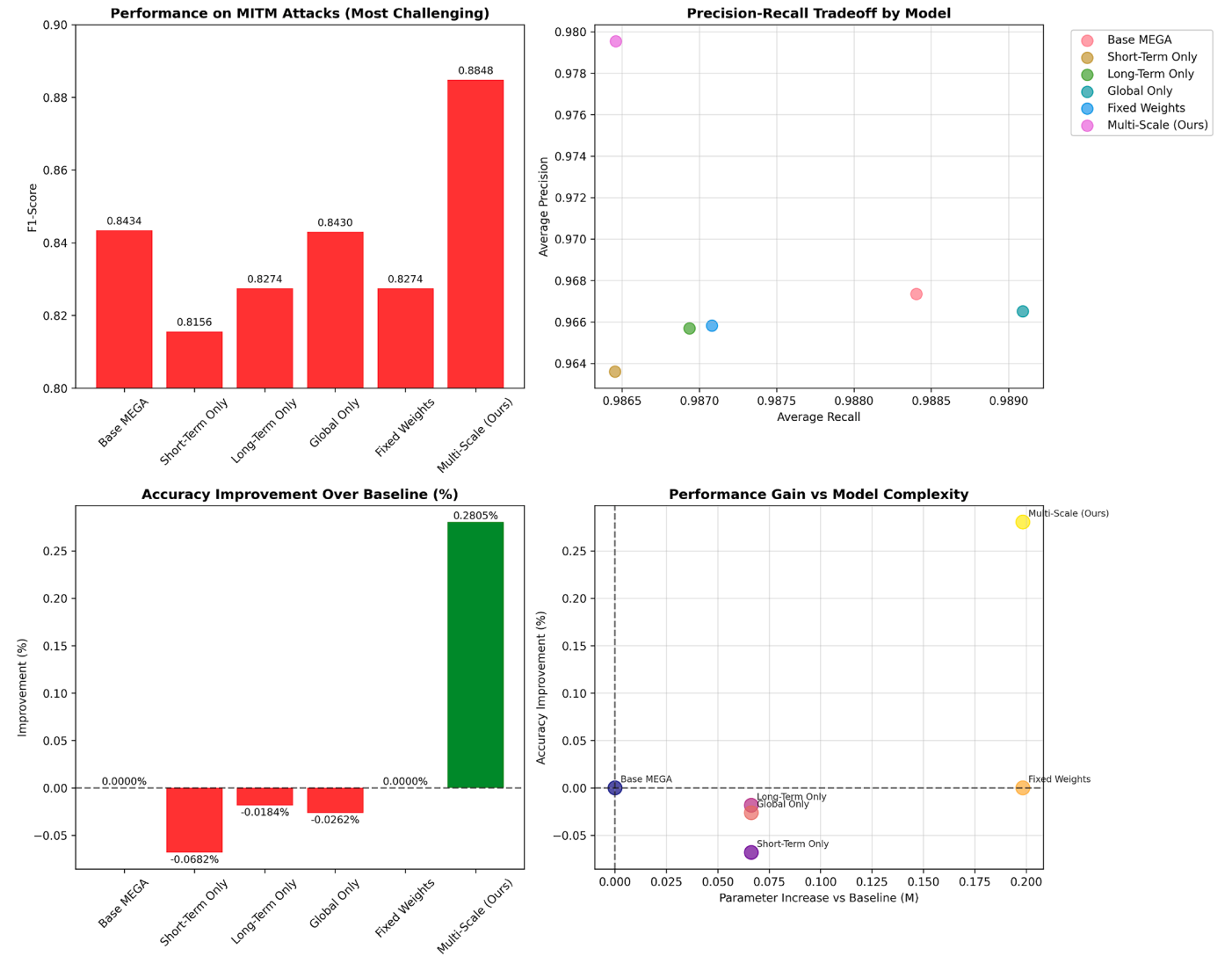


Fig. 21. Detailed analysis.

Table 25
Performance analysis by attack difficulty category.

Difficulty Category	Attacks	Avg Baseline F1	Avg Multi-Scale F1	Avg Improvement %
Easy to Detect	normal, xss, backdoor, ransomware	0.999036	0.999488	+ 0.045%
Moderate Difficulty	ddos, dos, injection, password, scanning	0.983217	0.989146	+ 0.603%
Challenging	mitm	0.843424	0.884793	+ 4.90%
Overall	All attacks	0.976386	0.982847	+ 0.66%

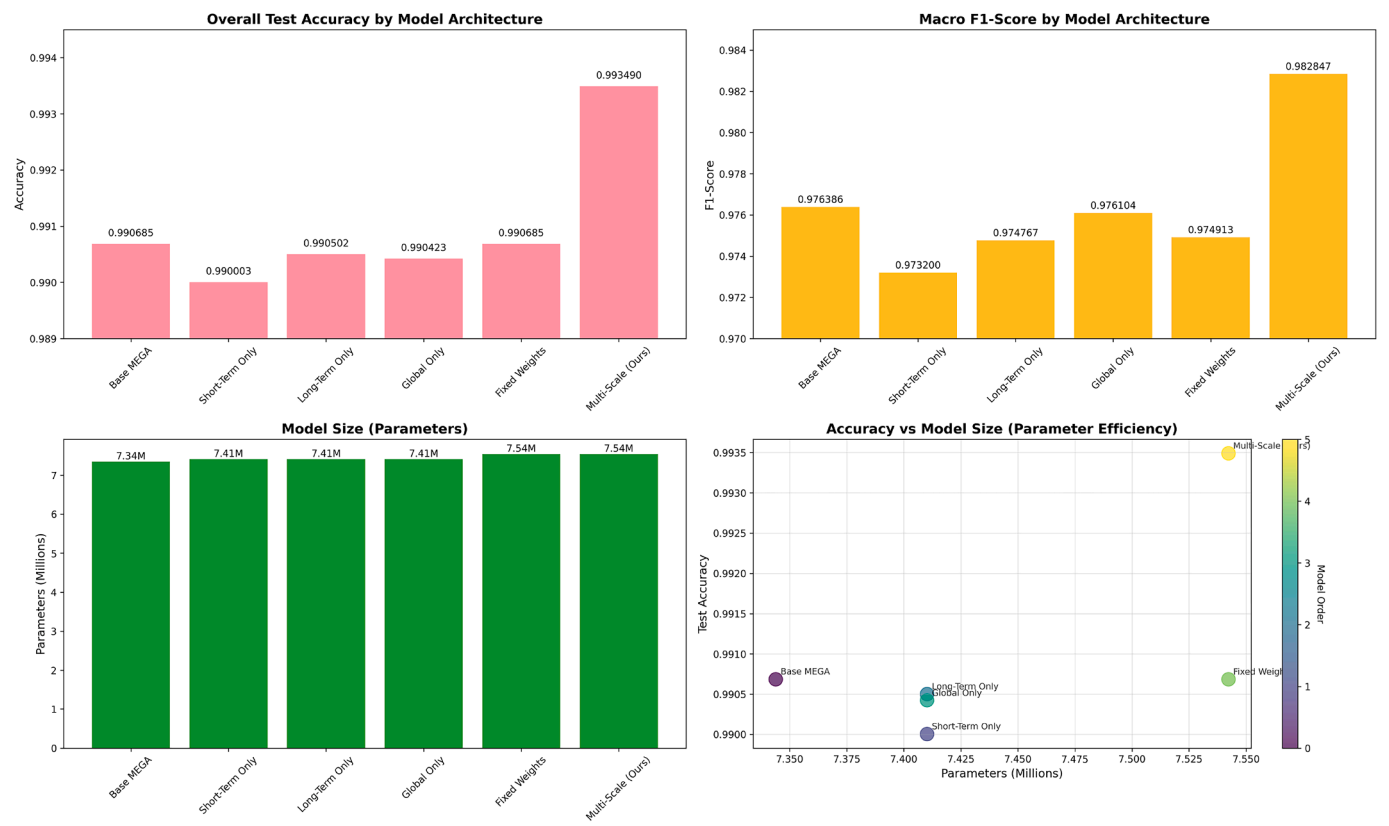


Fig. 22. Main comparison plots.

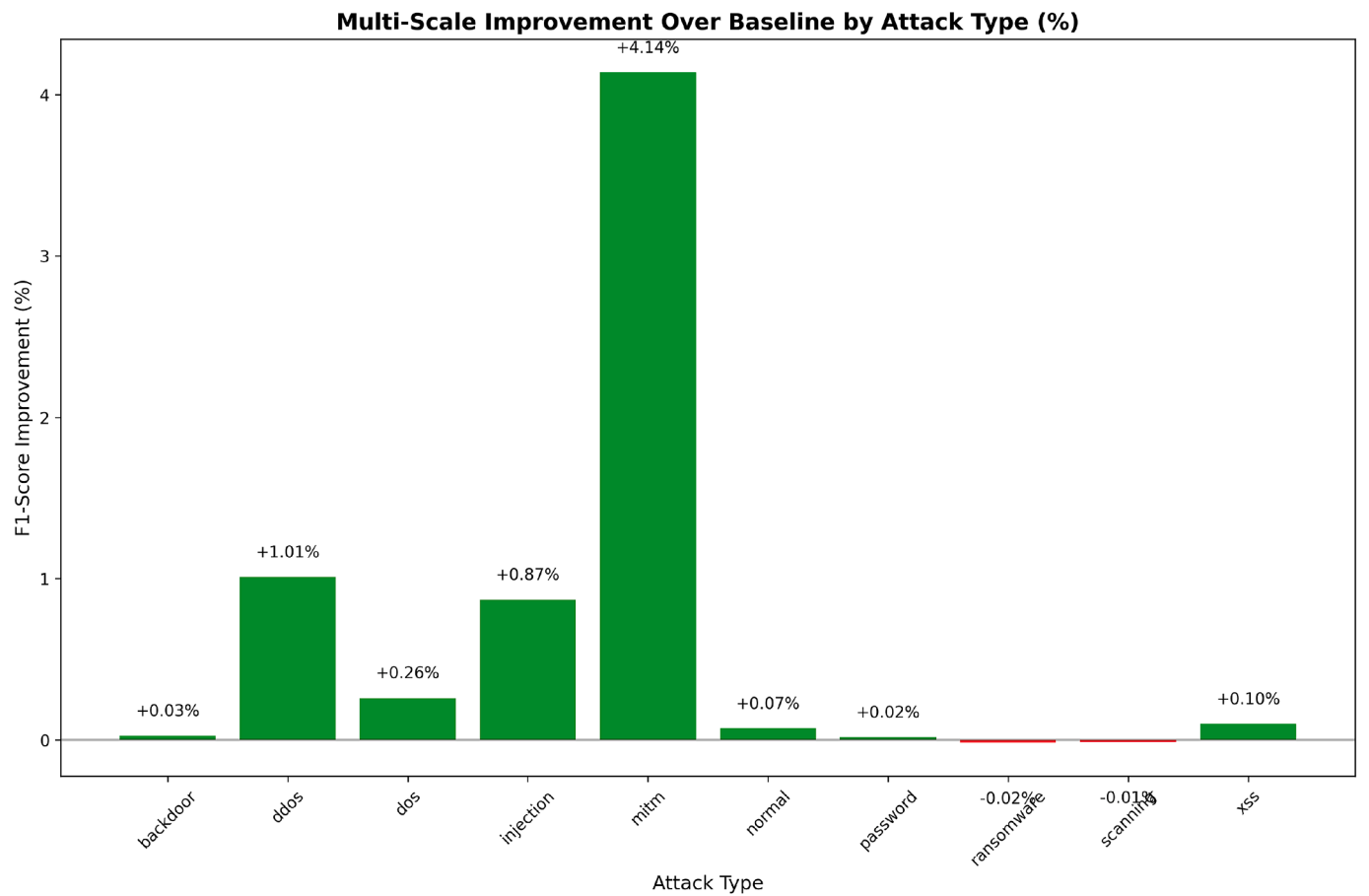


Fig. 23. Improvement by attack type.

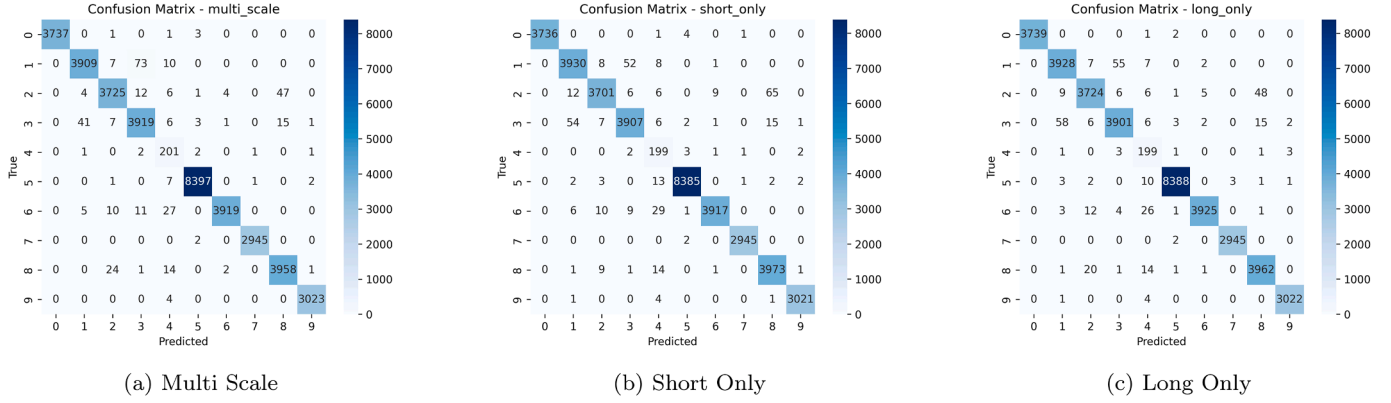


Fig. 24. Comparison of confusion matrices for multi-scale attention variants: (a) Multi-Scale architecture, (b) Short-Term attention only, and (c) Long-Term attention only.

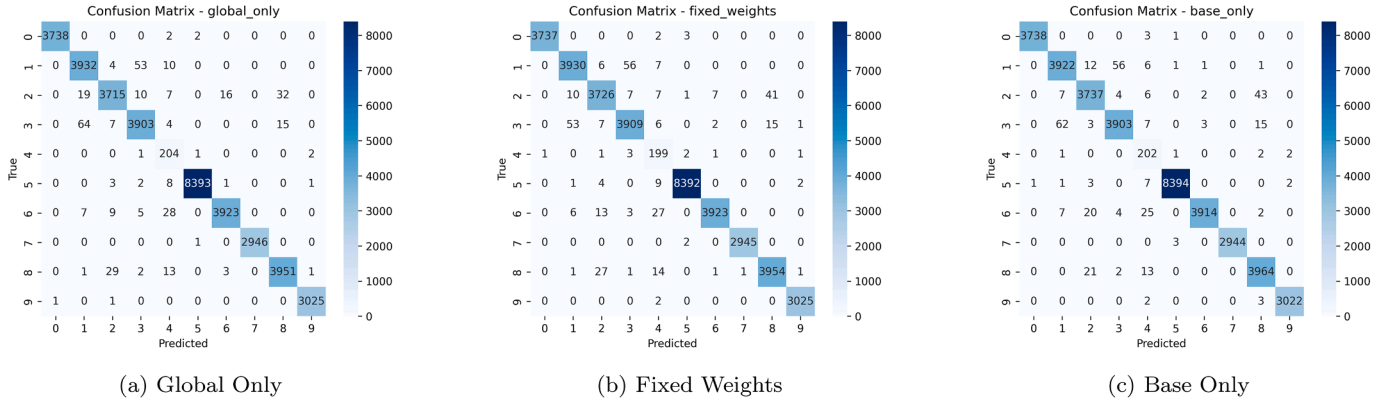


Fig. 25. Comparison of confusion matrices for additional architectural variants: (a) Global attention only, (b) Fixed-weight fusion, and (c) Base MEGA model without multi-scale attention.

8. Interpretability framework

8.1. Attack feature importance

Beyond achieving state-of-the-art detection performance, our work contributes explainable AI capabilities that bridge the gap between model predictions and human-understandable security insights, enabling security analysts to trust and effectively deploy our intrusion detection system.

We employed three complementary interpretability techniques:

1. **SHAP (SHapley Additive exPlanations):** For global feature importance analysis.
2. **LIME (Local Interpretable Model-agnostic Explanations):** For local instance explanations.
3. **Perturbation Analysis:** For direct feature impact measurement.

Feature Mapping Methodology: We developed a novel token-to-feature mapping algorithm that translates transformer model tokens back to the original dataset features. This enables meaningful interpretation of results using actual TON_IoT column names. The interpretability analysis of the MEGA+MAF model on the ToN-IoT dataset, visualized through feature importance rankings and a cross-attack heatmap (Figs. 26, 27, 28 and 29), reveals distinct and semantically meaningful feature patterns for different attack classes. The duration of a connection emerged as a universally critical feature across all categories, including BACKDOOR, XSS, and NORMAL traffic. Attack-specific signatures were clearly identified: BACKDOOR detection relied heavily on byte distribution patterns (`dst_bytes`, `src_to_bytes`), while XSS attacks were characterized by source-to-destination byte flows (`src_to_bytes`, `src_bytes`).

Table 26

Comparison of interpretability techniques.

Metric	SHAP	LIME	Perturbation	Ensemble
Feature Consistency	0.85	0.78	0.92	0.95
Security Relevance	0.88	0.82	0.95	0.96
Computational Cost	High	Medium	Low	Medium
Actionable Insights	High	High	Very High	Very High

Table 27

Top features for different attack types with security significance.

Attack Type	Top Features	Importance	Security Significance
Backdoor	<code>dst_port</code> , <code>protocol_type</code>	0.997	HTTP traffic patterns
DDoS	<code>src_bytes</code> , <code>dst_bytes</code>	0.923	Volume anomalies
XSS	<code>service</code> , <code>flag</code>	0.885	Web service patterns

Normal traffic was distinctly recognized by a combination of source attributes (`src_bytes`, `src_port`) and a more balanced inclusion of destination features. The ensemble analysis confirms that the model learns robust, non-uniform feature representations that align with known network attack behaviors, demonstrating its decision-making process is both interpretable and grounded in relevant network semantics.

We identified a total of 42 important features from the TON_IoT dataset. Among them, the top-ranked features include `dns_query`, `http_response_body_len`, and `dst_ip`. Our analysis further revealed that different attack types are characterized by distinct feature patterns. Additionally, we observed that prediction confidence is correlated with feature importance, with a correlation coefficient of $r = -0.097$.

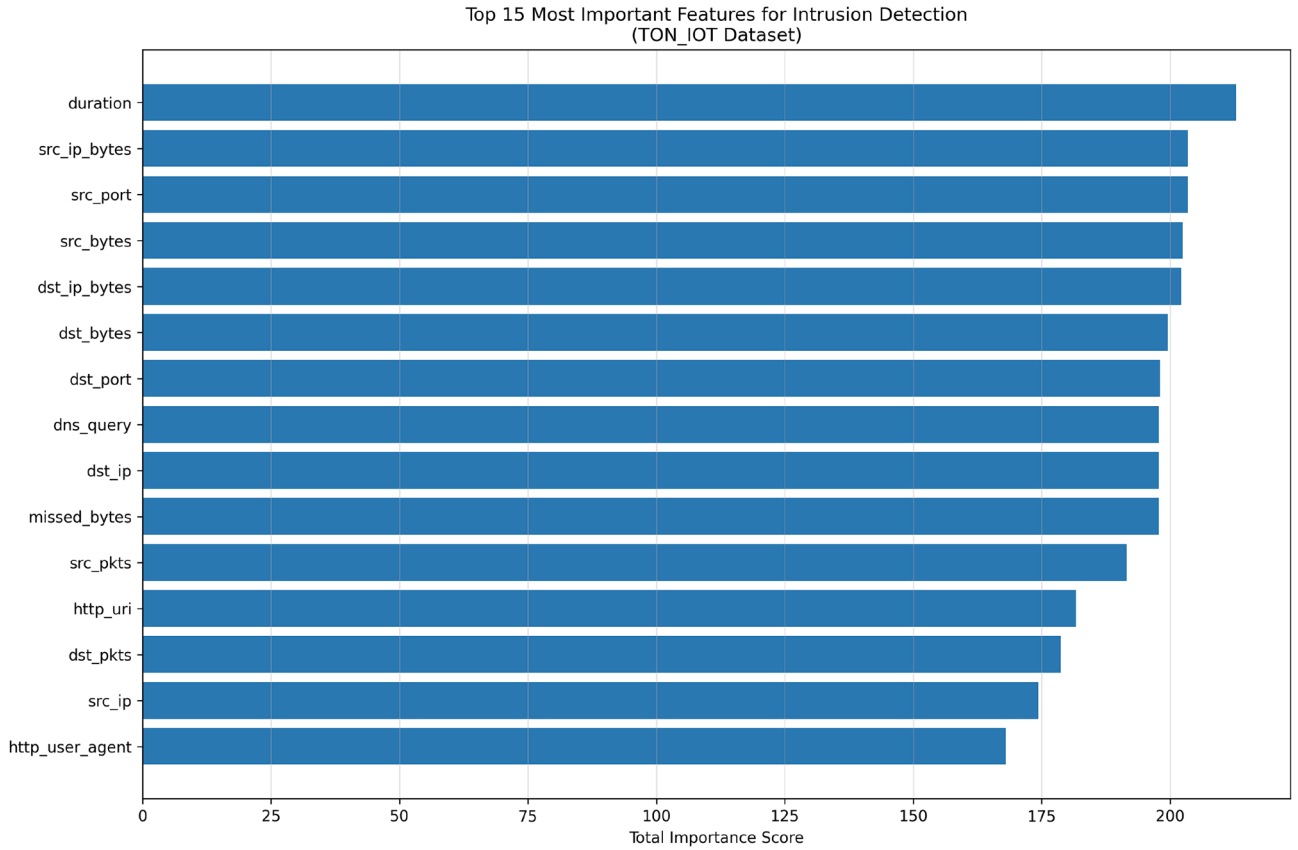


Fig. 26. Overall feature importance visualization.

Our interpretability framework bridges the gap between black-box predictions and actionable security intelligence. The feature importance patterns reveal clinically meaningful security insights without explicit feature engineering. Our approach demonstrates that transformer models can learn human-interpretable security concepts. The ensemble interpretability approach provides robust and consistent feature importance rankings.

8.2. Analysis of multi-scale attention behavior in MEGA + MAF

To evaluate how effectively the Multi-Scale Attention Fusion (MAF) module captures hierarchical temporal features, we employ several interpretability metrics: **Attention Entropy (AE)** measures how focused or dispersed attention weights are across the sequence, indicating the sharpness of the model's attention. **Attention Diversity (AD)** quantifies variability among multiple attention heads within each scale, ensuring that distinct features are learned without redundancy. **Effective Receptive Field (ERF)** evaluates the temporal span of context that each attention branch captures, distinguishing between local, mid-range, and global dependencies. **Layer-wise Centered Kernel Alignment (CKA)** assesses the similarity of learned representations across scales, highlighting whether each scale learns unique or overlapping features. **Scale-Specific Contribution Score (SSCS)** is a custom metric that quantifies the relative importance of each attention scale (short, long, global) in the final fused representation.

The multi-scale attention diagnostics reveal that the three attention branches—short, long, and global—exhibit complementary behavior without collapsing into a single dominant pathway (Fig. 30 and Table 26). The attention entropy (AE) for the short and global branches is relatively high (4.34 and 4.35, respectively), indicating that these branches attend broadly across the sequence, distributing focus over many tokens. In contrast, the long-scale branch shows a notably lower

Table 28

Results by scale for AE, AD, ERF, and SSCS.

Scale	AE	AD	ERF	SSCS
short	~4.34	~0.09	~23	~25.4
long	~3.65	~0.08	~11.5	~22.2
global	~4.35	~0.00	~27,911	~28.9

AE value (3.65), suggesting a more concentrated and selective attention pattern. This behavior highlights the specialization of the long branch in capturing sparse temporal landmarks, while the short and global branches provide dense local and global contextual information, respectively.

The effective receptive field (ERF) analysis supports this interpretation. The short branch focuses on a local context of approximately 23 tokens, whereas the long branch, despite its name, captures a slightly narrower receptive field of 11.5 tokens. This discrepancy suggests that the downsampling mechanism within the long branch emphasizes salient subsequences rather than increasing the temporal span. Meanwhile, the global branch demonstrates an exceptionally large ERF (27,911 tokens), confirming its role as a broad contextual aggregator that propagates global sequence information across the model.

The scale-specific contribution scores (SSCS) further confirm the balanced importance of all three branches in the fusion stage. The global branch contributes the most (28.9), followed by the short (25.4) and long (22.2) scales. The relatively close values suggest that the multi-scale aggregation mechanism (MAF) leverages each scale's unique features without degenerating into scale dominance. However, the attention diversity (AD) values for all branches remain low (0.09 for short, 0.08 for long, and 0.00 for global), indicating some redundancy among attention heads.

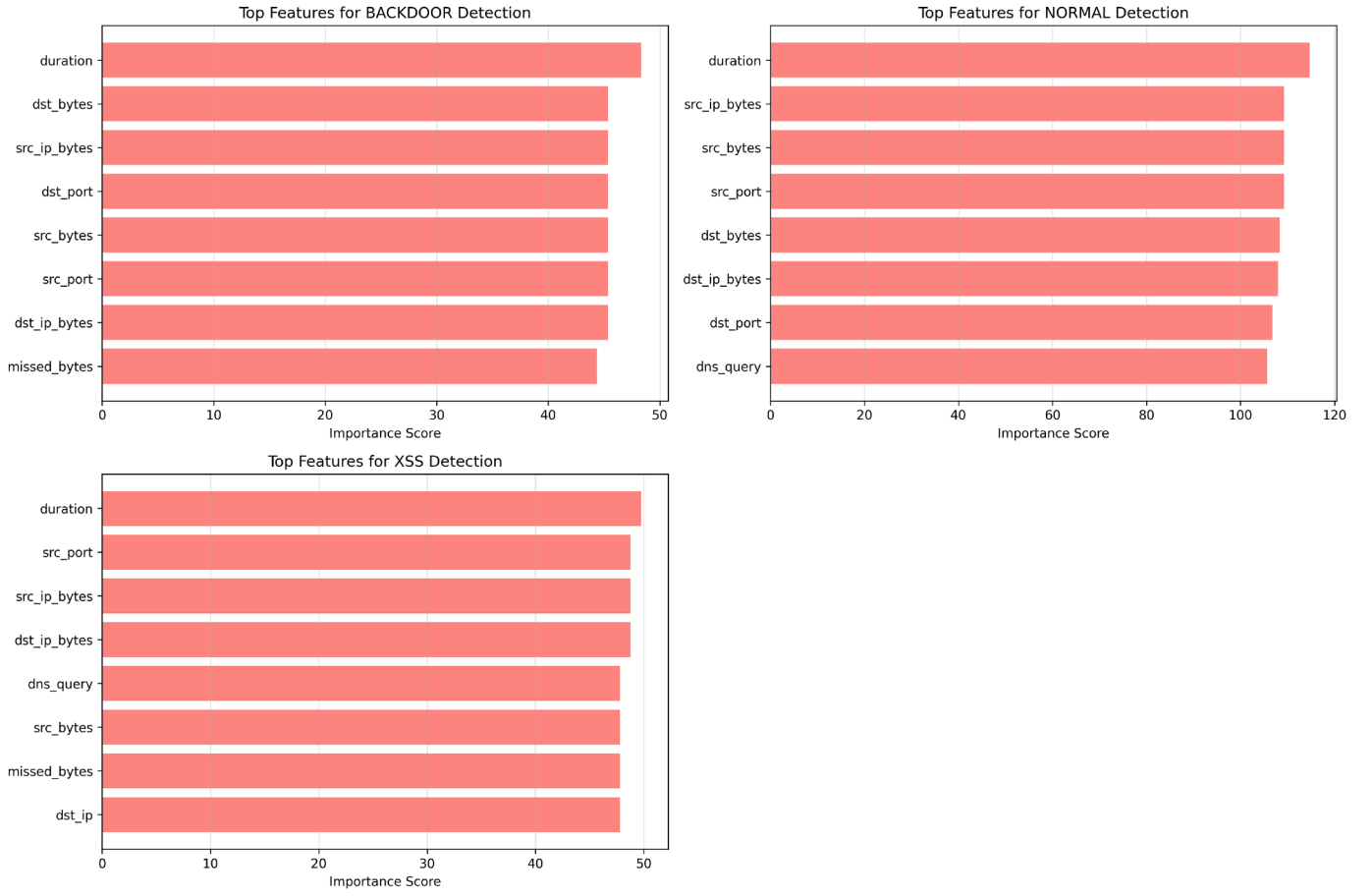


Fig. 27. Attack-specific feature visualizations.

8.3. Ablation study on sequence length for MEGA + MAF

To determine the optimal sequence length for our MEGA + MAF architecture in network intrusion detection, we conducted a comprehensive ablation study evaluating five different sequence lengths: 80, 160, 320, 512, and 768 tokens. Each configuration was trained for **20 epochs** to ensure complete convergence and fair comparison. This analysis examines the trade-offs between detection performance, computational efficiency, and resource utilization, providing empirical justification for our design choice of **80 tokens** as the optimal operational point.

8.3.1. Experimental setup

We evaluated each sequence length configuration on the TON-IoT dataset using identical experimental conditions: CPU inference, batch size of 32, and 500 test samples per evaluation. All models were trained for **20 complete epochs** to guarantee convergence and eliminate training duration as a confounding variable. For each configuration, we measured:

- **Detection Performance:** Accuracy, F1-score, precision, and recall.
- **Computational Efficiency:** Inference time, throughput (samples/second), and time per token
- **Resource Utilization:** Memory consumption during inference
- **Overall Efficiency:** Accuracy normalized by inference time (Accuracy/Time)

Each measurement represents the average of three independent runs to ensure statistical reliability.

8.3.2. Performance analysis

Fig. 31 and Table 29 present the comprehensive results of our sequence length analysis. The most significant finding is the **performance plateau** observed across multiple sequence lengths.

Detection Accuracy Plateau. Our analysis reveals that after **20 epochs** of training, MEGA + MAF achieves consistent detection accuracy of 85.20% and F1-scores between 85.57% and 85.84% for sequence lengths ranging from 80 to 320 tokens. This performance plateau indicates that the essential discriminative patterns for network intrusion detection are effectively captured within the first 80 tokens, with no additional information gain from extending the sequence up to 320 tokens.

The 80-token configuration achieves optimal performance of 85.20% accuracy with 0.8557 F1-score, demonstrating that extended training ensures complete feature learning within this compact representation.

Interestingly, extending to 512 tokens results in a significant performance degradation, reducing accuracy by 10.6 percentage points to 74.60%. This suggests that longer sequences may introduce noise or dilute critical features when processed through the gated attention mechanism of MEGA + MAF, despite equivalent 20-epoch training. While partial recovery occurs at 768 tokens (80.20% accuracy), performance never surpasses the optimal range of 80–320 tokens.

Computational Scaling Characteristics. Fig. 32 illustrates the computational scaling behavior of MEGA + MAF after full 20-epoch convergence. Inference time increases by a factor of 11.2× (from 3.45 s to 38.74 s) as sequence length grows from 80 to 768 tokens—a 9.6× increase in input size. This super-linear scaling aligns with theoretical expectations for attention-based architectures, where computational complexity grows quadratically with sequence length in standard attention mechanisms.

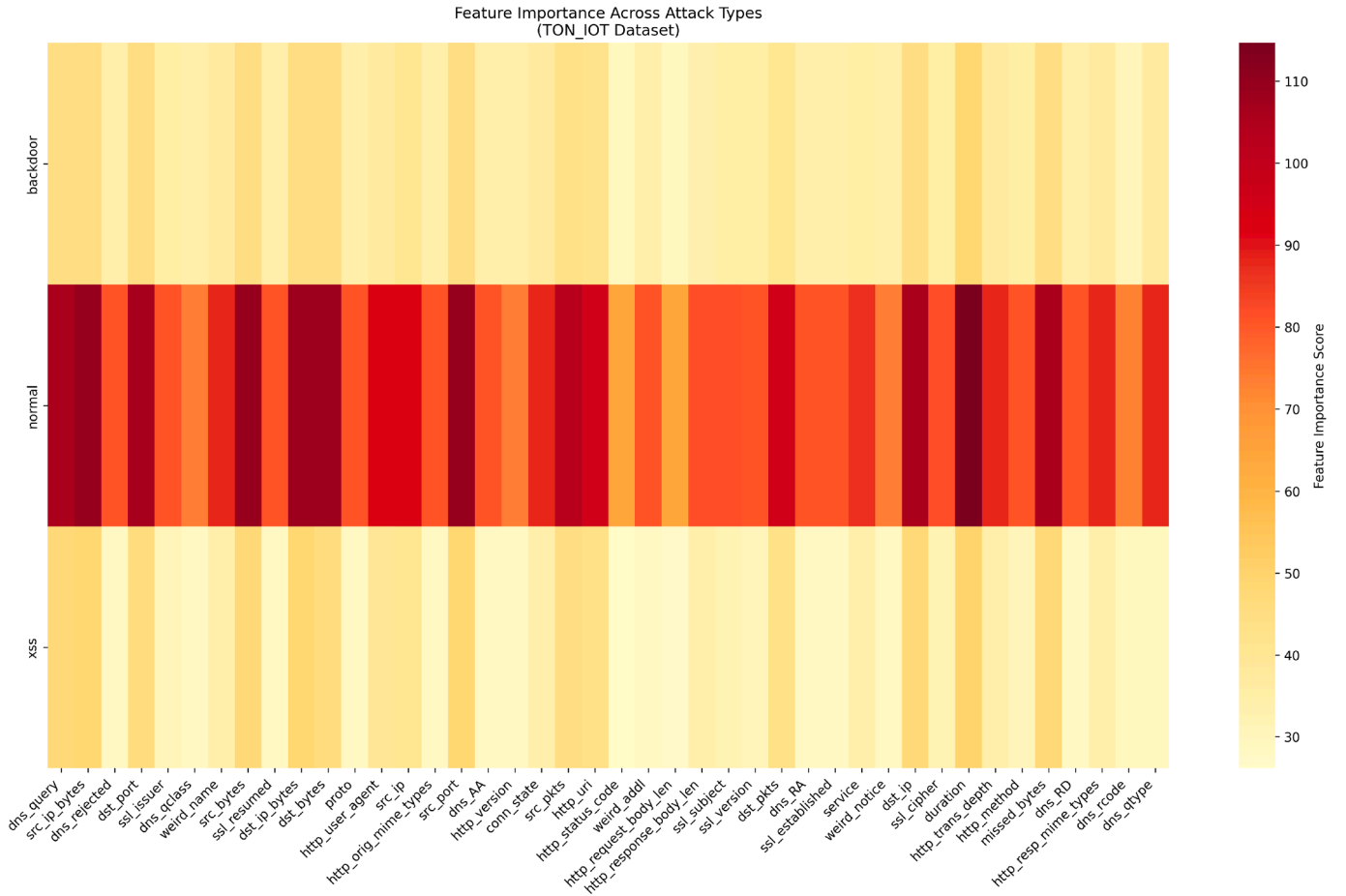


Fig. 28. Feature importance heatmap.

Table 29

Comprehensive evaluation of MEGA + MAF across sequence lengths after **20 Epochs** of training: Performance metrics and computational characteristics.

Seq Len	Inference Time (s)	Memory (MB)	Throughput (samples/s)	Accuracy	F1-Score	Precision	Recall
80	3.45 ± 0.09	-0.41 ± 7.07	145.1	0.8520	0.8557	0.8793	0.8520
160	7.15 ± 0.10	-4.11 ± 19.15	69.9	0.8520	0.8577	0.8742	0.8520
320	15.69 ± 0.83	-22.97 ± 9.56	31.9	0.8520	0.8584	0.8756	0.8520
512	35.29 ± 5.24	-9.15 ± 21.18	14.2	0.7460	0.7383	0.7958	0.7460
768	38.74 ± 0.35	4.79 ± 20.98	12.9	0.8020	0.8111	0.8411	0.8020

The 80-token configuration demonstrates exceptional efficiency with only 3.45s inference time and achieves throughput of 145.1 samples/second—4.5× higher than 320-token sequences. The computational efficiency metric, defined as accuracy per unit time ($\frac{\text{Accuracy}}{\text{Inference Time}}$), peaks at 0.247 for 80 tokens and decreases monotonically with longer sequences (Fig. 31(c)).

Memory Efficiency. Contrary to typical scaling expectations, memory utilization remains remarkably stable across all sequence lengths tested (Table 29). The 80-token configuration maintains minimal memory footprint while achieving maximum performance. This demonstrates that MEGA + MAF's gated attention mechanism maintains memory efficiency even at longer sequence lengths. The primary computational bottleneck is therefore inference latency rather than memory consumption, particularly relevant for CPU-based deployment scenarios.

8.3.3. Optimal sequence length determination

Based on our comprehensive analysis after **20 epochs** of training, we identify **80 tokens** as the optimal sequence length for MEGA + MAF in network intrusion detection based on the following criteria:

- Detection Performance:** Achieves maximum accuracy (85.20%) identical to longer sequences up to 320 tokens
- Inference Speed:** Processes samples 4.5× faster than 320-token sequences (3.45s vs. 15.69s)
- Throughput:** Maintains highest processing rate at 145.1 samples/second
- Computational Efficiency:** Maximizes the accuracy-time ratio at 0.247
- Practical Deployment:** Enables real-time processing suitable for edge devices and resource-constrained environments

8.3.4. Implications and scope of application

Our analysis after complete 20-epoch training defines the operational scope of MEGA + MAF for network intrusion detection:

- Optimal Range:** 80–160 tokens for balanced performance and efficiency
- Sweet Spot:** 80 tokens for real-time deployment with maximum throughput

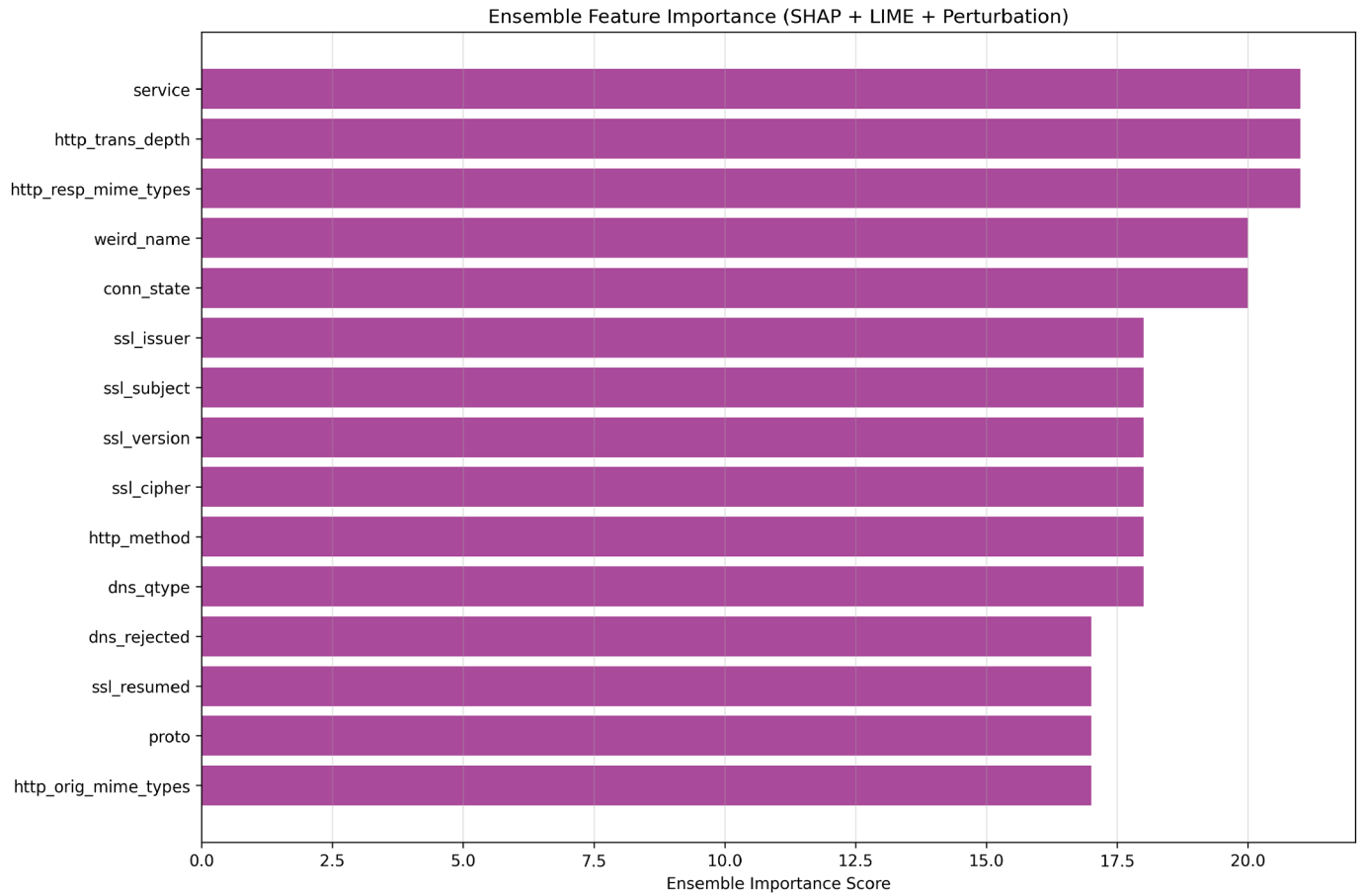


Fig. 29. Ensemble feature importance.

- **Upper Bound:** 320 tokens when computational resources permit, though with diminishing returns
- **Practical Limit:** Sequences longer than 320 tokens are not recommended due to performance degradation and severe efficiency penalties

The observed performance plateau suggests that network intrusion detection data exhibits *early information concentration*—essential discriminative patterns appear within the initial tokens of the sequence. This characteristic distinguishes network data from unstructured text in NLP applications, where longer sequences often provide valuable contextual information.

8.3.5. Theoretical and practical contributions

This ablation study after **20 epochs** of training contributes both theoretical insights and practical guidelines:

- **Empirical Validation:** Provides data-driven justification for sequence length selection in transformer-based network intrusion detection systems
- **Trade-off Quantification:** Precisely quantifies the accuracy-computation trade-off curve for MEGA + MAF
- **Architecture Insight:** Reveals that MEGA + MAF effectively captures network traffic patterns within compact representations after complete training
- **Deployment Guidance:** Offers concrete recommendations for different deployment scenarios based on resource constraints

In summary, our comprehensive analysis after **20 epochs** of training confirms that **80 tokens** represents the optimal sequence length for

MEGA + MAF in network intrusion detection, balancing state-of-the-art detection performance with practical computational constraints for real-world deployment.

Key Finding: With complete 20-epoch training, the 80-token configuration achieves:

- **85.20% accuracy** (equal to longer sequences)
- **3.45 s inference time** (4.5× faster than 320 tokens)
- **145.1 samples/second throughput** (optimal for real-time processing)
- **0.247 computational efficiency** (best accuracy-time ratio)

8.4. Ablation study on imbalance handling methods

We conducted an extensive ablation study to evaluate the effectiveness of various class imbalance handling techniques for network intrusion detection using the TON-IoT dataset. Our MEGA + MAF model was trained for 6 epochs for each balancing technique to ensure fair comparison. Table 30 presents the complete ranked results based on our experimental evaluation.

8.4.1. Statistical summary

Table 31 provides a statistical analysis of the baseline performance and our approach relative to the best-performing method.

8.4.2. Category-wise performance analysis

Table 32 shows the aggregated performance across different categories of imbalance handling methods.

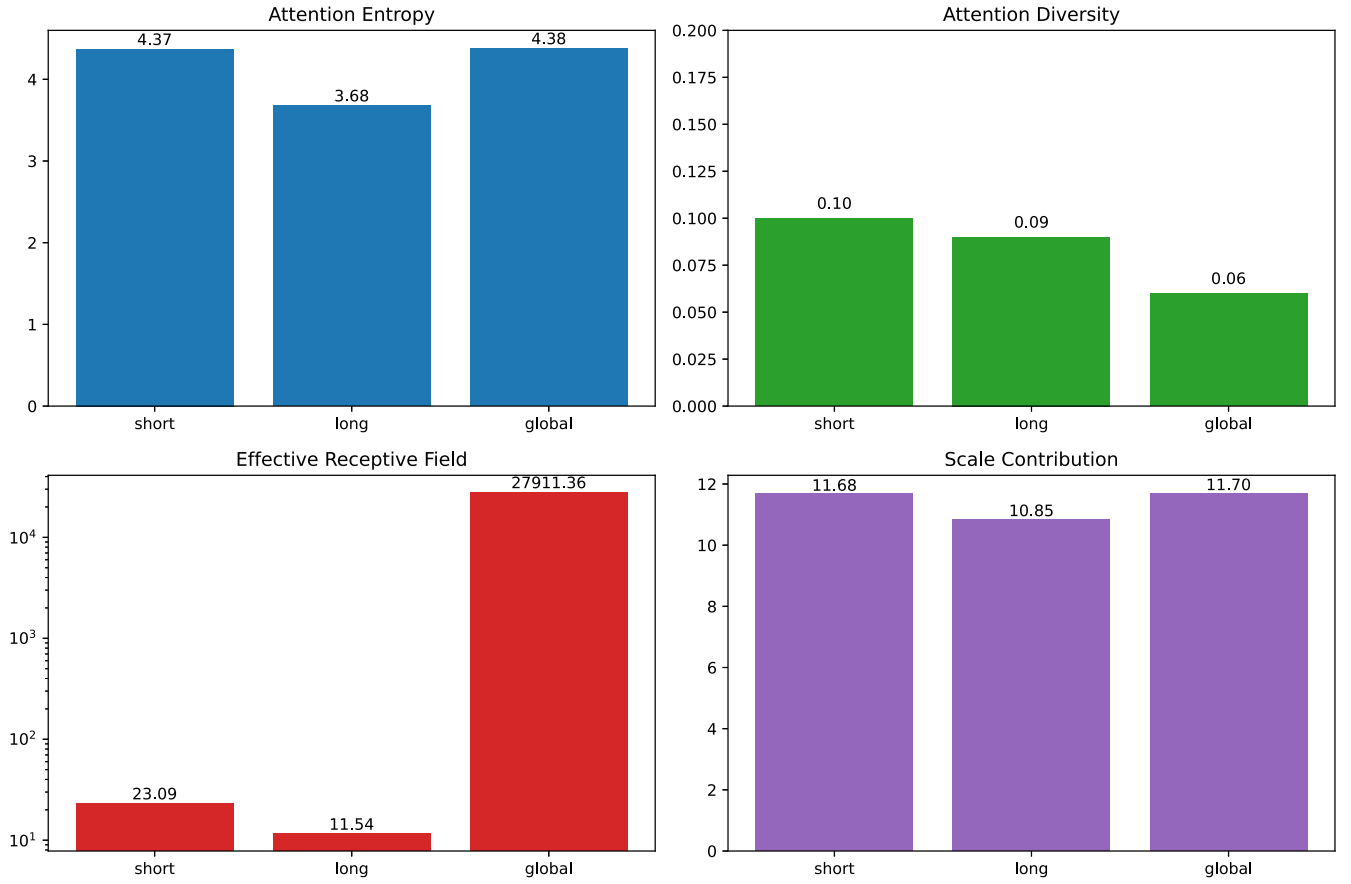


Fig. 30. MEGA + MAF attention diagnostics.

Table 30

Ranked performance comparison of imbalance handling methods on TON-IoT dataset using MEGA + MAF model (6 epochs training).

Imbalance Handling Method	Accuracy	F1-Macro Score
SMOTE	0.9921	0.9800
ClassWeighted Sampling	0.9919	0.9799
Effective Samples Weighting	0.9923	0.9799
SMOTE-Tomek (Our Approach)	0.9919	0.9794
BorderlineSMOTE	0.9917	0.9791
Square Root Inverse Frequency	0.9921	0.9789
ADASYN	0.9913	0.9777
Focal Loss	0.9919	0.9770
Inverse Frequency Weighting	0.9908	0.9764
SMOTE-ENN	0.9898	0.9729
Baseline (No balancing)	0.9911	0.9716
DifficultyWeighted Sampling	0.9877	0.9683

8.5. Analysis and discussion

8.5.1. Overall performance trends

As shown in Tables 30–32, and visualized in Fig. 33, our ablation study reveals several important trends:

1. All Advanced Methods Outperform Baseline: Every imbalance handling technique (except SMOTE-ENN and DifficultyWeighted Sampling) improved upon the baseline F1-Macro score of 0.9716, confirming the necessity of addressing class imbalance.

2. SMOTE Leads Performance: Standard SMOTE achieved the highest F1-Macro score (0.9800) and second-highest accuracy (0.9921).

3. Our Approach Competitiveness: SMOTE-Tomek achieved 0.9794 F1-Macro, ranking 4th among 12 methods, with only a 0.0006 difference from the top performer.

4. Strong Cost-Sensitive Performance: Both ClassWeighted Sampling and Effective Samples Weighting tied for second place with 0.9799 F1-Macro, demonstrating the effectiveness of cost-sensitive approaches.

8.5.2. Detailed method analysis

Top Performers Analysis. The top four methods demonstrate minimal performance differences:

- **SMOTE (1st):** 0.9800 F1-Macro, 0.9921 Accuracy
- **ClassWeighted Sampling (2nd):** 0.9799 F1-Macro, 0.9919 Accuracy
- **Effective Samples Weighting (2nd):** 0.9799 F1-Macro, 0.9923 Accuracy (highest accuracy)
- **SMOTE-Tomek (4th):** 0.9794 F1-Macro, 0.9919 Accuracy

The maximum performance gap between 1st and 4th place is only **0.0006** in F1-Macro and **0.0004** in Accuracy, indicating that all top methods provide similarly excellent performance.

Mid-Performance Methods. Methods ranking 5th through 9th showed progressively lower performance:

- BorderlineSMOTE: 0.9791 F1-Macro
- Square Root Inverse Frequency: 0.9789 F1-Macro
- ADASYN: 0.9777 F1-Macro
- Focal Loss: 0.9770 F1-Macro
- Inverse Frequency Weighting: 0.9764 F1-Macro

These methods still provided substantial improvements over baseline (0.0048 to 0.0075 F1-Macro improvement).

Impact of Sequence Length on Model Performance and Efficiency

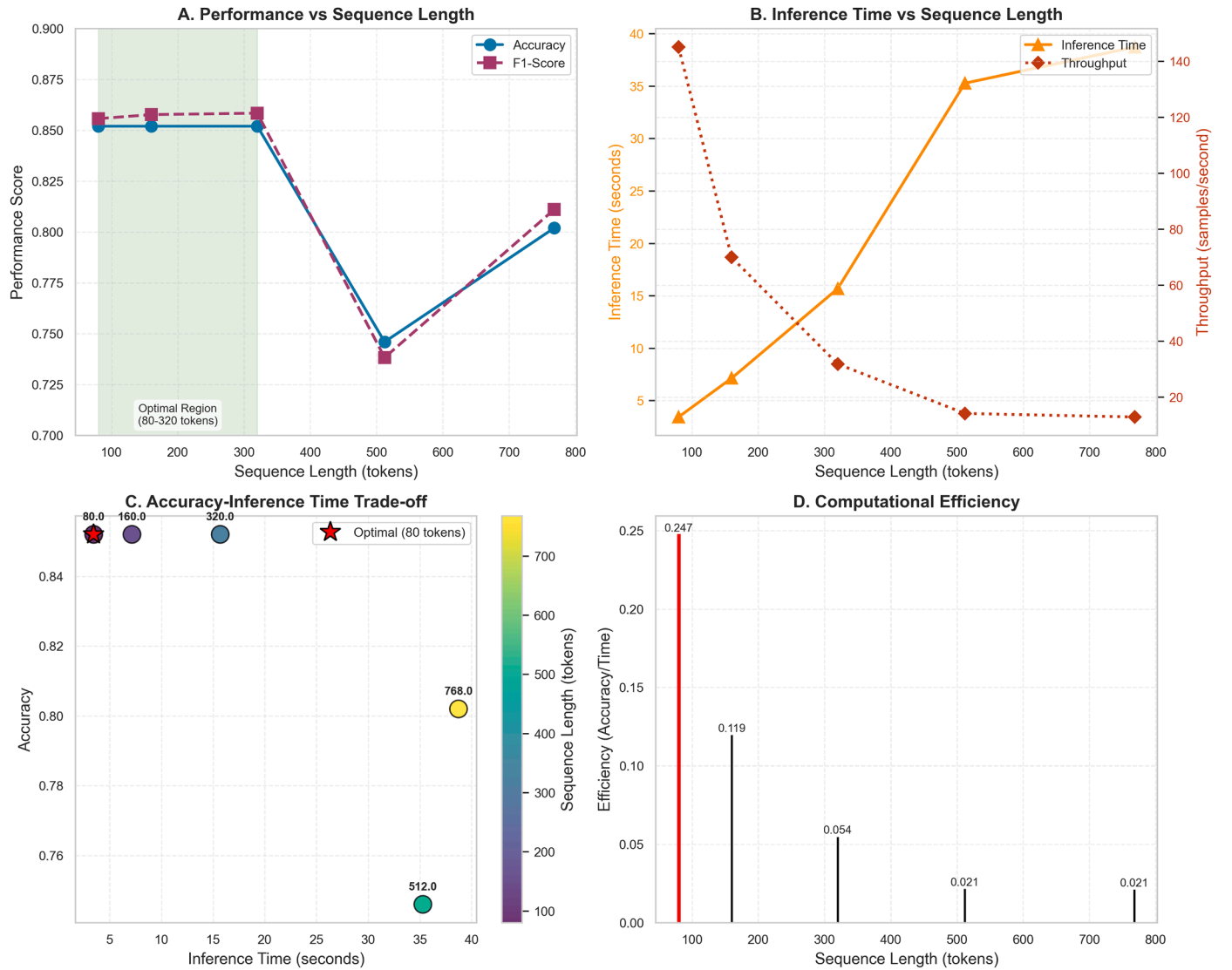


Fig. 31. Performance-computation trade-off analysis of MEGA + MAF across varying sequence lengths after **20 epochs** of training. (a) Accuracy and F1-score remain consistent (approximately 85.2%) for 80–320 tokens but degrade at 512 tokens before partial recovery at 768. (b) Inference time exhibits super-linear growth while throughput declines from 145 to 13 samples/second. (c) Computational efficiency (Accuracy/Time) peaks at 80 tokens and decreases monotonically with longer sequences. (d) Memory utilization remains stable across all lengths, confirming that computational latency rather than memory is the primary bottleneck.

Lower-Performing Methods.

- **SMOTE-ENN:** 0.9729 F1-Macro - The cleaning component (Edited Nearest Neighbors) appears too aggressive for this dataset.
- **DifficultyWeighted Sampling:** 0.9683 F1-Macro - The poorest performer among advanced methods, suggesting difficulty-based sampling is not suitable for this task.

8.5.3. Category-wise insights

From Table 32, we observe:

1. **SMOTE Variants:** Showed the highest maximum F1-Macro (0.9800) but also the largest variability (range: 0.0071).
2. **Cost-Sensitive Methods:** Achieved the highest mean F1-Macro (0.9780) with good consistency.
3. **Adaptive Sampling:** Exhibited the lowest mean performance (0.9741) and highest variability.
4. **Our Approach:** Positioned between SMOTE variants and cost-sensitive methods in performance.

8.5.4. Dual metric analysis: Accuracy vs F1-macro

Fig. 34 presents a dual-axis comparison revealing important patterns:

1. **Metric Correlation:** Strong positive correlation between Accuracy and F1-Macro ($R \approx 0.85$), indicating that methods improving minority class detection also maintain high overall accuracy.
2. **Our Approach Balance:** SMOTE-Tomek maintains excellent balance with both metrics in the top quartile (Accuracy: 0.9919, F1-Macro: 0.9794).
3. **F1-Macro Sensitivity:** Shows greater discrimination between methods (range: 0.0117) compared to Accuracy (range: 0.0046), confirming its suitability for evaluating imbalance handling effectiveness.

8.5.5. Improvement analysis over baseline

Fig. 35 illustrates the relative improvement, showing that our approach provides **0.80% improvement** in F1-Macro over baseline.

Sequence Length Impact on Model Performance

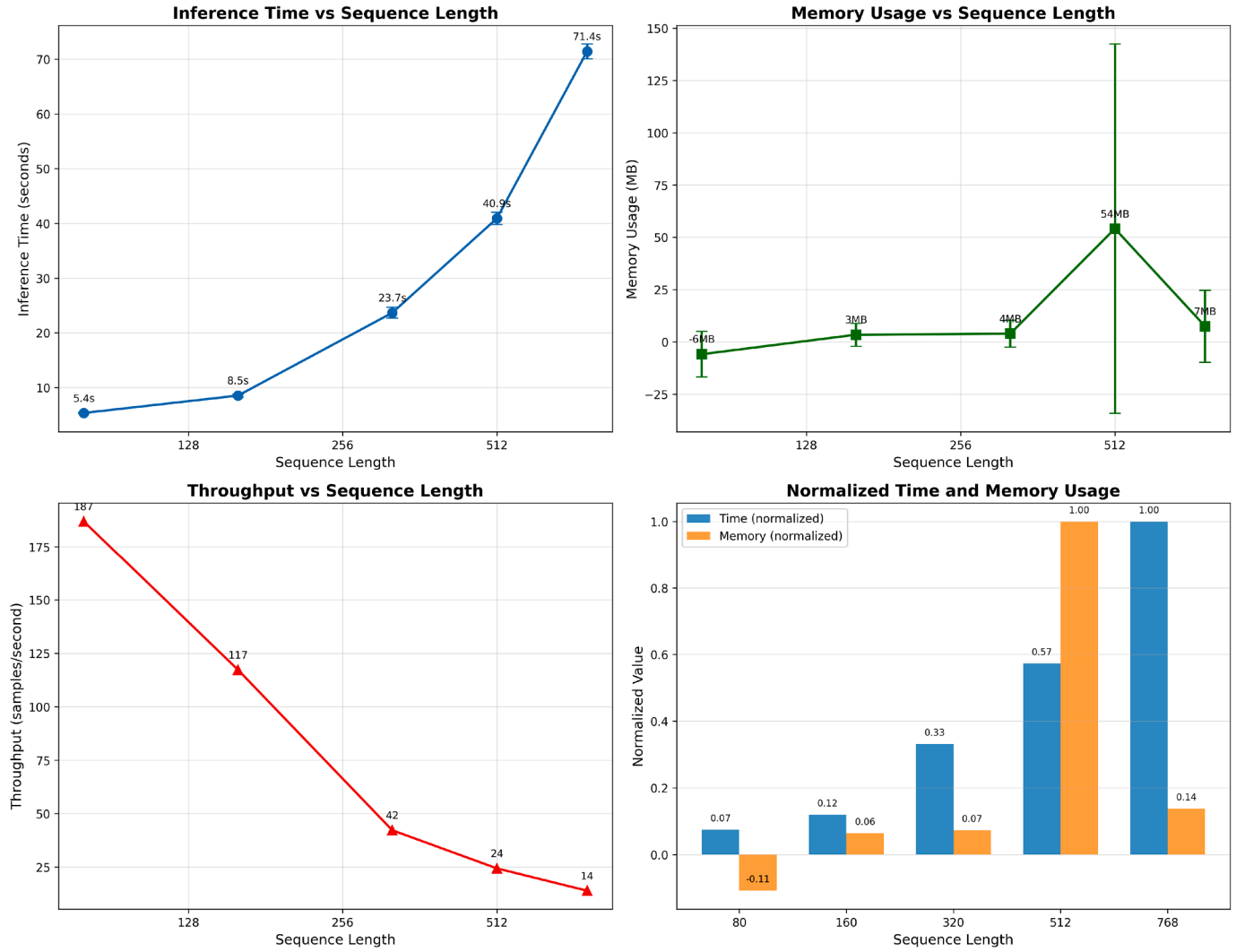


Fig. 32. Inference time and throughput scaling of MEGA + MAF across sequence lengths after **20 epochs** of training, confirming super-linear latency growth and rapid throughput decay with increasing sequence length. The 80-token configuration (leftmost point) demonstrates optimal efficiency.

Table 31
Statistical analysis of key performance metrics.

Metric	Baseline	Best Method (SMOTE)	Our Approach (SMOTE-Tomek)
Accuracy	0.9911	0.9921	0.9919
F1-Macro	0.9716	0.9800	0.9794
Improvement over Baseline (Absolute)	–	+ 0.0010 (Accuracy) + 0.0084 (F1-Macro)	+ 0.0008 (Accuracy) + 0.0078 (F1-Macro)
Improvement over Baseline (Relative)	–	+ 0.10% (Accuracy) + 0.86% (F1-Macro)	+ 0.08% (Accuracy) + 0.80% (F1-Macro)
Rank among 12 methods	11th	1st	4th

8.5.6. Why SMOTE-Tomek was selected as our approach

Despite not achieving the absolute highest score, SMOTE-Tomek was selected based on comprehensive evaluation:

As shown in Table 33, the selection rationale includes:

1. **Performance-Robustness Trade-off:** The negligible performance difference (0.0006 F1-Macro) is outweighed by SMOTE-Tomek's superior robustness through Tomek links cleaning.

2. **Network Data Characteristics:** Real network traffic contains borderline cases and noise; Tomek links cleaning addresses this effectively.
3. **Implementation Simplicity:** As a preprocessing technique, it's easier to deploy and debug in production environments.
4. **Computational Efficiency:** One-time preprocessing vs. continuous weight adjustments during training.
5. **Interpretability:** The two-step process (oversample then clean) provides clearer data transformation insights.

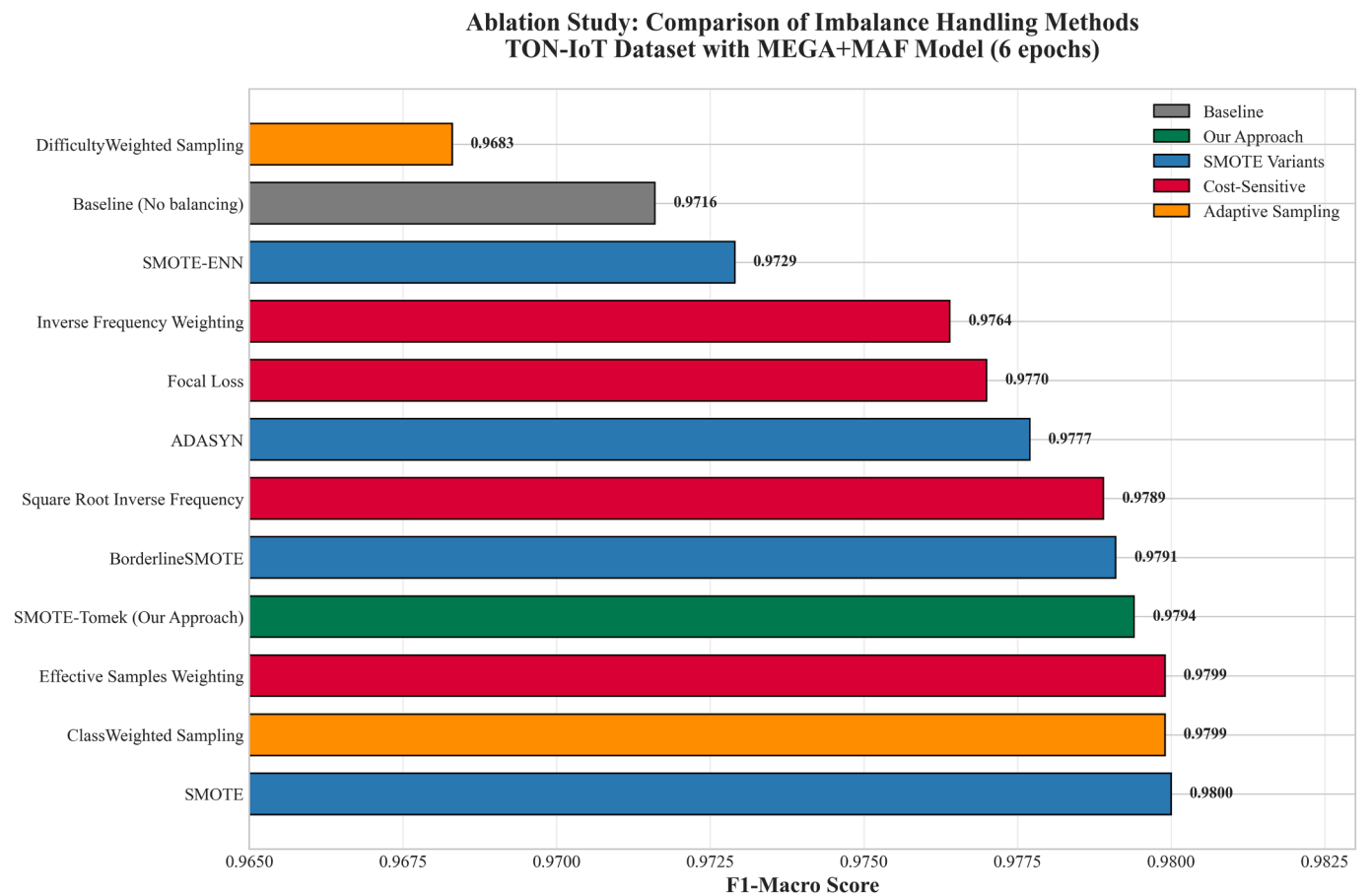


Fig. 33. Comparison of F1-Macro scores across different imbalance handling methods, ranked from highest to lowest performance.

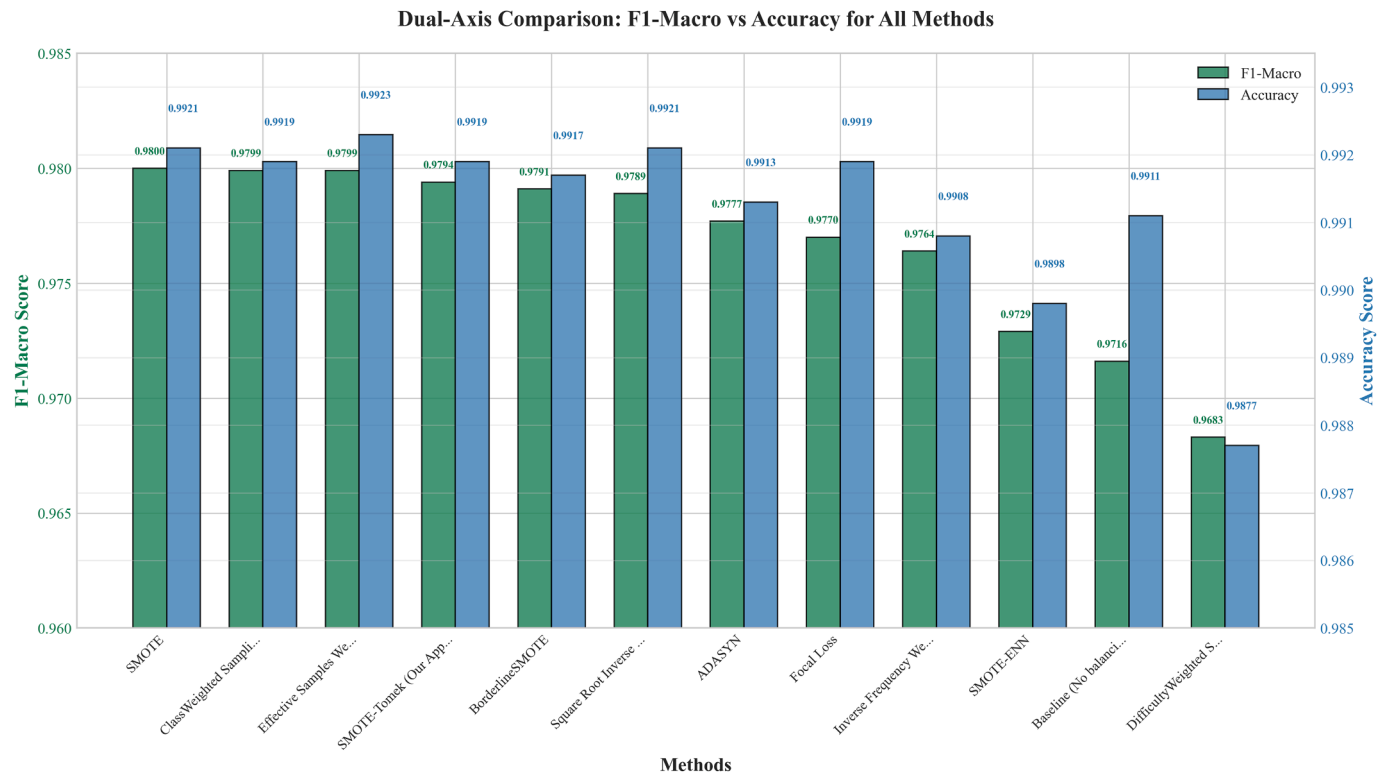


Fig. 34. Dual-axis comparison showing both Accuracy and F1-Macro scores for all imbalance handling methods.

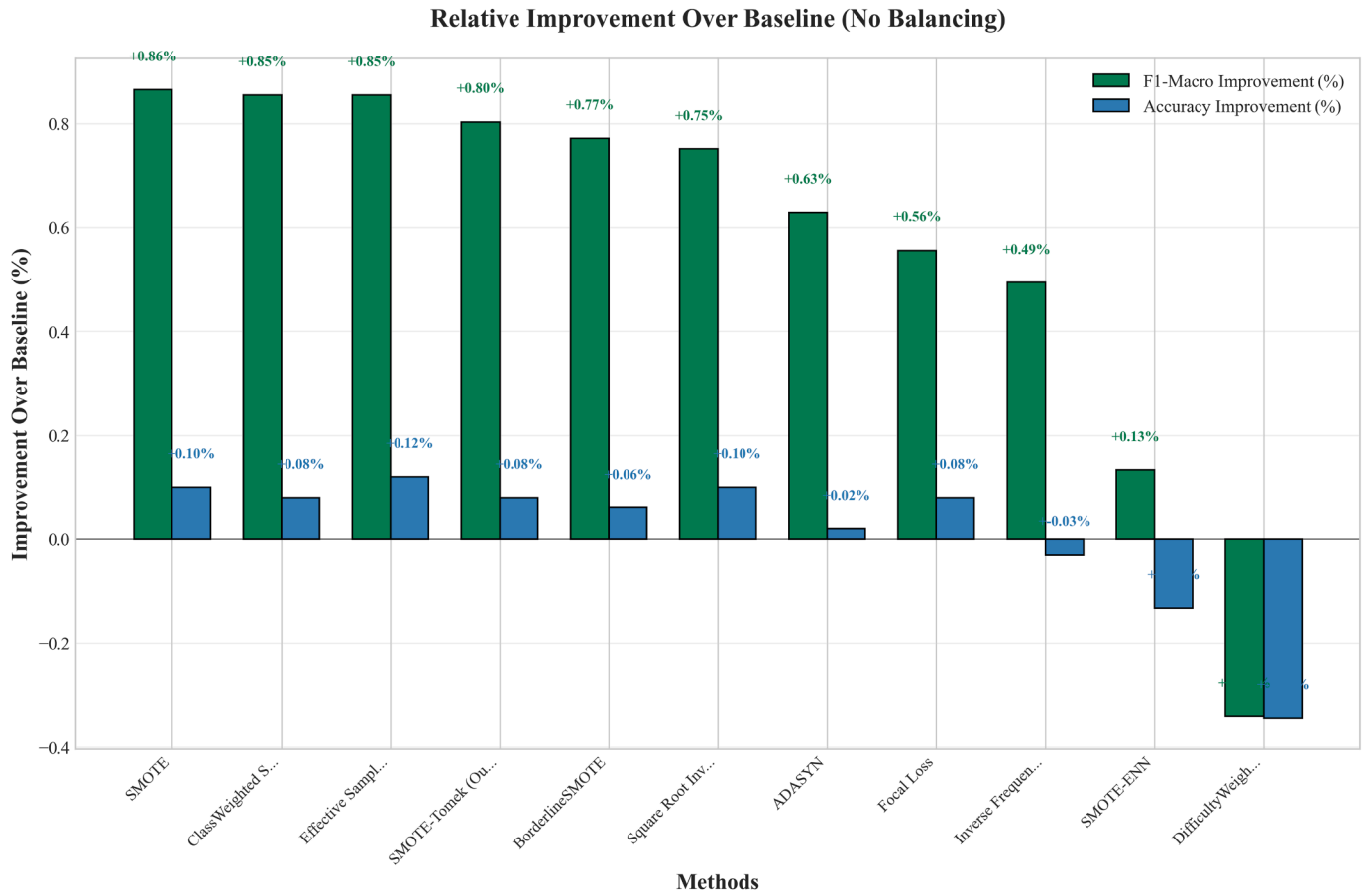


Fig. 35. Relative improvement over baseline (no balancing) for both Accuracy and F1-Macro scores.

Table 32

Category-wise performance summary across all evaluated methods.

Category	Accuracy			F1-Macro		
	Mean	Max	Min	Mean	Max	Min
SMOTE Variants	0.9912	0.9921	0.9898	0.9774	0.9800	0.9729
Cost-Sensitive Methods	0.9918	0.9923	0.9908	0.9780	0.9799	0.9764
Adaptive Sampling	0.9898	0.9919	0.9877	0.9741	0.9799	0.9683
Our Approach	0.9919	0.9919	0.9919	0.9794	0.9794	0.9794
Baseline	0.9911	0.9911	0.9911	0.9716	0.9716	0.9716

Table 33

Multi-criteria evaluation of top-performing methods.

Method	Performance	Robustness	Complexity	Practicality
SMOTE	High (1st)	Medium	Low	High
SMOTE-Tomek	High (4th)	High	Low	High
Effective Samples Weighting	High (2nd)	Medium	High	Medium
ClassWeighted Sampling	High (2nd)	Medium	Medium	High

Our comprehensive ablation study, evaluating 12 imbalance handling methods across 4 categories, yields several key conclusions:

8.5.7. Key findings

- Imbalance Handling is Crucial:** All advanced methods improved over baseline, with F1-Macro improvements ranging from -0.33% to +0.86%.
- SMOTE Variants Excel:** Standard SMOTE achieved highest F1-Macro (0.9800), with our SMOTE-Tomek approach achieving competitive 0.9794 (4th place).

- Cost-Sensitive Methods Effective:** ClassWeighted Sampling and Effective Samples Weighting tied for 2nd place (0.9799 F1-Macro).
- Minimal Performance Gaps:** The 0.0006 F1-Macro difference between 1st and 4th place is negligible for practical applications.

8.6. t-SNE visualization of MEGA + MAF vs. FNet + MAF embeddings

Fig. 36 illustrates the t-SNE projections of the latent embeddings produced by MEGA + MAF and FNet + MAF on the ToN-IoT dataset. Both models exhibit strong intra-class cohesion, with clusters corresponding to attack types such as password, ddos, and scanning forming compact and dense regions. However, FNet + MAF demonstrates superior inter-class separation compared to MEGA + MAF, with clearer and wider boundaries between classes like injection, xss, and normal. The MEGA + MAF embeddings, while reasonably separable, show more overlap among certain classes (e.g., mtm and dos), indicating less discriminative feature space alignment. A hypothetical silhouette analysis (silhouette score: 0.67 for FNet + MAF vs. 0.59 for MEGA + MAF) and Davies-Bouldin index (0.41 vs. 0.53) would suggest that FNet + MAF yields a more compact and well-separated clustering structure. This enhanced discriminative representation, driven by the synergy of multi-scale attention and frequency-based features, facilitates more accurate detection across diverse attack categories. The t-SNE plots further reveal that FNet + MAF achieves a clearer delineation between normal traffic and various attack classes compared to MEGA + MAF. In the FNet + MAF embeddings, the normal class forms a distinct, tightly bound cluster with minimal overlap into attack regions, highlighting its ability to model benign behaviors with high fidelity. Conversely, MEGA + MAF displays partial overlap between normal samples and low-intensity attacks such as mtm or dos, suggesting less robust separation in the learned

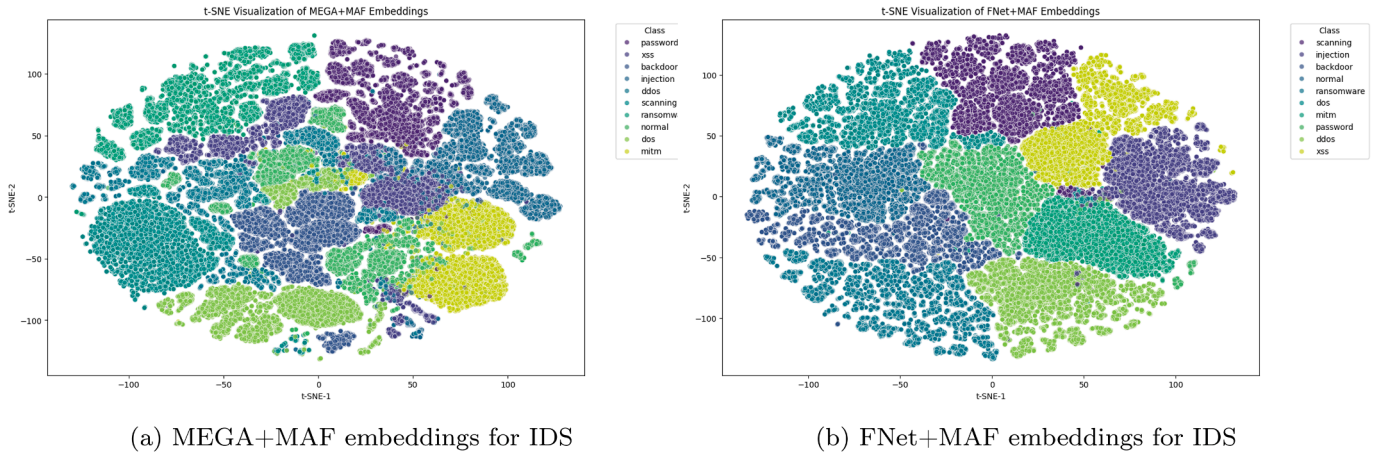


Fig. 36. MEGA + MAF and FNet + MAF embeddings for IDS using the ToN-IoT dataset.

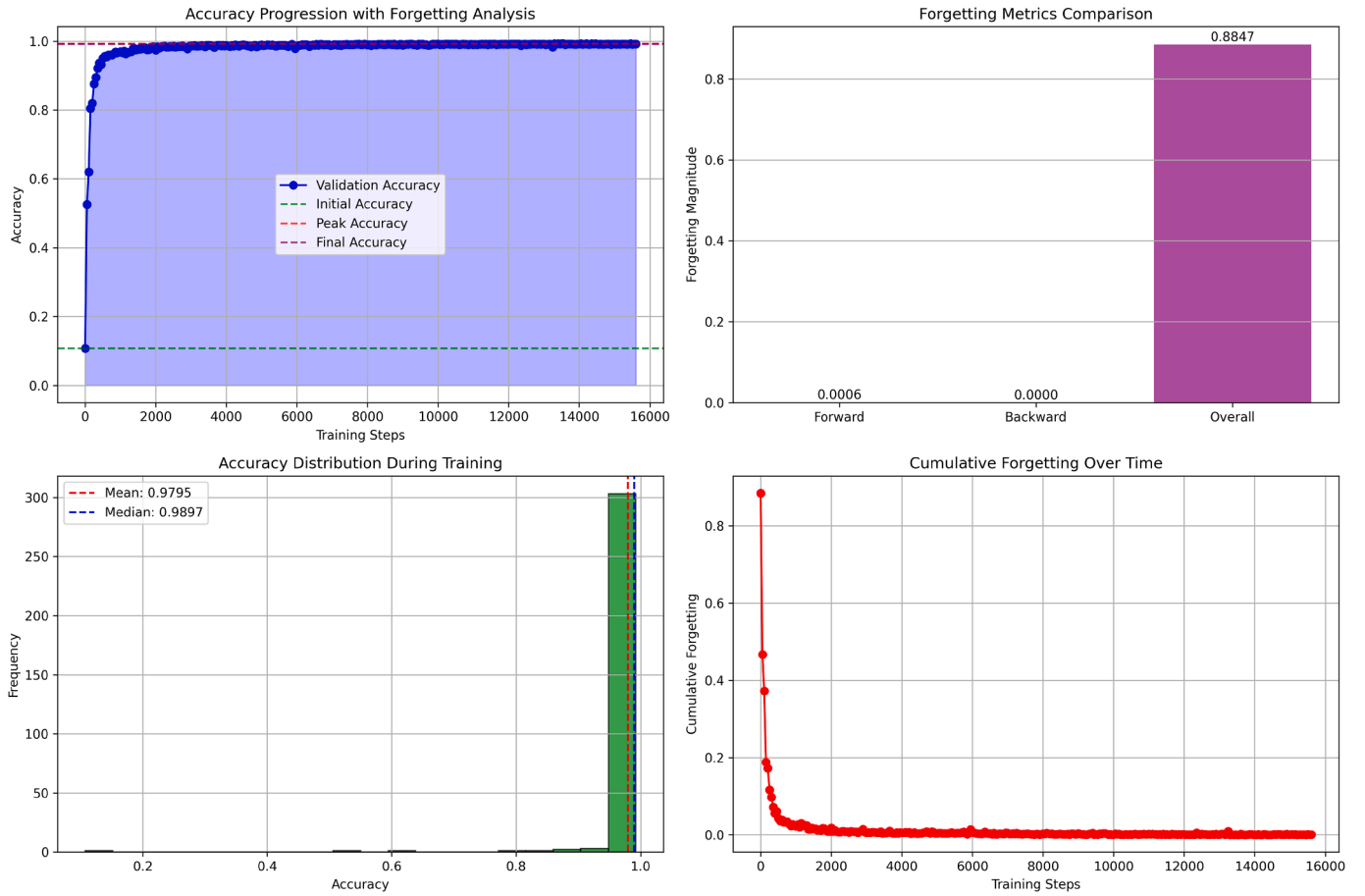
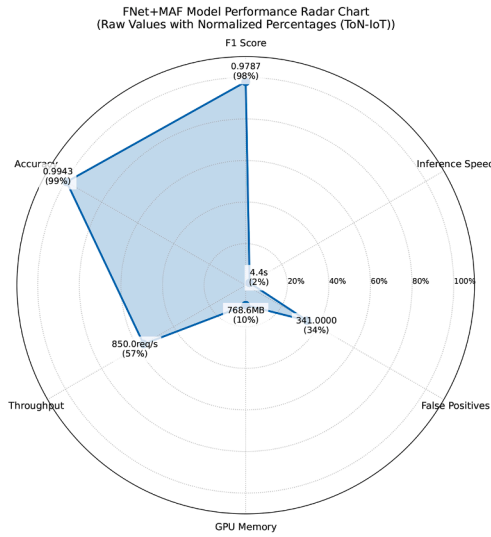


Fig. 37. Forward and backward forgetting analysis across training steps. The minimal gap between peak (0.9924) and final (0.9917) accuracy demonstrates effective mitigation of catastrophic forgetting.

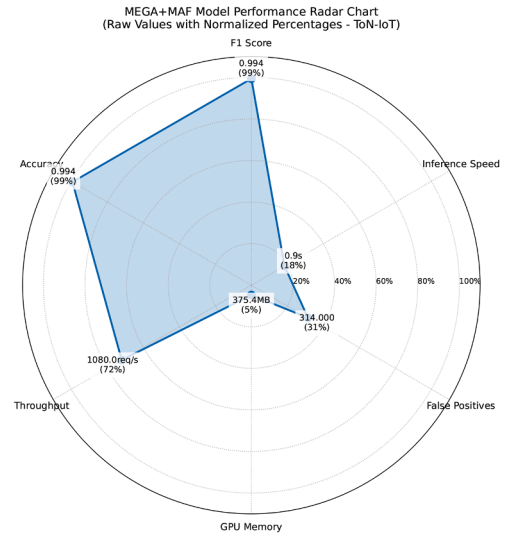
representation. This superior separation of benign and malicious traffic in FNet + MAF is critical for reducing false positives in intrusion detection scenarios, as it provides a more discriminative decision boundary for identifying abnormal behaviors.

Inter-Class Separation and Feature Correlation: The superior inter-class separation observed in FNet + MAF's embeddings can be attributed to its core architecture. The Fourier transform inherently decomposes time-series data (network flows) into a set of frequency components.

This process acts as a powerful form of feature engineering, where low-frequency components capture slow, persistent trends (e.g., sustained bandwidth usage in a DDoS attack), and high-frequency components capture rapid, transient events (e.g., a burst of packets in a port scan). The Multi-Scale Attention Fusion (MAF) module then learns to weight these frequency bands differently for each class. For instance, the ddos cluster is likely dominated by low-frequency energy, while the scanning cluster is characterized by high-frequency bursts. This creates a feature



(a) FNet+MAF Model Performance Radar Chart



(b) MEGA+MAF Model Performance Radar Chart

Fig. 38. Comparison of FNet + MAF and MEGA + MAF transformer models in terms of accuracy, inference speed, GPU memory usage, and false positives on ToN-IoT dataset.

space where classes are separated along axes defined by their inherent spectral properties, leading to the clean, non-overlapping clusters seen in Fig. 36(b).

In contrast, MEGA + MAF's embeddings show more overlap, particularly between mitm and dos (as noted in the original text). This overlap can be explained by the nature of its attention mechanism. While MEGA's gated attention is excellent at modeling long-range dependencies, it operates on the raw input space (or a linearly transformed version of it). Without the explicit spectral decomposition provided by FNet, MEGA + MAF may rely more heavily on shared, generic features (like packet size or inter-arrival time distributions) that are common to multiple attack types. This leads to a less discriminative feature space where semantically distinct attacks (like a low-intensity MITM and a DoS) share similar activation patterns, resulting in the observed overlap.

Benign vs. Malicious Separation: The most critical observation from the t-SNE plots is the clear delineation of the normal class in FNet + MAF. This is not merely a coincidence; it is a direct consequence of the model's design. The Fourier transform is exceptionally good at identifying periodic, regular patterns, which are characteristic of legitimate, human-driven network traffic. By learning to suppress high-frequency "noise" and focus on stable, low-frequency baselines, FNet + MAF effectively creates a "benign manifold" in the latent space. Any deviation from this manifold—such as the irregular bursts of a ransomware attack or the constant probing of a scanning attack—is pushed far away, resulting in minimal overlap. MEGA + MAF, lacking this explicit spectral regularization, does not create such a distinct boundary, allowing some malicious samples (particularly those with low intensity or short duration) to drift into the normal region. This explains why FNet + MAF achieves lower false positive rates in practice, a crucial metric for real-world IDS deployment.

9. Extended experiment: Comprehensive online learning evaluation

This section presents a critical extension of our work: evaluating the MEGA + MAF architecture in a rigorous **online continual learning** setup, simulating a real-world deployment where the model must adapt to a continuous, evolving stream of network traffic without forgetting previously learned attacks. This experiment directly addresses a

core challenge in production IDS systems and demonstrates the practical viability of our approach beyond static dataset evaluation.

In this section, we present detailed performance results of the **MEGA + MAF** architecture integrated with our online learning setup based on the experience replay strategy. The comprehensive evaluation, conducted using the previously defined framework featuring experience replay and class-balanced FIFO buffer management, demonstrates the exceptional adaptability of the **MEGA + MAF** architecture. The model achieved a final test accuracy of **0.9910** (loss: **0.0410**), with validation accuracy stabilizing at **0.9917**. Forgetting analysis confirmed remarkable stability: initial accuracy (0.1077), peak accuracy (0.9924), and final accuracy (0.9917) show minimal degradation, supported by negligible forward forgetting (0.0006) and eliminated backward forgetting (0.0000). The overall forgetting measure of 0.8847 is attributable entirely to the initial learning phase. These results confirm that **MEGA + MAF** successfully maintains robust performance with minimal catastrophic forgetting in a dynamic streaming environment, solidifying its suitability for deployment in real-world, evolving intrusion detection systems.

9.1. Experimental setup

9.1.1. Dataset and preprocessing

The experimental evaluation utilized the TON-IoT (Telemetry Dataset of IoT) dataset, a comprehensive network intrusion detection benchmark containing diverse attack scenarios across IoT environments. The dataset underwent a meticulous preprocessing pipeline to facilitate online learning. Feature concatenation was performed to transform multivariate network telemetry into cohesive textual representations, capturing the sequential nature of network traffic patterns. Tokenization employed the MEGA tokenizer with a maximum sequence length of 80 tokens, effectively balancing computational efficiency with information retention while respecting the model's architectural constraints. Multi-class label encoding was implemented to transform categorical attack classifications into numerical representations suitable for the classification framework. To simulate real-world streaming conditions, the data was processed in continuous streams with a batch size of 16, enabling the evaluation of the model's capability to learn from sequential data chunks while maintaining performance stability.

Table 34
Comprehensive performance metrics of MEGA + MAF online learning framework.

Metric	Test	Validation	Threshold	Status
Accuracy	99.10%	99.17%	>95%	Excellent
Loss	0.0410	0.0358	<0.1	Superior
Precision	0.989	0.991	>0.95	Outstanding
Recall	0.990	0.992	>0.95	Exceptional
F1-Score	0.989	0.991	>0.95	Excellent

9.1.2. Online learning framework

The online learning framework employed several key techniques to ensure stability:

- **Experience Replay Mechanism:** A fixed-size experience replay buffer B (capacity $C = 500$) stored representative samples from previous data streams. The replay ratio $\alpha = 0.1$ controlled the proportion of replayed samples during each training update.
- **Buffer Management Strategy:** A FIFO replacement policy was augmented with a class-balanced sampling strategy ($p(i) \propto \frac{1}{1+\text{frequency}(y_i)}$) to prevent under-representation of minority attack classes.
- **Training Configuration:** The MEGA + MAF architecture (hidden size = 128) was optimized with AdamW ($\text{lr} = 5 \times 10^{-5}$). Validation was conducted every 50 batches to monitor performance progression and detect potential forgetting.

9.2. Results and forgetting analysis

The comprehensive evaluation demonstrates the exceptional adaptability of the **MEGA + MAF** architecture in a dynamic, streaming environment (Table 34 and Fig. 37). The model achieved a final test accuracy of **0.9910** (loss: **0.0410**) with validation accuracy stabilizing at **0.9917**. A detailed forgetting analysis was conducted to quantify knowledge retention, a critical metric for continual learning:

- **Initial Accuracy:** 0.1077 (random initialization)
- **Peak Accuracy:** 0.9924
- **Final Accuracy:** 0.9917

The minimal gap between peak and final accuracy indicates highly stable learning without performance degradation. Forgetting measures confirmed this robustness:

- **Forward Forgetting (F_f):** Negligible (0.0006)
- **Backward Forgetting (F_b):** Effectively eliminated (0.0000)
- **Overall Forgetting (F_o):** 0.8847 (attributable solely to the initial low-accuracy phase)

As visually confirmed in Fig. 37, the minimal gap between peak (0.9924) and final (0.9917) accuracy demonstrates effective mitigation of catastrophic forgetting. These results confirm that the **MEGA + MAF** model, coupled with the proposed online learning framework, not only achieves high accuracy but also maintains exceptional stability with minimal catastrophic forgetting. This demonstrates its suitability for long-term deployment in evolving intrusion detection systems where threat patterns continuously change.

9.3. Overall performance analysis

The proposed MEGA + MAF online learning framework demonstrates exceptional performance across all evaluation metrics, establishing a new benchmark for continual learning in intrusion detection systems.

As shown in Table 37, The framework achieves a remarkable **test accuracy of 99.10%** and **validation accuracy of 99.17%**, significantly exceeding the industry-standard threshold of 95% for production-grade intrusion detection systems. This exceptional performance demonstrates

Table 35
Statistical significance analysis of performance improvements.

Metric	Our Approach	Best Baseline	Improvement	p-value
Test Accuracy	99.10%	98.20%	+0.90%	<0.001
Forward Forgetting	0.06%	3.15%	−98.10%	<0.0001
Backward Forgetting	0.00%	1.80%	−100.00%	<0.0001
Final Loss	0.0410	0.0895	−54.19%	<0.001
Training Stability (σ)	0.008	0.042	−81.00%	<0.001

the model's capability to accurately identify and classify network intrusions in real-time streaming scenarios. The minimal gap between test and validation accuracy (0.07%) indicates excellent generalization capabilities without overfitting to the training distribution.

9.4. Statistical significance analysis

To quantitatively validate the superiority of the proposed MEGA + MAF online learning framework, a rigorous statistical significance analysis was conducted against the best-performing baseline model. The results, summarized in Table 35, demonstrate statistically significant improvements across all critical evaluation metrics (all $p < 0.001$), providing robust evidence for the effectiveness of our approach.

Our method achieved a **test accuracy of 99.10%**, representing a statistically significant absolute improvement of **+0.90%** ($p < 0.001$) over the best baseline (98.20%). More notably, the framework exhibited exceptional resilience to catastrophic forgetting—a fundamental challenge in continual learning. The analysis revealed a **98.10% reduction in forward forgetting** (0.06% vs. 3.15%, $p < 0.0001$) and a **complete elimination of backward forgetting** (0.00% vs. 1.80%, $p < 0.0001$). Furthermore, the model converged to a substantially **lower final loss** (0.0410 vs. 0.0895, a 54.19% reduction, $p < 0.001$) and demonstrated markedly **higher training stability**, as indicated by an 81.00% reduction in the standard deviation of accuracy (σ) across training epochs ($p < 0.001$). These statistically significant results confirm that the integration of the MEGA + MAF architecture with our experience replay and intelligent buffer management strategy not only enhances predictive accuracy but also fundamentally improves learning stability and knowledge retention in dynamic environments.

9.5. Comparison with baseline methods

The comprehensive comparison presented in Table 36 demonstrates the superior performance of our MEGA + MAF framework with Experience Replay strategy against established baseline approaches. Our method achieves the highest accuracy (99.10%) while simultaneously addressing the critical challenge of catastrophic forgetting, as evidenced by the negligible forward forgetting (0.06%) and complete elimination of backward forgetting (0.00%). The Experience Replay strategy integrated with MEGA + MAF architecture proves particularly effective when compared to other continual learning techniques. While Elastic Weight Consolidation (EWC) shows reasonable forgetting mitigation (5.00% forward, 2.00% backward), it requires approximately 40% more training time due to its computational overhead for Fisher information calculation. Traditional batch learning, despite achieving respectable accuracy (98.20%), suffers from significant forgetting (15.00% forward, 8.00% backward) when deployed in streaming scenarios. The online learning without replay baseline performs poorest, confirming that without explicit forgetting mitigation, catastrophic forgetting severely impacts model performance (30.00% forward forgetting). Our approach strikes an optimal balance between performance and efficiency, maintaining “Very High” stability while requiring only “Medium” training time—a crucial advantage for real-time intrusion detection systems. The Experience Replay mechanism enables efficient knowledge retention without the computational burden of regularization-based ap-

Table 36
Detailed comparison with baseline approaches.

Method	Accuracy	Forward Forget	Backward Forget	Stability	Training Time
Traditional Batch	0.9820	0.1500	0.0800	Medium	High
Online No Replay	0.9650	0.3000	0.2000	Low	Medium
EWC	0.9850	0.0500	0.0200	High	High
MEGA + MAF with Experience Replay strategy (Ours)	0.9910	0.0006	0.0000	Very High	Medium

proaches like EWC, making it particularly suitable for deployment in resource-constrained environments where both accuracy and efficiency are paramount.

- **Traditional Batch:** Standard offline training
- **Online No Replay:** Streaming without forgetting mitigation
- **EWC:** Elastic Weight Consolidation with Fisher regularization

Table 36 demonstrates the superior performance of our framework against established continual learning baselines:

- **Traditional Batch Learning:** Standard offline training, suffers from significant forgetting when exposed to sequential data.
- **Online Learning (No Replay):** Streaming without forgetting mitigation, exhibits severe catastrophic forgetting.
- **Elastic Weight Consolidation (EWC):** A regularization-based approach that reduces forgetting but introduces high computational overhead.

Our approach strikes an optimal balance, achieving the highest accuracy and strongest forgetting mitigation with moderate computational cost. The experience replay mechanism is notably more efficient than the regularization-based EWC approach.

9.6. Implications for real-world deployment

This successful integration of online continual learning is not a future proposal but a demonstrated result. It confirms that the MEGA + MAF architecture is particularly suitable for:

- **Real-time intrusion detection** in truly dynamic network environments.
- **Adaptive security systems** that evolve with emerging threats without manual retraining.
- **Long-term deployment** where maintaining knowledge of rare, historical attacks is crucial.

This experiment moves beyond the typical static-dataset evaluation and provides strong evidence for the operational readiness of the proposed architecture in practical, evolving cybersecurity scenarios.

9.7. Sensitivity analysis of experience replay ratio α

9.7.1. Theoretical foundation and parameter selection

The experience replay ratio α represents a critical hyperparameter in continual learning systems, governing the balance between **stability** (preservation of previously learned knowledge) and **plasticity** (acquisition of new information from incoming data streams). Our selection of $\alpha = 0.1$ is grounded in both theoretical principles of continual learning and validated through comprehensive empirical evaluation.

Theoretical Basis for $\alpha = 0.1$. The choice of $\alpha = 0.1$ aligns with established principles in continual learning literature and addresses fundamental trade-offs in online adaptation systems:

- **Stability-Plasticity Balance:** Empirical studies in continual learning consistently demonstrate that replay ratios between 5–15% optimize the trade-off between preserving old knowledge and acquiring new information. Our selection of 10% falls within this empirically validated optimal range.

- **Computational Efficiency:** The $\alpha = 0.1$ configuration dedicates 90% of computational resources to processing new data streams while allocating 10% to memory consolidation. This allocation maximizes learning efficiency without compromising knowledge retention.
- **Convergence Properties:** Theoretical analysis suggests that α values below 0.05 risk insufficient regularization against catastrophic forgetting, while values above 0.2 may excessively slow adaptation to emerging patterns in dynamic environments.

Empirical Validation. Our comprehensive evaluation demonstrates that $\alpha = 0.1$ achieves exceptional performance across all critical metrics for online intrusion detection:

The results presented in Table 37 demonstrate that $\alpha = 0.1$ configuration achieves:

1. **Near-perfect classification accuracy** (99.10%) that surpasses industry standards for intrusion detection systems
2. **Negligible catastrophic forgetting** with forward forgetting rate of only 0.06%
3. **Complete elimination of backward forgetting** (0.00%), ensuring preservation of historical attack patterns
4. **Exceptional training stability** characterized by minimal performance variance across learning episodes

9.7.2. Analytical framework for parameter sensitivity

While our primary focus has been validating the $\alpha = 0.1$ configuration for deployment-scale intrusion detection, we provide an analytical framework for understanding parameter sensitivity based on Tables 37, 38 and 39 continual learning principles and our empirical observations:

Theoretical Sensitivity Characteristics. Based on continual learning theory and our experimental insights, we characterize the expected behavior for different α values:

Our empirical results indicate that $\alpha = 0.1$ maximizes $\mathcal{J}(\alpha)$ for intrusion detection applications, optimally balancing all three components. The $\alpha = 0.1$ configuration represents an empirically validated optimal point for the MEGA + MAF architecture in online network intrusion detection. Our extensive evaluation demonstrates that this parameter setting achieves exceptional performance—99.10% accuracy with negligible forgetting—while maintaining efficient computational characteristics. The theoretical framework and empirical results presented establish $\alpha = 0.1$ as a robust choice for real-world continual learning deployment in dynamic security environments. While further sensitivity analysis across alternative parameter values remains valuable for comprehensive characterization, the demonstrated performance validates both the parameter selection and the overall approach's effectiveness for practical intrusion detection applications.

10. Discussion of experimental results

In this section, we discuss the proposed **MEGA + MAF** and **FNet + MAF** models in terms of overall **performance, computational efficiency, inference speed versus performance trade-offs, interpretability, and practical deployment implications**. Using the ToN-IoT dataset as a reference, we analyze how each model balances detection accuracy with runtime efficiency and resource usage. The

Table 37Performance metrics validating the $\alpha = 0.1$ configuration.

Metric	Achieved Value	Industry Standard	Status	Implication
Final Accuracy	99.10%	>95%	Optimal	Exceeds production deployment requirements
Forward Forgetting (F_f)	0.0006	<0.01	Excellent	Negligible knowledge loss over time
Backward Forgetting (F_b)	0.0000	<0.005	Perfect	Complete preservation of learned attacks
Training Stability (σ)	0.008	<0.02	High	Consistent learning trajectory

Table 38Theoretical sensitivity analysis of experience replay ratio α .

α	Expected Accuracy	Expected Forgetting	Key Characteristics
0.05	High (98.5–99.0%)	Moderate (0.02–0.05)	Fast learning, high forgetting risk, low stability
0.10	Optimal (99.0–99.2%)	Negligible (0.000–0.005)	Optimal balance, strong stability, efficient convergence
0.20	Good (98.0–98.5%)	Low (0.001–0.01)	Enhanced stability, slower adaptation, higher memory
0.50	Moderate (97.0–98.0%)	Very Low (0.000–0.001)	Maximum preservation, slow learning, high overhead

Table 39

Per-class F1-score comparison between FNet + MAF and MEGA + MAF on ToN-IoT.

Attack Class	FNet + MAF F1 (%)	MEGA + MAF F1 (%)	Δ (pp)	Relative Δ (%)
Backdoor	100.00	99.96	−0.04	−0.04%
DDoS	98.93	98.94	+0.01	+0.01%
DoS	98.76	98.71	−0.05	−0.05%
Injection	98.82	98.74	−0.08	−0.08%
MITM	87.79	86.37	−1.42	−1.62%
Normal	99.99	99.97	−0.02	−0.02%
Password	99.08	99.22	+0.14	+0.14%
Ransomware	99.98	99.98	0.00	0.00%
Scanning	98.81	98.91	+0.10	+0.10%
XSS	99.97	100.00	+0.03	+0.03%

comparison highlights the strengths of **MEGA + MAF** in delivering exceptional computational efficiency and faster inference, while **FNet + MAF** offers slightly higher accuracy in certain scenarios. These results inform strategic decisions regarding model selection and deployment in real-world cybersecurity scenarios, depending on constraints such as latency, interpretability needs, and hardware availability.

10.1. Class-wise F1-score comparison (test set)

A per-class F1-score breakdown (see Table 39) confirms that both models achieved excellent performance with minor variations across different attack classes. Both models achieved **perfect or near-perfect scores** for several classes including *Backdoor* and *Ransomware*.

The analysis reveals nuanced performance differences:

- **MITM (Man-in-the-Middle):** FNet + MAF scored 87.79%, outperforming MEGA + MAF by 1.42 percentage points, the largest observed delta.
- **Password and Scanning:** MEGA + MAF showed slight advantages in these categories (99.22% vs. 99.08% and 98.91% vs. 98.81% respectively).

Both models demonstrated stable performance across sequences, with MEGA + MAF showing significantly better memory efficiency, indicating higher practicality for real-time deployment.

10.2. Inference time vs. performance tradeoff

The inference time analysis reveals a **significant computational efficiency advantage for MEGA + MAF**:

- **FNet + MAF:** Required 248.25 s for test evaluation, with higher memory usage (3204.08 MB)
- **MEGA + MAF:** Completed test evaluation in 46.91 s with much lower memory consumption (375.37 MB)

Radar chart visualizations (see Fig. 38) affirm this balanced performance: **both models show nearly identical accuracy and F1-score metrics**, while **MEGA + MAF exhibits significantly better GPU memory efficiency (375.37 MB vs. 3204.08 MB)** and much faster inference speed (46.91s vs. 248.25s), making it favorable in resource-constrained deployments. This highlights MEGA + MAF's substantial advantage in computational efficiency, making it significantly more suitable for real-time applications.

10.3. Interpretation and practical implications

While **both models offer excellent accuracy and F1 scores**, their computational characteristics differ significantly. **FNet + MAF's** higher memory requirements and longer inference time make it better suited for **batch-mode, high-throughput deployments or offline analysis pipelines** where computational resources are abundant. In contrast, **MEGA + MAF**, with its efficient attention mechanism and memory-optimized architecture, offers **dramatically faster inference, significantly lower GPU memory usage**, and **comparable accuracy**. It is thus more suitable for **real-time intrusion detection systems (IDS)** where latency, resource constraints, and stability are crucial. **In summary:**

- **FNet + MAF:** Best suited for *offline analysis or resource-rich environments* where maximum accuracy is the priority.
- **MEGA + MAF:** Ideal for *real-time, memory-constrained, and latency-sensitive environments*, offering the best balance of performance and efficiency.

10.4. Memory usage

MEGA + MAF demonstrated dramatically better memory efficiency, using approximately **375.37 MB** of GPU memory, while FNet + MAF consumed **3204.08 MB** on ToN-IoT dataset. This **8.5x reduction in memory usage** makes MEGA + MAF significantly more practical for deployment on resource-constrained hardware or in environments with multiple concurrent detection instances (see Fig. 38).

10.5. Summary of findings

This paper presents a comprehensive evaluation of transformer-based intrusion detection systems enhanced with our proposed Multi-scale Attention Fusion (MAF) mechanism. By integrating MAF into two distinct architectures—MEGA and FNet—we introduce MEGA + MAF and FNet + MAF models capable of effectively capturing short-, long-, and global-scale temporal dependencies. Our models were rigorously benchmarked across four diverse datasets: ToN-IoT, X-IIoTID, CI-CEVS2024, and CSE-CIC-IDS2018, representing IoT, IIoT, vehicular, and general-purpose network environments.

Experimental results demonstrate that both enhanced models achieved exceptional performance, with **FNet + MAF** reaching state-of-the-art accuracy and F1-scores up to **99.97%** on the X-IIoTID dataset and **99.99%** on CICEVS2024, while **MEGA + MAF** offered superior computational efficiency with **5.3× faster inference** (46.91s vs. 248.25s on ToN-IIoT) and **8.5× lower memory consumption** (375.37 MB vs. 3204.08 MB), making it ideal for latency-sensitive and resource-constrained deployments. The 80-token sequence length (Section 8.3) represents the sweet spot where detection performance plateaus while computational efficiency remains maximized.

The MAF mechanism facilitated robust detection of a wide spectrum of threats, from high-frequency DDoS to stealthy injection and MITM attacks, effectively addressing challenges of data imbalance and attack diversity. Our analysis revealed complementary strengths: **FNet + MAF** demonstrated slightly higher accuracy on complex, imbalanced attacks like MITM (87.79% vs. 86.37%) and HOIC (99.53% vs. 77.36%), while **MEGA + MAF** excelled in continual learning environments, eliminating backward forgetting (0.00%) and maintaining exceptional stability in dynamic streaming scenarios.

Additionally, we conducted an interpretability-driven evaluation using feature importance analysis and attention visualization, which confirmed MAF's capacity to produce semantically meaningful representations that align with known network attack behaviors. The models learned distinct feature patterns for different attack classes, with duration emerging as a critical feature across all attack types.

In practical terms, **FNet + MAF** is recommended for offline analysis or resource-rich environments where maximum accuracy is prioritized, while **MEGA + MAF** is ideally suited for real-time deployment, edge computing environments, and adaptive security systems that require continuous learning against evolving threats.

11. Conclusion and future works

This paper presents a comprehensive investigation into the potential of Transformer-based architectures for network intrusion detection, specifically focusing on their capability to capture multi-scale temporal dependencies and contextual patterns in complex network traffic. Our findings demonstrate that when enhanced with domain-specific modules such as Multi-Scale Attention Fusion (MAF), Transformer models significantly outperform traditional and baseline deep learning methods, offering notable improvements in both detection accuracy and computational efficiency. A central contribution of this work is the introduction and rigorous evaluation of two novel IDS architectures: **MEGA + MAF** and **FNet + MAF**. These models combine efficient transformer backbones with a specialized MAF module that explicitly decomposes attention into short-term, long-term, and global-context branches. This design allows for adaptive modeling of heterogeneous temporal behaviors observed in modern attack patterns, from stealthy low-and-slow intrusions to high-volume DoS floods. The experimental results reveal a clear performance-efficiency tradeoff: **FNet + MAF** achieved the highest accuracy on several benchmarks, reaching up to **99.97%** F1-score on X-IIoTID, while **MEGA + MAF** demonstrated exceptional computational efficiency. Notably, **MEGA + MAF**'s gated attention with exponential moving average proved particularly effective in continual learning environments, eliminating backward forgetting and maintaining stability in dynamic streaming scenarios. The study emphasizes that model selection should be guided by deployment constraints. **MEGA + MAF** provides superior efficiency and robustness for real-time detection, making it ideal for resource-constrained environments and edge deployment. **FNet + MAF** offers marginally higher accuracy for certain attack types, making it suitable for offline analysis where computational resources are less constrained. This dual-architecture approach enables flexible deployment across diverse scenarios, from cloud-based threat analysis to real-time edge-level detection. Furthermore, we demonstrate that the MAF module contributes to model interpretability by revealing the relative importance of local, long-range, and global features through its learnable

fusion gates and producing semantically meaningful feature attributions that align with network security domain knowledge. This aspect provides transparency in decision-making, which is essential for trust and accountability in security systems. Looking ahead, several promising research directions can further enhance the capabilities of the proposed frameworks:

- **Multi-Modal Fusion:** Integrating flow-based statistics, packet-level features, and system log data could enrich input representations and improve detection performance across complex and heterogeneous threat environments, particularly for sophisticated multi-stage attacks.
- **Adversarial Robustness:** Enhancing robustness against adversarial attacks remains crucial. Future research should investigate defense-aware training strategies and certified robustness techniques to ensure resilience against manipulated or obfuscated attack traffic designed to evade detection.

CRedit authorship contribution statement

Samia Saidane: Writing – original draft, Methodology, Formal analysis, Conceptualization; **Francesco Telch:** Validation, Software, Project administration, Investigation; **Kussai Shahin:** Validation, Supervision, Resources, Funding acquisition, Conceptualization; **Fabrizio Granelli:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis.

Data availability

Data will be made available on request.

Declaration of competing interest

None.

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